Information Management Handbook

ISO 19650 information management in practice — process, documents, CDE, level of information need, asset information and handover

BIM Management Portal · ISO 19650 & ISO/IEC 42001
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A practical reference for appointing parties, lead appointed parties and task teams. It explains the standards in our own words; refer to the published standards for their exact requirements.

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1. The ISO 19650 series

ISO 19650 sets out how information about built assets is specified, produced, shared, checked and handed over using building information modelling. This handbook summarises the practice as implemented in the platform; it explains the standards in our own words and does not reproduce their text.

Part 1 — Concepts and principles 2018

Defines the concepts and principles of information management using BIM across the whole asset life cycle: information requirements, information models, the information delivery cycle, the CDE and the roles of appointing and appointed parties.

Key topics: OIR / AIR / PIR / EIR hierarchy · PIM and AIM · Level of information need · Information container · CDE concept · Delivery team structure

Part 2 — Delivery phase of the assets 2018

Specifies the information management process during the delivery phase through eight activities, from assessment and need to project close-out.

Key topics: Information management process (5.1–5.8) · EIR and BEP · MIDP / TIDP · Responsibility matrix · Information protocol · Authorization and acceptance

Part 3 — Operational phase of the assets 2020

Specifies the information management process during operation, including trigger events, the AIM and its maintenance, and integration with asset management (ISO 55000).

Key topics: Trigger events · AIM maintenance · OIR to AIR · Asset management integration

Part 4 — Information exchange 2022

Details the process and criteria for each information exchange, including CDE state transitions and the review criteria for sharing, authorization and acceptance.

Key topics: Exchange criteria · Status and revision · Review criteria · Open data / interoperability

Part 5 — Security-minded approach 2020

Specifies a security-minded approach to information management: sensitivity assessment, security strategy, security management plan and breach/incident management.

Key topics: Sensitivity assessment · Security triage · Security management plan · Incident management plan

Part 6 — Health and safety information 2025

Specifies requirements for collaborative sharing of structured health and safety information throughout the project and asset life cycle.

Key topics: H&S information requirements · Risk information · Hazard communication

2. The information management process (ISO 19650-2)

End-to-end information management process for a project delivery, with stage gates, owners and the documents each stage produces.

Assessment & need

Appointing party sets up requirements, standards and the CDE.

> **Gate:** Project information requirements, standard, methods, CDE and protocol established.

Activity	Owner	Outputs
Appoint information management function	Appointing party	Information management appointment
Define OIR / AIR / PIR	Appointing party	OIR, AIR, PIR
Set information delivery milestones	Appointing party	Milestone schedule
Publish information standard & naming convention	Appointing party	Project information standard
Configure CDE and workflow	Appointing party	CDE procedure
Draft information protocol	Appointing party	Information protocol

Invitation to tender

EIR and tender requirements issued to prospective lead appointed parties.

> **Gate:** Invitation to tender issued with EIR and evaluation criteria.

Activity	Owner	Outputs
Write appointing party EIR	Appointing party	EIR
Assemble reference information	Appointing party	Reference information register
Define tender response & evaluation criteria	Appointing party	Tender evaluation criteria
BIM scope of services for each appointment	Appointing party	Scope of services

Tender response

Delivery team proves capability and proposes its approach.

> **Gate:** Tender response submitted: pre-appointment BEP, capability assessment, mobilization plan, risk register.

Activity	Owner	Outputs
Pre-appointment BEP	Prospective lead appointed party	Pre-appointment BEP
Capability & capacity assessment	Task teams	Capability assessment
Mobilization plan	Prospective lead appointed party	Mobilization plan
Information delivery risk register	Prospective lead appointed party	Risk register

Appointment

BEP confirmed, responsibilities and delivery plans agreed.

> **Gate:** Appointment documents complete incl. post-appointment BEP, responsibility matrix, MIDP and TIDPs.

Activity	Owner	Outputs
Post-appointment BEP	Lead appointed party	Post-appointment BEP
Detailed responsibility matrix	Lead appointed party	Responsibility matrix
TIDPs	Task teams	TIDP
MIDP	Lead appointed party	MIDP

Mobilization

People, IT and methods mobilized and tested.

> **Gate:** Resources inducted, IT mobilized, production methods tested.

Activity	Owner	Outputs
Induct and train delivery team	Lead appointed party	Training record
Mobilize IT & CDE access	Lead appointed party	IT mobilization record
Test production methods (naming, federation)	Lead appointed party	Test report

Collaborative production

Information produced, checked, reviewed and shared.

> **Gate:** Information models coordinated, QA checked and shared at the required suitability.

Activity	Owner	Outputs
Generate information in WIP	Task teams	Information containers
QA check (naming / metadata / format)	Task teams	QA checklist

Activity	Owner	Outputs
Review & approve for sharing	Task team manager	Shared containers
Federated model review / clash coordination	Lead appointed party	Coordination report

Information model delivery

Authorization by lead appointed party, acceptance by appointing party.

> **Gate:** Information model accepted and Published.

Activity	Owner	Outputs
Submit for authorization	Task teams	Submission
Review & authorize	Lead appointed party	Authorization record
Review & accept	Appointing party	Acceptance record

Project close-out

Archive the PIM and capture lessons learned.

> **Gate:** PIM archived, AIM updated, lessons learned recorded.

Activity	Owner	Outputs
Archive PIM / transfer to AIM	Appointing party	Archive record
Lessons learned	All parties	Lessons learned report

Clause by clause

5.1 Assessment and need

The appointing party establishes the project's information requirements, standards, methods and the CDE before any appointment.

5.1.1 Name who will manage information for the client

Assign internal or external resources responsible for information management on behalf of the appointing party.

5.1.2 Set out the project's information requirements

Define PIR derived from OIR/AIR, covering purpose, key decision points and the information needed.

5.1.3 Set the information delivery milestones

Align information delivery milestones with the project programme and key decision points.

5.1.4 Set the project information standard

Define exchange formats, structure, classification, naming, metadata and level of information need.

5.1.5 Set the production methods and procedures

Capture, generation, review, approval, security and delivery procedures.

5.1.6 Gather reference information and shared resources

Existing asset information, libraries, templates and shared resources to be provided to tenderers.

5.1.7 Set up the common data environment

Implement, configure and support the CDE before inviting tenders.

5.1.8 Agree the information protocol

Define the information-related obligations, licences and liabilities incorporated into appointments.

5.2 Invitation to tender

The appointing party compiles the EIR and tender response requirements for each prospective lead appointed party.

5.2.1 Write the client's exchange information requirements

EIR per appointment: requirements, acceptance criteria, supporting information and milestones.

5.2.2 Collect reference information and shared resources for tenderers

Issue relevant reference information and shared resources to tenderers.

5.2.3 Decide what tenders must contain and how they are scored

Define what the tender response shall contain and how it will be evaluated.

5.2.4 Put together the tender package

Bundle EIR, information standard, methods, protocol and resources into the invitation.

5.3 Tender response

The prospective lead appointed party demonstrates capability and capacity, producing the pre-appointment BEP.

5.3.1 Name who will manage information in the delivery team

Nominate the delivery team's information management roles.

5.3.2 Prepare the pre-appointment BEP

Proposed approach, high-level responsibility matrix, federation strategy and proposed amendments to standards.

5.3.3 Check each task team's capability and capacity

Assess skills, IT and capacity of each task team.

5.3.4 Summarise the delivery team's capability and capacity

Aggregate task team assessments into a delivery team summary.

5.3.5 Plan how the delivery team will mobilize

How information technology and processes will be mobilized and tested.

5.3.6 Record information delivery risks

Risks relating to timely delivery of information and their mitigation.

5.3.7 Submit the tender response

Submit the pre-appointment BEP, capability summary, mobilization plan and risk register.

5.4 Appointment

Confirm the BEP, produce detailed responsibility matrix, TIDPs and MIDP, and complete appointment documents.

5.4.1 Confirm the BEP

Update and agree the BEP, including names of individuals and confirmed federation strategy.

5.4.2 Build the detailed responsibility matrix

Who produces which information container, when, and to which level of information need.

5.4.3 Write the lead appointed party's EIR for each appointed party

EIR issued by the lead appointed party to each appointed party.

5.4.4 Prepare the task information delivery plans

Each task team's TIDP listing containers, LOIN, durations and dependencies.

5.4.5 Prepare the master information delivery plan

Aggregate TIDPs into the MIDP aligned to delivery milestones.

5.4.6 Finalise the lead appointed party's appointment

Incorporate EIR, information standard, methods, protocol, BEP and MIDP.

5.4.7 Finalise each appointed party's appointment

Incorporate the lead appointed party's EIR, protocol, BEP and TIDP.

5.5 Mobilization

Mobilize resources and IT, and test the proposed information production methods and procedures.

5.5.1 Get people ready

Confirm resource availability and deliver training/induction.

5.5.2 Get IT and CDE access ready

Procure, implement, configure and test IT and CDE access.

5.5.3 Trial the production methods and procedures

Test container breakdown, naming, federation, shared resources and delivery.

5.6 Collaborative production of information

Task teams generate information in WIP, check it, review it and share it through the CDE.

5.6.1 Confirm reference information and shared resources are available

Confirm required information is accessible before production.

5.6.2 Produce the information

Produce information to the information standard and LOIN, coordinating with other task teams.

5.6.3 Run the quality assurance check

Check conformance with the information standard (naming, metadata, format) before review.

5.6.4 Review and approve information for sharing

Review content against requirements; approve and assign suitability for the Shared state.

5.6.5 Review the federated information model

Lead appointed party reviews the federated information model for coordination and completeness.

5.7 Information model delivery

Submit the information model for authorization by the lead appointed party and acceptance by the appointing party.

5.7.1 Submit the model for authorization

Task teams submit the information model for authorization.

5.7.2 Check and authorize the model

Lead appointed party reviews against EIR and authorizes, or rejects with reasons.

5.7.3 Submit the model for client acceptance

Authorized model is submitted to the appointing party.

5.7.4 Check and accept the model

Appointing party reviews and accepts into the Published state, or rejects.

5.8 Project close-out

Archive the project information model and capture lessons learned for future projects.

5.8.1 Archive the PIM

Archive containers with access, retention and future-use requirements; hand over to the AIM.

5.8.2 Record lessons learned

Capture lessons learned from appointing and appointed parties.

3. Documents and who produces them

A single reference for every key ISO 19650 document: who writes it, when, what it must contain and the mistakes to avoid. Start with the workflow matrix to see which party produces what at each stage, then open the section for your role, or use the document guide for a full checklist of contents.

Workflow overview — Document table

How to read the table

Each row is a stage of the ISO 19650-2 delivery process, from assessment and need (5.1) to project close-out (5.8). Each column is a party. A card shows a document that party produces or owns at that stage, so reading across a row shows how information passes between parties, and reading down a column shows one party's full set of documents.

Select a card to see its purpose, key content and the clause it relates to. Use the stage and party filters to focus on your own role. Empty cells mean the party has no document of its own at that stage, usually because it is not yet appointed or is reviewing another party's work.

Documents by stage and party

Assessment & need ISO 19650-2 5.1

Document	Party	Purpose	Must contain	Clause
Information management assignment matrix	Appointing party	Records which individuals or organisations perform the appointing party's information management function. It is updated as parties are appointed and tasks are delegated.	Information management tasks; Party and named individual responsible; Tasks delegated to third parties; Authority to accept information; Revision history	ISO 19650-2 5.1.1
Organizational Information Requirements (OIR) [OIR]	Appointing party	Sets out the information the organisation needs at a strategic level, independent of any single project. It is a key input to the AIR and the project information requirements.	Business and asset management objectives; Decisions the information must support; Reporting and regulatory needs; Links to asset management systems; Owner and review cycle	ISO 19650-2 5.1.2
Asset Information Requirements (AIR) [AIR]	Appointing party	States the asset information the operator needs, so that the delivery team can produce it during the project rather than collect it afterwards.	Asset types and systems covered; Attributes required for each asset type; Documents required, such as manuals and certificates; Formats and target systems; Timing of delivery	ISO 19650-2 5.1.2

Document	Party	Purpose	Must contain	Clause
Project information requirements (PIR)	Appointing party	Defines the information the appointing party needs to make its key decisions on this project. It links the organisational and asset needs to the requirements in each EIR.	Project scope and purpose; Key decision points; Questions the information must answer; Information purposes, such as approvals, cost and programme; Links to the OIR and AIR	ISO 19650-2 5.1.2
Information delivery milestones	Appointing party	Establishes the points at which information must be delivered so that it is available for each key decision. Every EIR and delivery plan is built around these milestones.	Milestone identifiers and dates; Linked key decision points; Stage boundaries; Information purpose at each milestone; Alignment with the project programme	ISO 19650-2 5.1.3
Project Information Standard (PIS) [PIS]	Appointing party	Defines how information is structured, classified and exchanged on the project, so that every appointment produces consistent information.	Exchange formats and versions; Container naming convention; Classification system; Metadata, status and suitability codes; Method for defining the level of information need; Units, coordinates and origin	ISO 19650-2 5.1.4
Information production methods and procedures	Appointing party	Describes the procedures every task team follows to produce and deliver information, so that review, approval and security are handled the same way across the project.	Capture of existing asset information; Container breakdown structure; Review and approval procedure; Security and access procedure; Delivery to the appointing party	ISO 19650-2 5.1.5
Reference information and shared resources register	Appointing party	Lists the existing information, libraries and templates the appointing party will make available, with notes on their status and reliability.	Surveys and existing models; Existing asset records; Templates, title blocks and object libraries; Status and reliability of each item; How and when each item is made available	ISO 19650-2 5.1.6
CDE Workflow & Procedure (CDE) [CDE]	Appointing party	Describes the common data environment the appointing party establishes before tender, so that every party can share and manage information in it from the start.	States: work in progress, shared, published, archived; Workflows between states; Roles and permissions; Metadata and revision rules; Onboarding of new parties; Archive and retention	ISO 19650-2 5.1.7
Information Protocol (IP) [IP]	Appointing party	Sets the information-related obligations of each party and is incorporated into every appointment, so the requirements are contractually binding at every tier.	Parties and appointments covered; Obligations of each party; Licences to use information and their limits; Liabilities relating to information; Security requirements; References to the standard and methods	ISO 19650-2 5.1.8

Document	Party	Purpose	Must contain	Clause
Security Triage & Security Management Plan (SEC) [SEC]	Appointing party	Assesses whether the project or asset is sensitive and, where it is, sets proportionate security measures for handling information.	Outcome of the sensitivity assessment; Security measures required; Access controls and vetting; Handling of sensitive containers; Incident management and reporting	ISO 19650-5; ISO 19650-2 5.1.5

Invitation to tender ISO 19650-2 5.2

Document	Party	Purpose	Must contain	Clause
Exchange Information Requirements (EIR) [EIR]	Appointing party	Defines the information requirements for a specific appointment. It is the document the lead appointed party tenders and contracts against.	Information requirements for the appointment; Level of information need; Acceptance criteria for each requirement; Information delivery milestones; Supporting information provided; Interfaces with other appointments	ISO 19650-2 5.2.1
Tender response requirements and evaluation criteria	Appointing party	Tells tenderers which information management submissions are required and how they will be evaluated, so responses can be compared fairly.	Required submissions, such as the pre-appointment BEP; Scoring method and weightings; Minimum thresholds; Format and page limits; Clarification process	ISO 19650-2 5.2.3
BIM Specification & Scope of Services (SPEC) [SPEC]	Appointing party	Describes the information management tasks and deliverables the appointment includes, so they can be priced and do not fall between appointments.	Information management tasks; Deliverables; Interfaces with other appointments; Links to the EIR; Exclusions	ISO 19650-2 5.2.4
Invitation to tender pack	Appointing party	Bundles the requirements, standards, methods, protocol and reference information into one invitation, so all tenderers respond to the same information.	EIR; Project information standard; Production methods and procedures; Information protocol; Reference information and shared resources; Tender response requirements and evaluation criteria	ISO 19650-2 5.2.4
Tender queries and EIR review notes	Lead appointed party	Records the prospective lead appointed party's review of the invitation to tender and the clarifications it needs before it can commit to the EIR.	Requirements that are unclear or conflicting; Questions on the standard, methods and protocol; Gaps in reference information; Proposed amendments to raise in the BEP; Responses received	ISO 19650-2 5.2.4

Tender response ISO 19650-2 5.3

Document	Party	Purpose	Must contain	Clause
Tender clarification responses	Appointing party	Resolves ambiguity in the invitation to tender and keeps the competition fair by giving every tenderer the same answers.	Question received; Response; Documents amended as a result; Date issued; Distribution to all tenderers	ISO 19650-2 5.2.4
Nomination of information management individuals	Lead appointed party	Names the individuals who will carry out information management for the delivery team and shows they have the experience and authority to do it.	Names and roles; Relevant experience; Authority within the delivery team; Availability for the appointment; Deputies	ISO 19650-2 5.3.1
BIM Execution Plan (Pre-appointment) (BEP-PRE) [BEP-PRE]	Lead appointed party	Explains how the delivery team proposes to meet the EIR. It is scored at tender and becomes the starting point for the confirmed BEP.	Proposed information management individuals; Information delivery strategy; Federation strategy; High-level responsibility matrix; Proposed amendments to the standard and methods; Proposed software and IT	ISO 19650-2 5.3.2
High-level responsibility matrix	Lead appointed party	Shows how responsibility for the information deliverables will be split across task teams, so the appointing party can see the delivery team covers the whole EIR.	Information deliverables; Task team responsible; Milestones; Interfaces between task teams	ISO 19650-2 5.3.2
Delivery team capability and capacity summary	Lead appointed party	Combines the task team assessments into one view of whether the delivery team can deliver the information required, and how any gaps will be closed.	Skills and experience; Software and IT; Capacity against the programme; Gaps identified; Actions to close gaps	ISO 19650-2 5.3.4
Mobilization Plan (MOB) [MOB]	Lead appointed party	Sets out how the delivery team will get ready to produce information, including how it will test the production methods before production starts.	Resources and timescales; Training and induction; IT and CDE setup; Tests of the production methods; Responsibilities	ISO 19650-2 5.3.5
Information Delivery Risk Register (RR) [RR]	Lead appointed party	Identifies the risks that could stop the delivery team meeting the EIR and milestones, with owners and mitigations. It is maintained throughout delivery.	Risk description; Owner; Likelihood and impact; Mitigation; Assumptions about client inputs; Status	ISO 19650-2 5.3.6
Delivery team tender response	Lead appointed party	Brings together the delivery team's information management documents into the response required by the invitation to tender.	Pre-appointment BEP; Capability and capacity summary; Mobilisation plan; Risk register; Any other required responses	ISO 19650-2 5.3.7

Document	Party	Purpose	Must contain	Clause
Capability & Capacity Assessment (CCA) [CCA]	Task teams / sub-appointed	Assesses the task team's skills, IT and capacity against the requirements, so the lead appointed party can build an honest delivery team summary.	Skills and experience; Software and hardware; IT infrastructure; Capacity against the programme; Gaps and training needs	ISO 19650-2 5.3.3

Appointment ISO 19650-2 5.4

Document	Party	Purpose	Must contain	Clause
Tender evaluation record	Appointing party	Records the evaluation of each tender against the published criteria and any amendments agreed before appointment. It is the appointing party's decision record for selection.	Scores against each criterion; Clarifications relied on; Amendments agreed with the preferred tenderer; Decision and rationale; Decision maker and date	ISO 19650-2 5.2.3
Lead appointed party appointment documents	Appointing party	Incorporates the information management documents into the lead appointed party's appointment, so the requirements and plans are binding.	EIR; Project information standard; Production methods and procedures; Information protocol; Confirmed BEP; MIDP	ISO 19650-2 5.4.6
BIM Execution Plan (Post-appointment) (BEP-POST) [BEP-POST]	Lead appointed party	Confirms the delivery team's approach after appointment, replacing the tender proposal with an agreed commitment that is kept up to date.	Named information management individuals; Information delivery strategy; Confirmed federation strategy; Agreed amendments to the standard and methods; Schedule of software, hardware and IT	ISO 19650-2 5.4.1
Detailed Responsibility Matrix (DRM) [DRM]	Lead appointed party	Refines the high-level matrix so that each deliverable has one responsible task team, a level of information need and a milestone.	Information deliverables; Responsible task team; Level of information need; Milestone; Interfaces and exclusions; Link to the EIR requirement	ISO 19650-2 5.4.2
Model Production and Delivery Table (MPDT) [MPDT]	Lead appointed party	Sets out who models which elements at each stage and to what level of information need, supporting the detailed responsibility matrix.	Elements or systems; Responsible task team by stage; Level of information need by stage; Delivery milestone; Notes on interfaces	ISO 19650-2 5.4.2
Lead appointed party EIR [EIR]	Lead appointed party	Passes the relevant client requirements, plus the lead appointed party's own requirements, to each appointed party in the delivery team.	Requirements for the appointed party; Level of information need; Acceptance criteria; Milestones; Supporting information	ISO 19650-2 5.4.3

Document	Party	Purpose	Must contain	Clause
Master Information Delivery Plan (MIDP) [MIDP]	Lead appointed party	Combines every task team's plan into one schedule of information containers, so the delivery team and client can track delivery against the milestones.	Containers from every TIDP; Responsible task team; Planned delivery dates; Milestone served; Dependencies; Baseline and forecast	ISO 19650-2 5.4.5
Appointed party appointment documents	Lead appointed party	Incorporates the information management documents into each appointed party's appointment, so the requirements are binding at every tier.	Lead appointed party EIR; Project information standard; Production methods and procedures; Information protocol; Confirmed BEP; TIDP	ISO 19650-2 5.4.7
Acknowledgement of the lead appointed party EIR	Task teams / sub-appointed	Records that the appointed party has reviewed its EIR, raised any queries and accepts the requirements it will deliver against.	Requirements accepted; Queries raised; Clarifications received; Scope boundaries confirmed; Date and signatory	ISO 19650-2 5.4.3
Task Information Delivery Plan (TIDP) [TIDP]	Task teams / sub-appointed	Plans the information containers one task team will produce and when, including what it needs from other task teams. It feeds the MIDP.	Container names following the naming convention; Level of information need; Author; Production duration; Dependencies and need-by dates; Planned delivery dates	ISO 19650-2 5.4.4

Mobilisation ISO 19650-2 5.5

Document	Party	Purpose	Must contain	Clause
CDE access and permissions record	Appointing party	Records the CDE accounts and permissions granted to the delivery team, so access matches roles and can be withdrawn when people leave.	Organisations and users; Roles and permissions; Date granted; Approver; Review and removal process	ISO 19650-2 5.5.2
Training and induction records	Lead appointed party	Shows that the delivery team's resources are confirmed and that everyone has been inducted into the project's requirements and methods.	Resources confirmed against the mobilisation plan; Induction content; Attendance; Training completed; Outstanding actions	ISO 19650-2 5.5.1
IT and CDE mobilisation record	Lead appointed party	Records that the information technology the delivery team needs has been procured, configured and tested, including access to the CDE.	Software and versions; Hardware and network; CDE access for each task team; Interoperability checks; Issues and resolutions	ISO 19650-2 5.5.2

Document	Party	Purpose	Must contain	Clause
Production methods test report	Lead appointed party	Records the outcome of testing the production methods and procedures before production starts, and the changes made to the BEP as a result.	Container breakdown structure tested; Naming and metadata tested; Federation tested; Delivery workflow tested; Issues found and actions; Resulting BEP updates	ISO 19650-2 5.5.3
Training and competence records	Task teams / sub-appointed	Shows that each task team member understands the project's standard, methods and CDE before producing information.	Team members; Induction attended; Training completed; Competence gaps; Planned actions	ISO 19650-2 5.5.1
Test containers and configuration record	Task teams / sub-appointed	Provides the sample information and software configuration that the delivery team uses to test the production methods.	Sample containers produced; Templates and settings used; Software versions; Test results; Corrections made	ISO 19650-2 5.5.3

Collaborative production ISO 19650-2 5.6

Document	Party	Purpose	Must contain	Clause
Client review comments and change instructions	Appointing party	Records the appointing party's comments on shared information at agreed review points and any changes it instructs, so they can be traced into the plans.	Containers reviewed; Comments and responses; Change instructed; Impact on the EIR, BEP and MIDP; Date and author	ISO 19650-2 5.6
Information model review record and issue log	Lead appointed party	Records the lead appointed party's review of the federated information model and tracks the coordination issues it raises through to closure.	Containers federated; Coordination and completeness checks; Issues and owners; Due dates and status; Actions agreed	ISO 19650-2 5.6.5
BIM Progress Report (BPR) [BPR]	Lead appointed party	Reports information delivery progress against the plan, so that milestones at risk are visible to the appointing party early.	Progress against the MIDP; Milestone forecast; Open issues; Risks; Actions	ISO 19650-2 5.6
Reference information availability check	Task teams / sub-appointed	Confirms that the reference information and shared resources the task team needs are available, so production does not start on missing inputs.	Information required; Source and status; Date confirmed; Gaps; Actions raised	ISO 19650-2 5.6.1
Task information model	Task teams / sub-appointed	The information containers the task team produces, generated to the information standard and level of information need and coordinated with other task teams.	Containers named to the standard; Level of information need met; Status and revision; Coordination with shared information; Metadata	ISO 19650-2 5.6.2

Document	Party	Purpose	Must contain	Clause
Information Container QA Checklist (QA) [QA]	Task teams / sub-appointed	Records that each container conforms to the information standard and methods before its content is reviewed.	Container checked; Naming; Metadata; Format; Status and revision; Result and checker	ISO 19650-2 5.6.3
Review and approval record	Task teams / sub-appointed	Records the review of container content against the requirements and the approval to move it to the shared state.	Container reviewed; Reviewer, other than the author; Checks against requirements; Suitability assigned; Decision and date	ISO 19650-2 5.6.4

Information model delivery ISO 19650-2 5.7

Document	Party	Purpose	Must contain	Clause
Acceptance record	Appointing party	Records the appointing party's review of the submitted information model against the EIR and its decision to accept it into the published state.	Milestone; Containers accepted, with revision; Acceptance criteria checked; Decision and date; Decision maker	ISO 19650-2 5.7.4
Rejection record (appointing party)	Appointing party	Records why a submission was rejected, so the lead appointed party knows exactly what to correct and resubmit.	Containers rejected; Reasons, linked to acceptance criteria; Corrections required; Resubmission date; Decision maker	ISO 19650-2 5.7.4
Authorisation record	Lead appointed party	Records that the lead appointed party has reviewed the information model against the EIR and authorised it for submission to the appointing party.	Milestone; Containers authorised; Checks against the EIR; Decision and date; Decision maker	ISO 19650-2 5.7.2
Rejection record (lead appointed party)	Lead appointed party	Records why task team information was rejected at authorisation, so it can be corrected before the client milestone.	Containers rejected; Reasons; Corrections required; Task team responsible; Resubmission date	ISO 19650-2 5.7.2
Submission for appointing party acceptance	Lead appointed party	Formally submits the authorised information model to the appointing party for review and acceptance at the milestone.	Milestone; Containers submitted, with revision and suitability; Link to the authorisation record; Transmittal; Date	ISO 19650-2 5.7.3
Submission for authorisation	Task teams / sub-appointed	Submits the task team's information containers due at the milestone for the lead appointed party to review and authorise.	Milestone; Containers due; QA and review records; Suitability for authorisation; Transmittal	ISO 19650-2 5.7.1

Project close-out ISO 19650-2 5.8

Document	Party	Purpose	Must contain	Clause
Archive record	Appointing party	Records the archiving of the project information model with its access, retention and future-use rules, and the information transferred to the asset information model.	Containers archived; Retention period; Access rules; Future-use requirements; Information transferred to the AIM	ISO 19650-2 5.8.1
Lessons Learned Report (LL) [LL]	Appointing party	Captures what worked and what did not, from the appointing party and the delivery teams, and records the changes to make to requirements and standards.	What worked; What did not; Root causes; Changes to the OIR, standard, methods and EIR; Owners of follow-up actions	ISO 19650-2 5.8.2
Final information for archive	Lead appointed party	Confirms that the delivery team's information is complete, in the correct states and ready for the appointing party to archive.	Containers in the published state; Superseded containers identified; Metadata complete; Outstanding items; Confirmation of completeness	ISO 19650-2 5.8.1
Delivery team lessons learned input	Lead appointed party	Brings together the delivery team's lessons for the appointing party's lessons learned report.	Successes; Problems and causes; Recommendations; Task team contributions	ISO 19650-2 5.8.2
Final containers for archive	Task teams / sub-appointed	Provides the task team's final information in the agreed state and with complete metadata, ready to be archived.	Final containers; Revisions and suitability; Metadata complete; Superseded containers withdrawn	ISO 19650-2 5.8.1
Task team lessons learned input	Task teams / sub-appointed	Shares the task team's lessons with the lead appointed party for inclusion in the delivery team's input.	What worked; What did not; Recommendations; Contact for follow-up	ISO 19650-2 5.8.2

Practical guidance, not a fixed RACI
ISO 19650-2 defines activities and parties, not a document list or job titles. The stage and party mapping here is practical guidance: some records, such as clarification responses or progress reports, are common practice rather than named outputs of the standard. Agree the actual responsibilities in the information management assignment matrix and the BEP, as job titles and tiers vary between projects.

Workflow overview — Summary matrix

One table, every document

The matrix shows the documents each party produces at each stage of ISO 19650-2. Read across a row to see how documents pass between parties at a stage; read down a column to see everything one party is responsible for across the project.

Where a party mainly reviews rather than produces at a stage, the cell starts with Reviews. Names match the templates in this portal where one exists.

Documents by stage and party

Stage	Appointing party	Lead appointed party	Task teams / sub-appointed
5.1 Assessment and need	Information management assignment matrix; Organizational Information Requirements; Asset Information Requirements; project information requirements; information delivery milestones; Project Information Standard; production methods and procedures; reference information register; CDE Workflow & Procedure; Information Protocol; Security Triage & Security Management Plan	Not yet appointed	Not yet appointed
5.2 Invitation to tender	Exchange Information Requirements; tender response requirements; evaluation criteria; BIM Specification & Scope of Services; invitation to tender pack	Reviews the invitation to tender, EIR, standard, methods and protocol	Reviews the EIR and standards passed on by the prospective lead appointed party
5.3 Tender response	Clarification responses; Reviews tender responses	Nomination of information management individuals; BIM Execution Plan (Pre-appointment); high-level responsibility matrix; delivery team capability and capacity summary; Mobilization Plan; Information Delivery Risk Register; tender response	Capability & Capacity Assessment; input to the pre-appointment BEP
5.4 Appointment	Tender evaluation record; lead appointed party appointment documents; Reviews the confirmed BEP and MIDP	BIM Execution Plan (Post-appointment); Detailed Responsibility Matrix; Model Production and Delivery Table; lead appointed party EIR for each appointed party; Master Information Delivery Plan; appointed party appointment documents	Task Information Delivery Plan; acknowledgement of the lead appointed party EIR
5.5 Mobilisation	CDE access permissions; client induction material; Reviews mobilisation test results	Training and induction records; IT and CDE mobilisation record; production methods test report; updated Information Delivery Risk Register	Training attendance; test containers; software configuration records

Stage	Appointing party	Lead appointed party	Task teams / sub-appointed
5.6 Collaborative production	Responses to queries; change instructions; Reviews shared information at agreed review points	Federated information model; coordination reports; issue log; BIM Progress Report; BIM Meeting Minutes; updated MIDP	Information containers in work in progress; Information Container QA Checklist records; review records; shared containers with suitability codes
5.7 Information model delivery	Acceptance records; rejection records with reasons	Authorisation records; rejection records with reasons; submission of the authorised model for acceptance	Submission for authorisation; revised containers after rejection
5.8 Project close-out	Archive record; updated asset information model; Lessons Learned Report	Lessons learned input; Information Handover Strategy contributions; final information for archive	Final containers for archive; lessons learned input

Tiered delivery teams
Where an appointed party appoints its own sub-consultants or specialists, it takes the lead role for that tier. It issues an EIR, collects TIDPs and authorises information in the same way, so the middle column repeats one level down.

Documents that must appear in every appointment

- The relevant exchange information requirements
- The project information standard
- The information production methods and procedures
- The information protocol
- The confirmed BIM execution plan
- The MIDP for a lead appointed party, or the TIDP for an appointed party

Workflow overview — Document flow

How the documents build on each other

1. **1. OIR and AIR set the organisation's needs.** The appointing party records what information its organisation and its assets need, independent of any single project.
2. **2. PIR and milestones focus on this project.** Project information requirements and information delivery milestones say which decisions this project must support and when.
3. **3. Standard, methods, CDE and protocol set the rules.** The project information standard, production methods, CDE procedure and information protocol apply to every appointment on the project.

4. **4. The EIR defines one appointment.** Each appointment gets its own EIR, drawn from the PIR and AIR, with acceptance criteria and milestones.
5. **5. The pre-appointment BEP responds.** The prospective lead appointed party explains how it would meet the EIR, with a high-level responsibility matrix, capability summary, mobilisation plan and risk register.
6. **6. The confirmed BEP and responsibility matrix commit.** After appointment the BEP is confirmed and the detailed responsibility matrix assigns every deliverable to a task team.
7. **7. Lead EIRs flow down.** The lead appointed party issues an EIR to each appointed party so the client requirements reach every task team.
8. **8. TIDPs and the MIDP schedule delivery.** Each task team plans its containers in a TIDP; the lead appointed party combines them into the MIDP.
9. **9. Containers, QA and review records prove quality.** Task teams produce containers, record QA checks and reviews, and share them through the CDE.
10. **10. Authorisation and acceptance records close each milestone.** The lead appointed party authorises and the appointing party accepts, each with records and reasons for any rejection.
11. **11. Archive and lessons learned complete the loop.** The PIM is archived, relevant content goes to the AIM, and lessons learned improve the next project's OIR, standards and EIR.

Golden thread

Every document should be traceable to the one before it. An auditor should be able to pick a container in the MIDP and follow it back through the TIDP, the responsibility matrix, the BEP and the EIR to the client requirement it satisfies.

Keep these links explicit with requirement identifiers, so that a change to the EIR can be traced to every plan and container it affects.

> **Change control**
> When a requirement changes, update the documents in order: EIR, BEP, responsibility matrix, TIDPs and MIDP. Changing the MIDP alone hides the reason for the change.

Workflow overview — Colour legend

Who each colour represents

Amber: appointing party

The client side. Sets requirements, standards, methods, the CDE and the protocol, and accepts information at each milestone.

Green: lead appointed party

Receives the appointment from the client, plans and coordinates the delivery team, and authorises the information model.

Teal: task teams and sub-appointed parties

Produce the information: plan in TIDPs, generate, check, review and share containers.

Blue: shared activities

Activities several parties carry out together, such as mobilisation and close-out, under one lead.

Parties and roles are not the same

ISO 19650 talks about parties: the appointing party, the lead appointed party and appointed parties. It does not prescribe job titles. One person can fill several roles on a small project, and a large organisation can split one role across a team.

Record who actually does each task in the information management assignment matrix and the BEP, rather than relying on job titles.

Common job titles and the party they usually sit in

- Client information manager, asset manager, project sponsor: appointing party
- Lead information manager, design manager, project manager: lead appointed party
- Task team manager, task information manager, BIM coordinator: task teams
- BIM author, modeller, engineer, designer: task teams
- CDE administrator: often appointing party, sometimes delegated

Appointing party — Requirements

The requirements hierarchy

The appointing party's requirements run from the general to the specific: OIR for the organisation, AIR for the asset, PIR for the project, and an EIR for each appointment. The EIR is the only one the delivery team contracts against, so everything the client needs must eventually land in an EIR.

Requirements documents

Document	Purpose	When	Key content	Common mistakes
Organizational Information Requirements (OIR)	Records the information the organisation needs for its strategic objectives	Before the project, reviewed periodically	Business objectives; decisions supported; reporting needs; links to asset management	Written as a wish list with no link to real decisions
Asset Information Requirements (AIR)	States the information needed to operate and maintain the asset	Before tender, informed by the operator	Asset types; attributes; documents; formats; handover timing; links to CAFM or CMMS systems	Asking for every attribute of every object, including ones nobody will use
Project information requirements (PIR)	Defines what information the project needs to support its key decisions	During assessment and need (5.1.2)	Key decision points; questions to answer; purposes such as approvals, cost and programme	Confusing PIR with the EIR and skipping it
Information delivery milestones	Sets when information is required across the project	During assessment and need (5.1.3)	Milestone dates; linked decision points; stage boundaries	Milestones that do not match the project programme

Document	Purpose	When	Key content	Common mistakes
Exchange Information Requirements (EIR)	Defines what one appointment must deliver, when, how and to what level of information need	Before tender (5.2.1)	Requirements; acceptance criteria; milestones; supporting information; level of information need	No acceptance criteria, or copied from another project
EIR per appointment (several lead appointed parties)	Gives each lead appointed party requirements specific to its scope	Before each tender	Scope-specific requirements; interfaces with other appointments; shared milestones	One generic EIR for every appointment, leaving interfaces undefined
Tender response requirements and evaluation criteria	Explains what tenderers must submit and how information management will be scored	With the invitation to tender (5.2.3)	Required submissions; scoring method; weightings; minimum thresholds	Information management not scored at all

Requirements quality check

- ☐ Every EIR requirement traces to the PIR or AIR
- ☐ Every requirement has an acceptance criterion
- ☐ Level of information need stated for each deliverable
- ☐ Milestones match the project programme
- ☐ Interfaces between appointments are defined
- ☐ Evaluation criteria include information management

> **Write it once, flow it down**
> The appointing party writes the EIR for the lead appointed party. The lead appointed party then writes its own EIR for each appointed party. Both are needed in a tiered delivery team.

Appointing party — Standards & methods

The project rulebook

These documents apply to every appointment on the project. They are established during assessment and need (5.1.4 to 5.1.8) and issued with the invitation to tender, so tenderers can price and plan against them.

Standards, methods and environment

Document	Purpose	When	Key content	Common mistakes
Project Information Standard (PIS)	Defines how information is structured and exchanged	Before tender (5.1.4)	Exchange formats; naming convention; classification; metadata; status and suitability codes; level of information need method; units and coordinates	Leaving out coordinates and units, causing federation errors
Information production methods and procedures	Explains how information is captured, produced, reviewed, approved, secured and delivered	Before tender (5.1.5)	Capture of existing information; container breakdown; review and approval; security; delivery to the client	Describing software features instead of procedures
Reference information and shared resources register	Lists existing information and resources made available to the delivery team	Before tender (5.1.6, 5.2.2)	Surveys; existing models; asset records; templates; object libraries; title blocks	Issuing reference information without status or reliability notes
CDE Workflow & Procedure (CDE)	Describes how the common data environment works on the project	Before tender (5.1.7)	States; workflows; permissions; metadata; revision rules; archive; access for new parties	CDE configured after appointment
Information Protocol (IP)	Sets the contractual obligations, licences and liabilities for information	Before tender, incorporated into appointments (5.1.8)	Licences to use information; obligations of each party; liabilities; security requirements; references to the standard and methods	Included only in the top-tier appointment
Security Triage & Security Management Plan (SEC)	Assesses sensitivity and sets security measures where needed	During assessment and need, following ISO 19650-5	Sensitivity assessment outcome; security measures; access controls; incident management	Skipping the triage because the project seems routine
BIM Specification & Scope of Services (SPEC)	Describes the information management services required in the appointment	With the invitation to tender	Information management tasks; deliverables; interfaces; links to the EIR	Duplicating or contradicting the EIR

Ready to issue

- ☐ Naming convention tested with real examples
- ☐ Coordinate system and units defined
- ☐ Status and suitability codes listed
- ☐ CDE configured and tested with a sample workflow
- ☐ Information protocol reviewed by legal advisers
- ☐ Reference information has status and reliability notes

National annexes
If the project adopts a national annex, such as the UK National Annex, state it in the information standard and record any deviations from it.

Appointing party — Tender, acceptance & close-out

Decisions and records

After the tender is issued, the appointing party's documents are mostly decision records: who it appointed and why, which information it accepted at each milestone, and what it archived. These are short documents, but they carry the most weight in audits and disputes.

Decision and close-out documents

Document	Purpose	When	Key content	Common mistakes
Information management assignment matrix	Records who carries out each information management task	Assessment and need (5.1.1), updated as parties are appointed	Tasks; party responsible; named individuals; delegated tasks	Never updated after appointment
Invitation to tender pack	Bundles everything tenderers need	Invitation to tender (5.2.4)	EIR; standard; methods; protocol; reference information; tender requirements	Missing the CDE details or protocol
Tender evaluation record	Shows how information management responses were scored	Appointment (5.4)	Scores against criteria; clarifications; agreed amendments	No record of why amendments were accepted
Appointment documents	Makes the information requirements contractual	Appointment (5.4.6)	EIR; standard; methods; protocol; confirmed BEP; MIDP	Signing before the confirmed BEP and MIDP are attached, with no route to add them
Acceptance record	Records acceptance of the information model at a milestone	Information model delivery (5.7.4)	Milestone; containers accepted; criteria checked; date; decision maker	Acceptance by email with no link to the containers
Rejection record	Records why a submission was rejected	Information model delivery (5.7.4)	Containers rejected; reasons; required corrections; resubmission date	Rejecting without reasons
Archive record	Records what was archived and how it may be used	Close-out (5.8.1)	Containers archived; retention period; access rules; future-use requirements	Archiving everything with no retention or access rules

Document	Purpose	When	Key content	Common mistakes
Lessons Learned Report (LL)	Captures lessons for future projects	Close-out (5.8.2), with input throughout	What worked; what did not; changes to standards and requirements	Written only at the end, by one party

Acceptance record essentials

- ☐ Milestone identified
- ☐ Containers listed with revision and suitability
- ☐ Acceptance criteria checked
- ☐ Decision and date recorded
- ☐ Decision maker named
- ☐ Reasons recorded for any rejection

> **Keep decisions in the CDE**
> Store acceptance, rejection and archive records in the CDE alongside the containers they relate to. Decisions kept in email inboxes are lost when people move on.

Lead appointed party — Tender documents

Winning the appointment

At tender, the prospective lead appointed party shows the appointing party how its delivery team will meet the EIR and proves it has the capability and capacity to do so (5.3). These documents are scored, so they must answer the specific requirements rather than describe the company in general.

Tender response documents

Document	Purpose	When	Key content	Common mistakes
Nomination of information management individuals	Names who will perform the information management function	Tender (5.3.1)	Names; roles; experience; authority	Naming people who will not actually be available
BIM Execution Plan (Pre-appointment) (BEP-PRE)	Proposes how the delivery team will meet the EIR	Tender (5.3.2)	Information delivery strategy; federation strategy; high-level responsibility matrix; proposed amendments to standard and methods; software and IT	Generic company BEP that ignores the EIR
High-level responsibility matrix	Shows which task team delivers which information at a high level	Tender (5.3.2)	Information deliverables; task teams; milestones	Too detailed to be useful or too vague to score

Document	Purpose	When	Key content	Common mistakes
Delivery team capability and capacity summary	Aggregates task team assessments	Tender (5.3.4)	Skills; IT; capacity; gaps and how they will be closed	Summary that hides weak task teams
Mobilization Plan (MOB)	Explains how people, IT and methods will be mobilised and tested	Tender (5.3.5)	Resources; training; IT setup; CDE access; test plan; timescales	No testing of methods planned
Information Delivery Risk Register (RR)	Identifies risks to timely, compliant information delivery	Tender (5.3.6), maintained throughout	Risks; owners; likelihood; impact; mitigations; assumptions	Only generic risks, nothing about this EIR
Tender response	Compiles and submits the delivery team's offer	Tender (5.3.7)	All of the above, plus any responses required by the tender	Missing items required by the tender response requirements

Tender submission check

- ☐ Every EIR requirement answered in the BEP
- ☐ Proposed amendments to the standard clearly listed
- ☐ Capability assessment for every task team
- ☐ Mobilisation plan includes method testing
- ☐ Risk register covers client-side dependencies
- ☐ Named individuals confirmed available

> **Answer line by line**
> Number your BEP responses to match the EIR requirements. Evaluators can then score you quickly, and you have a ready-made traceability record for the confirmed BEP.

Lead appointed party — Appointment documents

From offer to plan

Once appointed, the lead appointed party confirms its BEP, assigns every deliverable, flows the requirements down and builds the master plan (5.4). These documents are incorporated into appointments, so they carry contractual weight.

Appointment documents

Document	Purpose	When	Key content	Common mistakes
BIM Execution Plan (Post-appointment) (BEP-POST)	Confirms how the delivery team will deliver the information	Appointment (5.4.1)	Named individuals; information delivery strategy; confirmed federation strategy; agreed amendments; software schedule	Pre-appointment BEP resubmitted unchanged
Detailed Responsibility Matrix (DRM)	Assigns each information deliverable to a task team	Appointment (5.4.2)	Deliverables; task team; level of information need; milestone	Deliverables in the EIR left unassigned
Model Responsibility Matrix (MRM)	Assigns model elements and systems to authors by stage	Appointment, alongside the DRM	Elements or systems; author by stage; level of information need	Overlaps where two teams model the same element
Model Production and Delivery Table (MPDT)	Details model production responsibilities and delivery stages	Appointment, alongside the DRM	Elements; responsible party; stage; level of information need	Out of step with the DRM
Lead appointed party EIR	Flows the client requirements down to each appointed party	Appointment (5.4.3)	Requirements; acceptance criteria; milestones; supporting information	Client requirements not flowed down
Master Information Delivery Plan (MIDP)	Combines all TIDPs into one delivery plan	Appointment (5.4.5), maintained throughout	Containers; task teams; dates; milestones; dependencies	Not re-baselined after change
Appointed party appointment documents	Makes the requirements contractual for each appointed party	Appointment (5.4.7)	Lead EIR; standard; methods; protocol; BEP; TIDP	Missing the information protocol

Appointment documents check

- ☐ Confirmed BEP agreed with the appointing party
- ☐ Every EIR deliverable appears in the DRM
- ☐ DRM, MRM and MPDT consistent with each other
- ☐ Lead EIR issued to every appointed party
- ☐ MIDP aligned with the client milestones
- ☐ Information protocol in every appointed party appointment

Proportionality
On small appointments the DRM, MRM and MPDT can be combined into one table, as long as each required piece of information is still clearly identifiable.

Lead appointed party — Delivery & close-out

Running delivery

During mobilisation, production and delivery the lead appointed party keeps the plans live and records the decisions it makes. Authorisation records are the most important: they show that the delivery team checked its own work before passing it to the client.

Delivery and close-out documents

Document	Purpose	When	Key content	Common mistakes
Mobilisation records	Shows people, IT and methods are ready	Mobilisation (5.5)	Training attendance; CDE access list; software versions; test results	No evidence that tests were actually run
Production methods test report	Records the outcome of method tests	Mobilisation (5.5.3)	Naming; breakdown structure; federation; delivery workflow; issues found	Problems found but not fixed in the BEP
Coordination reports and issue log	Records federated model reviews and issues	Collaborative production (5.6.5)	Clashes; issues; owners; due dates; status	Issues tracked outside the agreed channel
BIM Progress Report (BPR)	Reports delivery progress against the MIDP	Regularly during production	Progress; risks; issues; milestone forecast	Reporting activity rather than progress against the plan
BIM Meeting Minutes (MOM)	Records information management decisions and actions	Throughout	Attendees; decisions; actions; owners; dates	Decisions not reflected in the BEP or MIDP
Authorisation record	Records authorisation of the information model	Information model delivery (5.7.2)	Milestone; containers; criteria checked; decision maker; date	Authorising without checking against the EIR
Rejection record	Records why a task team submission was rejected	Information model delivery (5.7.2)	Containers; reasons; corrections required; resubmission date	Verbal rejections with no record
Lessons Learned Report (LL) input	Captures the delivery team's lessons	Throughout, compiled at close-out (5.8.2)	Successes; problems; recommendations	Left until the team has disbanded

Milestone readiness

- ☐ All containers due at the milestone are in the shared state
- ☐ Federated model reviewed
- ☐ Open issues closed or agreed
- ☐ Authorisation checked against the EIR acceptance criteria
- ☐ Authorisation record stored in the CDE
- ☐ MIDP updated with actual delivery

> **Plan for resubmission**
> Allow time in the MIDP for corrections between authorisation and acceptance at critical milestones. Rejection is a normal part of the process, not a failure of it.

Task teams / sub-appointed — Planning documents

Planning your part

Task teams and sub-appointed parties contribute to the tender and then plan their own deliverables. Their planning documents feed directly into the lead appointed party's BEP and MIDP, so accuracy here prevents problems across the whole delivery team.

Task team planning documents

Document	Purpose	When	Key content	Common mistakes
Capability & Capacity Assessment (CCA)	Shows the task team can deliver the information required	Tender (5.3.3)	Skills; experience; software; hardware; capacity; gaps and training needs	Overstating capacity
Input to the pre-appointment BEP	Provides discipline-specific content for the BEP	Tender (5.3.2)	Proposed container breakdown; software; exchange formats; proposed amendments	Not reading the client EIR before contributing
Acknowledgement of the lead appointed party EIR	Confirms the task team understands its requirements	Appointment (5.4.3)	Requirements accepted; queries; clarifications	Accepting requirements without querying unclear ones
Task Information Delivery Plan (TIDP)	Plans the task team's information containers and dates	Appointment (5.4.4)	Container names; level of information need; author; production duration; dependencies; delivery dates	Missing dependencies on other task teams
Input to the responsibility matrix	Confirms scope boundaries	Appointment (5.4.2)	Deliverables accepted; interfaces; elements excluded	Leaving gaps at interfaces between disciplines

Document	Purpose	When	Key content	Common mistakes
Task team appointment documents	Makes the requirements contractual	Appointment (5.4.7)	Lead EIR; standard; methods; protocol; BEP; TIDP	Signing without the information protocol

TIDP ready to submit

- ☐ Every deliverable from the DRM listed
- ☐ Container names follow the naming convention
- ☐ Level of information need set for each container
- ☐ Author named for each container
- ☐ Dependencies and need-by dates recorded
- ☐ Dates align with the MIDP milestones

> **Be honest about capacity**
> Overstating capacity at tender leads to missed milestones later. Lead appointed parties value a realistic assessment that names gaps and a plan to close them.

Task teams / sub-appointed — Production documents

Producing and proving quality

During production the task team's documents are the information itself and the records that show it was checked. ISO 19650-2 separates the quality assurance check (5.6.3), which looks at conformance with the information standard, from the review (5.6.4), which looks at the content.

Task team production documents

Document	Purpose	When	Key content	Common mistakes
Task information model	The task team's information containers: models, drawings, schedules, documents	Collaborative production (5.6.2)	Containers to the information standard and level of information need	Using information from other teams that is still in their work in progress
Information Container QA Checklist (QA)	Records the check against the information standard	Before review (5.6.3)	Naming; metadata; format; status; revision; result	Checking only the file name
Review and approval record	Records content review and approval for sharing	Before sharing (5.6.4)	Reviewer; checks against requirements; suitability assigned; date	Author reviewing their own work
Shared containers	Information made available for coordination	After approval (5.6.4)	Container with status, suitability and revision	Wrong suitability code

Document	Purpose	When	Key content	Common mistakes
Coordination issues (for example BCF)	Raises and tracks issues with other task teams	Collaborative production	Issue; location; owner; status	Issues raised by email outside the agreed channel
Submission for authorisation	Submits information at a milestone	Information model delivery (5.7.1)	Containers due; suitability for authorisation; transmittal	Submitting containers that have not passed QA
Lessons learned input	Shares the task team's lessons	Throughout and at close-out (5.8.2)	What worked; what did not; recommendations	No input from task teams at all

Before sharing

- ☐ QA check passed and recorded
- ☐ Content reviewed by someone other than the author
- ☐ Suitability code correct
- ☐ Metadata complete
- ☐ Superseded versions withdrawn
- ☐ Coordination issues addressed or recorded

> **Use only shared or published information**
> When coordinating with another task team, base your work on their shared or published containers. Their work in progress may change without notice.

Document guide — EIR

Exchange Information Requirements at a glance

Purpose
Defines what information an appointment must deliver, when, in what form and how it will be accepted.

Owner
The appointing party for the lead appointed party (5.2.1); the lead appointed party for each appointed party (5.4.3).

Timing
Before the invitation to tender, and before each lower-tier appointment.

What a good EIR contains

- ☐ Project and appointment identification
- ☐ Reference to the project information requirements it serves
- ☐ Information requirements for the appointment
- ☐ Level of information need for each requirement

- ☐ Acceptance criteria for each requirement
- ☐ Information delivery milestones
- ☐ Supporting information the delivery team needs to meet the requirements
- ☐ Reference to the project information standard
- ☐ Reference to the production methods and procedures
- ☐ Reference information and shared resources provided
- ☐ Asset information requirements relevant to this appointment
- ☐ Tender response requirements for information management
- ☐ Security requirements, where applicable
- ☐ Interfaces with other appointments

Test every requirement
For each requirement ask: how will we know it has been met? If there is no answer, add an acceptance criterion or remove the requirement.

Document guide — BEP

BIM Execution Plan at a glance

Purpose
Explains how the delivery team will carry out the information management aspects of the appointment.

Owner
The lead appointed party, with input from every task team.

Timing
Pre-appointment BEP with the tender (5.3.2); confirmed BEP after appointment (5.4.1), then kept up to date.

What a confirmed BEP contains

- ☐ Names and roles of the individuals performing the information management function
- ☐ Information delivery strategy for meeting the EIR
- ☐ Federation strategy
- ☐ Information container breakdown structure
- ☐ High-level responsibility matrix, refined by the detailed matrix
- ☐ Agreed additions or amendments to the information standard
- ☐ Agreed additions or amendments to the production methods and procedures
- ☐ Schedule of software, hardware and IT infrastructure
- ☐ Exchange formats and versions
- ☐ Coordination and clash management process

- ☐ Review, authorisation and submission process
- ☐ CDE use within the delivery team
- ☐ Response to each EIR requirement
- ☐ Change control and revision history

> **Two BEPs, two purposes**
> The pre-appointment BEP is a proposal that is scored at tender. The confirmed BEP is a commitment that is incorporated into the appointment. Make sure the second really replaces the first.

Document guide — MIDP

Master Information Delivery Plan at a glance

Purpose
Brings all TIDPs together into one plan showing when each information container will be delivered and by whom.

Owner
The lead appointed party.

Timing
Established at appointment (5.4.5), baselined, and maintained under change control throughout delivery.

What an MIDP contains

- ☐ Every container from every TIDP
- ☐ Task team responsible for each container
- ☐ Container name or reference
- ☐ Level of information need
- ☐ Planned production start and duration
- ☐ Planned delivery date
- ☐ Information delivery milestone the container serves
- ☐ Dependencies between containers and task teams
- ☐ Required suitability at each milestone
- ☐ Baseline and current forecast dates
- ☐ Actual delivery dates
- ☐ Revision history and change references

Keep it live
An MIDP is only useful if it shows the current forecast. Update it regularly and tell the appointing party when a milestone is at risk.

Document guide — TIDP

Task Information Delivery Plan at a glance

Purpose
Lists the information containers one task team will produce, with dates and dependencies.

Owner
Each task team, agreed with the lead appointed party.

Timing
Established at appointment (5.4.4) and updated as the work develops.

What a TIDP contains

- ☐ Task team name and task team manager
- ☐ Container name following the naming convention
- ☐ Container title or description
- ☐ Container type, such as model, drawing or schedule
- ☐ Level of information need
- ☐ Author responsible
- ☐ Estimated production duration
- ☐ Dependencies on information from other task teams
- ☐ Need-by dates for incoming information
- ☐ Planned delivery date
- ☐ Milestone served
- ☐ Required suitability at delivery

Dependencies are the point
The most valuable part of a TIDP is often the dependencies column. It tells other task teams what you need from them and when.

Document guide — Responsibility matrix

Responsibility matrices at a glance

Purpose
Assigns information deliverables to task teams so nothing is missed and nothing is duplicated.

Owner
The lead appointed party, agreed with task teams. The appointing party keeps its own information management assignment matrix.

Timing
High-level matrix with the pre-appointment BEP (5.3.2); detailed matrix at appointment (5.4.2).

What a detailed responsibility matrix contains

- ☐ Every information deliverable from the EIR
- ☐ Task team responsible for each deliverable
- ☐ Element, system or document covered
- ☐ Level of information need for each deliverable
- ☐ Information delivery milestone
- ☐ Interfaces between task teams
- ☐ Elements explicitly excluded from a task team's scope
- ☐ Links to the TIDP entries
- ☐ Reference to the EIR requirement satisfied
- ☐ Stage-by-stage changes in responsibility
- ☐ Agreement by each task team
- ☐ Revision history

Look for gaps at the edges
Most responsibility problems sit at interfaces, such as builder's work in connection or fire stopping. Name an owner for each interface element explicitly.

Document guide — Information protocol

Information Protocol at a glance

Purpose
Sets the contractual obligations, licences and liabilities for information on the project.

Owner
The appointing party, with legal advice.

Timing

Established during assessment and need (5.1.8) and incorporated into every appointment.

What an information protocol covers

- ☐ Parties and the appointments it applies to
- ☐ Precedence relative to other appointment documents
- ☐ Obligations of the appointing party
- ☐ Obligations of the lead appointed party and appointed parties
- ☐ Licences to use information and their limits
- ☐ Use of information after the appointment ends
- ☐ Liabilities and limitations relating to information
- ☐ Reference to the EIR, information standard and production methods
- ☐ Reference to the BEP and delivery plans
- ☐ Use of the CDE
- ☐ Security requirements and incident reporting
- ☐ Process for changing the protocol

Flow it through every tier

The protocol should be referenced in each appointment in the chain, not only the top tier. Otherwise licences granted to the client may not be backed by licences from the parties who created the information.

4. Appointment: requirements, tender and BEP

Follow an information management appointment from the first question the client asks to the archived project information model. Click any step in the swimlane diagram to see who leads it, the clause it comes from, what goes in, what happens and which documents come out. Then switch to the view for your own role, check the decision gates, and test yourself with the quiz.

Interactive diagram

ISO 19650-2 information management process, by party

Assess need and set requirements Appointing party ISO 19650-2 5.1.2

Turn OIR and AIR into project information requirements

Inputs

- Organisational information requirements (OIR)
- Asset information requirements (AIR) for the asset being created or changed
- Business case, project brief and key decision points
- Existing asset information and lessons learned from earlier projects

Outputs

- Project information requirements (PIR)
- List of key decision points linked to the project programme
- Agreed information purposes

Activities

- Identify the decisions the client must make during the project and the information each decision depends on
- Derive project information requirements (PIR) from the OIR and AIR
- Agree the purposes the information will serve, such as design approval, cost planning, statutory submissions and handover
- Record who in the client organisation needs which information and why
- Check proportionality: scale the requirements to the size and risk of the project

Tip: Start from the questions the client must answer, not from a list of model elements. A requirement that does not support a decision is a candidate for deletion.

Appoint the information management function

Appointing party | ISO 19650-2 5.1.1

Decide who manages information on the client side

Inputs

- Client organisation structure and in-house skills
- Project scale, complexity and procurement route
- Information management assignment matrix (ISO 19650-2 Annex A) as a starting point

Outputs

- Named individuals for the appointing party's information management function
- Completed information management assignment matrix
- Scope of services for any external information manager

Activities

- Decide whether the function is performed in-house, by a third party, or by a mix of both
- Name the individuals and define their authority and scope
- Assign each information management task using an assignment matrix
- Record any tasks delegated to a prospective lead appointed party and make that clear in the appointment

Tip: Delegating tasks to a consultant does not delegate accountability. The appointing party still owns its requirements and its acceptance decisions.

Establish milestones, standards, methods, CDE and protocol

Appointing party

ISO 19650-2 5.1.3 to 5.1.8

Set the rules of the game before anyone tenders

Inputs

- Project information requirements and key decision points
- Organisational standards and any national annex the client adopts
- Existing survey data, asset records and shared libraries
- Client legal and commercial advice on licences and liabilities

Outputs

- Information delivery milestone schedule
- Project information standard and production methods
- Configured and tested CDE
- Information protocol ready to incorporate into appointments

Activities

- Set information delivery milestones aligned with the project programme (5.1.3)
- Establish the project information standard: formats, naming, classification, metadata and level of information need (5.1.4)
- Establish production methods and procedures for capture, review, approval, security and delivery (5.1.5)
- Assemble reference information and shared resources such as templates, object libraries and surveys (5.1.6)
- Implement, configure and test the common data environment, including states, permissions and workflows (5.1.7)
- Establish the information protocol with licences, obligations and liabilities for information (5.1.8)

Tip: Test the CDE with a dummy upload, review and share cycle before the tender goes out. Tenderers price what they see, so a half-built CDE leads to vague tender responses.

Develop the EIR and issue the invitation to tender

Appointing party | ISO 19650-2 5.2.1 to 5.2.4

Tell each prospective lead appointed party exactly what is needed

Inputs

- Project information requirements and milestones
- Information standard, production methods, CDE details and information protocol
- Reference information and shared resources
- Procurement strategy and appointment scope

Outputs

- Exchange information requirements (EIR) for the appointment
- Tender response requirements and evaluation criteria
- Invitation to tender information pack

Activities

- Write the appointing party's exchange information requirements for this appointment, including acceptance criteria (5.2.1)
- Select the reference information and shared resources to issue to tenderers (5.2.2)
- Define what the tender response must contain and how information management will be evaluated (5.2.3)
- Compile the invitation to tender pack and issue it to each tenderer (5.2.4)

Tip: Write each requirement so that a reviewer could later say yes or no. If you cannot describe how you would accept it, you have not finished writing it.

Prepare the pre-appointment BEP

Lead appointed party | ISO 19650-2 5.3.1 to 5.3.2

Show how the delivery team would meet the EIR

Inputs

- Invitation to tender pack including the EIR
- Project information standard and production methods
- Proposed delivery team members and their roles

Outputs

- Pre-appointment BIM execution plan
- High-level responsibility matrix
- Proposed federation strategy

Activities

- Nominate the individuals who will carry out the information management function in the delivery team (5.3.1)
- Set out the proposed information delivery strategy
- Propose the federation strategy and information container breakdown structure
- Draft a high-level responsibility matrix showing which task team delivers which information
- Propose any additions or amendments to the information standard and production methods
- List the proposed software, hardware and IT infrastructure

Tip: Respond to the EIR line by line. Evaluators score what they can trace, and a generic company BEP rarely answers the specific requirements.

Assess capability and capacity

Appointed parties / task teams | ISO 19650-2 5.3.3 to 5.3.4

Each task team proves it can deliver

Inputs

- EIR and pre-appointment BEP draft
- Task team skills, experience and certifications
- Available IT, software versions and CDE connectivity
- Workload and resource forecasts

Outputs

- Task team capability and capacity assessments
- Delivery team capability and capacity summary
- Training and resource needs list

Activities

- Each task team assesses its information management capability and its capacity to deliver on time (5.3.3)
- Identify skills or technology gaps and how they will be closed
- The lead appointed party aggregates the results into a delivery team summary (5.3.4)
- Flag any task team whose capacity depends on other commitments

Tip: Capacity is as important as capability. A highly skilled team that is fully booked on another project is a delivery risk, so say so honestly.

Mobilisation plan, risk register and tender response

Lead appointed party | ISO 19650-2 5.3.5 to 5.3.7

Compile and submit the delivery team's offer

Inputs

- Pre-appointment BEP
- Delivery team capability and capacity summary
- Tender response requirements and evaluation criteria

Outputs

- Proposed mobilisation plan
- Information delivery risk register
- Submitted tender response

Activities

- Prepare the proposed mobilisation plan: how people, IT and methods will be brought in and tested (5.3.5)
- Establish the information delivery risk register with owners and mitigations (5.3.6)
- Check the response against every tender requirement
- Compile and submit the tender response (5.3.7)

Tip: Risks about the client's own inputs, such as late survey data, belong in the register too. Raising them at tender is far easier than after appointment.

Evaluate tenders and appoint

Appointing party | ISO 19650-2 5.2.3 and 5.4.6

Select the delivery team and sign the appointment

Inputs

- Tender responses from each prospective lead appointed party
- Evaluation criteria issued with the invitation to tender
- Clarification responses

Outputs

- Tender evaluation record
- Signed lead appointed party appointment
- Agreed list of amendments to the information standard or methods

Activities

- Score each response against the published information management criteria
- Clarify gaps or proposed amendments to the standard and methods
- Agree the scope of any changes before contract award
- Complete the lead appointed party's appointment documents incorporating the EIR, standard, methods, protocol, BEP and MIDP (5.4.6)

Tip: The appointment documents are only complete when the confirmed BEP and MIDP are attached. Agree a short, dated route to finalise them if the contract is signed first.

Confirm the BEP, responsibility matrix and lead EIR

Lead appointed party | ISO 19650-2 5.4.1 to 5.4.3

Turn the offer into a working delivery plan

Inputs

- Pre-appointment BEP and tender clarifications
- Appointing party's EIR and agreed amendments
- Confirmed delivery team members

Outputs

- Post-appointment BIM execution plan
- Detailed responsibility matrix
- Lead appointed party EIR for each appointed party

Activities

- Confirm the BEP with named individuals, the agreed information delivery strategy and the confirmed federation strategy (5.4.1)
- Develop the detailed responsibility matrix assigning every information deliverable to a task team (5.4.2)
- Write the lead appointed party's EIR for each appointed party, flowing down the client requirements (5.4.3)
- Seek agreement of the confirmed BEP with the appointing party

Tip: Check that every requirement in the client EIR appears in at least one appointed party EIR. Requirements that are not flowed down are often never delivered.

Produce task information delivery plans

Appointed parties / task teams | ISO 19650-2 5.4.4

Each task team plans its own containers and dates

Inputs

- Detailed responsibility matrix
- Lead appointed party EIR
- Information delivery milestones
- Level of information need for each deliverable

Outputs

- Task information delivery plan (TIDP) for each task team
- Container list with planned names

Activities

- List every information container the task team will produce
- Set the level of information need, author, production duration and dependencies for each container
- Align delivery dates with the milestones and with other task teams' needs
- Agree the TIDP with the lead appointed party

Tip: Plan dependencies explicitly. A structural model that needs the architectural grid by a certain date should say so in the TIDP, not in an email.

Aggregate TIDPs into the MIDP and complete appointments

Lead appointed party

ISO 19650-2 5.4.5 and 5.4.7

One plan for the whole delivery team

Inputs

- All task information delivery plans
- Information delivery milestones
- Detailed responsibility matrix

Outputs

- Baselined master information delivery plan (MIDP)
- Signed appointed party appointments

Activities

- Aggregate the TIDPs into the master information delivery plan (5.4.5)
- Resolve clashes in dates, dependencies and responsibilities between task teams
- Baseline the MIDP and share it with the appointing party
- Complete each appointed party's appointment documents with the lead EIR, protocol, BEP and TIDP (5.4.7)

Tip: Treat the MIDP as a live programme, not a one-off submission. Re-baseline it under change control and tell the appointing party when milestones move.

Mobilise people, IT and methods

Shared

ISO 19650-2 5.5.1 to 5.5.3

Train, set up and test before live production

Inputs

- Mobilisation plan
- Confirmed BEP and information standard
- CDE access and software list

Outputs

- Training and induction records
- IT and CDE mobilisation record
- Production methods test report

Activities

- Confirm resource availability and deliver induction and training on the project standards (5.5.1)
- Procure, install and configure software and hardware, and grant CDE access (5.5.2)
- Test container naming, breakdown structure, federation and delivery workflows with sample information (5.5.3)
- Update the BEP and risk register with anything the tests reveal

Tip: Run a full rehearsal: upload, check, share, federate and submit a sample container. Problems found in week one cost hours; the same problems found at the first milestone cost weeks.

Collaborative production of information

Appointed parties / task teams | ISO 19650-2 5.6.1 to 5.6.4

Work in progress, check, review and share

Inputs

- TIDP and lead appointed party EIR
- Reference information and shared resources
- Shared information from other task teams

Outputs

- Shared information containers with status and suitability
- Quality assurance records
- Coordination issue log

Activities

- Confirm reference information and shared resources are available before starting (5.6.1)
- Generate information in the work in progress state, coordinating spatially and technically with other task teams (5.6.2)
- Carry out a quality assurance check against the information standard: naming, metadata, format and status (5.6.3)
- Review content against the requirements and approve it for sharing with a suitability code (5.6.4)
- Raise and close coordination issues through agreed channels such as BCF

Tip: The QA check (is it named and formatted correctly?) and the review (is the content right?) are separate steps. Skipping the first wastes reviewers' time on files that will bounce anyway.

Review and authorise the information model

Lead appointed party | ISO 19650-2 5.6.5, 5.7.1 and 5.7.2

The lead appointed party checks the whole model

Inputs

- Shared information containers from all task teams
- Federated information model
- Appointing party's EIR and acceptance criteria

Outputs

- Federated model review and coordination report
- Authorisation record or rejection with reasons
- Authorised information model ready for submission

Activities

- Review the federated information model for coordination, completeness and consistency (5.6.5)
- Task teams submit their information for authorisation at the milestone (5.7.1)
- Review the submission against the EIR and the MIDP and authorise it, or reject it with recorded reasons (5.7.2)
- Send rejected containers back to the task team for revision

Tip: Use the client's acceptance criteria as your authorisation checklist. If the lead appointed party authorises against a different test, the appointing party will reject.

Appointing party acceptance Appointing party ISO 19650-2 5.7.3 and 5.7.4

The client accepts the model into the published state

Inputs

- Authorised information model
- EIR acceptance criteria
- Information delivery milestone

Outputs

- Acceptance record for the milestone
- Published information model
- Rejection record with reasons, if applicable

Activities

- Receive the authorised information model submission (5.7.3)
- Review it against the EIR and acceptance criteria
- Accept the model so it moves to the published state, or reject it with recorded reasons (5.7.4)
- Record the decision and notify the lead appointed party

Tip: Give reasons whenever you reject. A rejection without reasons cannot be fixed efficiently and is a common source of dispute.

Project close-out: archive the PIM and learn Shared ISO 19650-2 5.8.1 and 5.8.2

Hand over to the asset information model

Inputs

- Accepted project information model (PIM)
- Asset information requirements (AIR)
- Lessons learned captured during delivery

Outputs

- Archived project information model
- Information transferred to the AIM
- Lessons learned report

Activities

- The appointing party archives the project information model with access, retention and future-use rules (5.8.1)
- Transfer the relevant information into the asset information model (AIM) for operation, following ISO 19650-3
- Compile lessons learned from the appointing party and the delivery teams (5.8.2)
- Feed lessons into the organisational standards and the next project's requirements

Tip: Capture lessons at every milestone, not just at the end. By close-out, many of the people who knew what went wrong have moved on.

How to read the diagram

Each row is a party and each box is a step. Click a box to see its clause, inputs, activities, outputs, documents and a practical tip. Steps run broadly left to right, from assessment and need through to close-out, but in real projects several steps overlap and the production and delivery loop (5.6 and 5.7) repeats at every information delivery milestone.

The Shared row holds activities where several parties work together under one lead. Mobilisation is led by the lead appointed party but involves every task team; close-out is led by the appointing party with input from the whole delivery team.

> **One process, many appointments**
> A project can have several lead appointed parties, each with its own delivery team. The process repeats for every appointment: each one has its own EIR, BEP, MIDP and acceptance decisions, all working to the same project information standard and CDE.

Appointing party view

You set the rules and make the final decision

The appointing party is the client side: the organisation that receives information about the works, goods or services. In ISO 19650-2 it does most of its work before anyone is appointed. If the requirements, standard, methods, CDE and protocol are weak, every later step is harder.

After appointment, the appointing party's main job is to keep its requirements stable, answer questions promptly, and review and accept information at each milestone against the criteria it published.

Your duties, stage by stage

1. **5.1 Assessment and need.** Appoint the information management function, derive the PIR from the OIR and AIR, set milestones, and publish the information standard, production methods, reference information, CDE and information protocol.
2. **5.2 Invitation to tender.** Write an EIR for each appointment with acceptance criteria, choose the reference information to share, set tender response requirements and evaluation criteria, and issue the tender pack.
3. **5.3 Tender response.** Answer clarifications fairly and consistently for all tenderers. Do not change requirements without reissuing them to everyone.
4. **5.4 Appointment.** Evaluate responses, agree amendments, and make sure the EIR, standard, methods, protocol, confirmed BEP and MIDP are incorporated into the appointment.
5. **5.5 Mobilisation.** Provide CDE access, reference information and any client-side training. Confirm the delivery team's test results do not reveal problems with your own standards.
6. **5.6 Collaborative production.** Respond to queries, attend coordination reviews where useful, and keep your requirements stable. Raise changes through formal change control.
7. **5.7 Information model delivery.** Review authorised submissions against the acceptance criteria and accept or reject them, always recording reasons for rejection.
8. **5.8 Project close-out.** Archive the PIM, move the relevant information into the AIM and collect lessons learned from all parties.

Appointing party documents by stage

Stage	Documents you produce	Documents you review	Decision you make
5.1 Assessment and need	OIR; AIR; PIR; information management assignment matrix; milestone schedule; Project Information Standard; production methods; CDE Workflow & Procedure; Information Protocol	Existing asset information; lessons learned from earlier projects	Who performs the information management function and what the project needs
5.2 Invitation to tender	Exchange Information Requirements; tender response requirements; evaluation criteria; reference information register; BIM Specification & Scope of Services	Legal and procurement review of the tender pack	What each appointment must deliver and how it will be judged
5.3 Tender response	Clarification responses issued to all tenderers	Clarification questions	Whether any requirement needs reissuing
5.4 Appointment	Tender evaluation record; lead appointed party appointment documents	Pre-appointment BEP; capability summary; mobilisation plan; risk register; confirmed BEP; MIDP	Who to appoint and which amendments to accept
5.5 Mobilisation	CDE access permissions; client-side induction material	Mobilisation test results; updated risk register	Whether mobilisation is complete enough to start production
5.6 Collaborative production	Responses to queries; change instructions	Shared information at review points; progress reports	Whether requirements need formal change
5.7 Information model delivery	Acceptance records; rejection records with reasons	Authorised information model submissions	Accept or reject at each milestone
5.8 Project close-out	Archive record; updated AIM; lessons learned report	Lead appointed party lessons learned; handover information	What to retain, for how long and who may access it

Before you issue the tender

- ☐ Information management function appointed and named
- ☐ PIR traced to the OIR and AIR
- ☐ Information delivery milestones match the project programme
- ☐ Project information standard published, including naming and level of information need
- ☐ Production methods and procedures published
- ☐ Reference information and shared resources assembled
- ☐ CDE configured and tested with a sample workflow
- ☐ Information protocol reviewed by legal advisers
- ☐ EIR written for this appointment, with acceptance criteria
- ☐ Evaluation criteria include information management capability

Acceptance is your decision alone
Only the appointing party can accept an information model into the published state for its milestones. Advisers can help you review, but the decision and its recorded reasons belong to the appointing party.

Lead appointed party view

You plan, coordinate and authorise

The lead appointed party receives the appointment from the appointing party and coordinates information from its own appointed parties. It is the hinge of the process: it translates the client's EIR into a BEP and a set of lower-tier EIRs, combines task team plans into the MIDP, and authorises the information model before the client sees it.

Good lead appointed parties treat the BEP and MIDP as living management tools rather than tender paperwork.

Your duties, stage by stage

1. **5.3 Tender response.** Nominate information management individuals, prepare the pre-appointment BEP, collect task team capability and capacity assessments, prepare the mobilisation plan and risk register, and submit the tender response.
2. **5.4 Appointment.** Confirm the BEP, develop the detailed responsibility matrix, write an EIR for each appointed party, aggregate TIDPs into the MIDP and complete appointed party appointments.
3. **5.5 Mobilisation.** Lead the mobilisation: confirm resources, deliver training, set up IT and CDE access, and test naming, federation and delivery workflows.
4. **5.6 Collaborative production.** Coordinate task teams, run federated model reviews, manage the issue log and keep the MIDP up to date.
5. **5.7 Information model delivery.** Review task team submissions against the EIR and MIDP, authorise or reject with reasons, and submit the authorised model to the appointing party.
6. **5.8 Project close-out.** Support archiving and handover, and contribute lessons learned from the delivery team.

Lead appointed party documents by stage

Stage	Documents you produce	Documents you receive	Watch out for
5.3 Tender response	BIM Execution Plan (Pre-appointment); delivery team capability summary; Mobilization Plan; Information Delivery Risk Register; tender response	Invitation to tender; EIR; task team Capability & Capacity Assessments	Generic BEPs that do not answer the specific EIR
5.4 Appointment (BEP)	BIM Execution Plan (Post-appointment); Detailed Responsibility Matrix; Model Production and Delivery Table	Tender clarifications; agreed amendments	Forgetting to name individuals or confirm the federation strategy

Stage	Documents you produce	Documents you receive	Watch out for
5.4 Appointment (flow-down)	Lead appointed party EIR for each appointed party; appointed party appointment documents	Appointed party acknowledgements	Client requirements that are not flowed down
5.4 Appointment (planning)	Master Information Delivery Plan	Task Information Delivery Plans	MIDP dates that do not match the client milestones
5.5 Mobilisation	Training records; IT mobilisation record; methods test report; updated risk register	Task team readiness confirmations	Skipping the rehearsal because the programme is tight
5.6 Collaborative production	Federated model; coordination reports; issue log; BIM Progress Report; BIM Meeting Minutes	Shared containers from each task team	Reviewing content before the QA check is passed
5.7 Information model delivery	Authorisation records; rejection records with reasons; submission to the appointing party	Task team submissions for authorisation; acceptance or rejection from the client	Authorising against criteria that differ from the client's
5.8 Project close-out	Lessons Learned Report input; handover information	Archive requirements from the appointing party	Leaving lessons learned until the team has disbanded

Before live production starts

- ☐ Confirmed BEP agreed with the appointing party
- ☐ Detailed responsibility matrix covers every deliverable in the EIR
- ☐ An EIR issued to every appointed party
- ☐ Every task team has an agreed TIDP
- ☐ MIDP baselined and aligned with client milestones
- ☐ Mobilisation tests passed and recorded
- ☐ Risk register updated after mobilisation

> **Authorisation is a quality gate, not a formality**
> When you authorise an information model, you are telling the client it meets their requirements. Use the client's acceptance criteria as your test and keep evidence of the review.

Task team view

You produce the information

Appointed parties and their task teams create almost all the information on a project: models, drawings, schedules, specifications and reports. ISO 19650-2 asks each task team to prove its capability at tender, plan its own deliverables in a TIDP, and then produce, check, review and share information in the CDE.

A task team can be a whole firm, a department, or a group drawn from several organisations. What matters is that it has a clear scope in the responsibility matrix and a named task team manager.

Your duties, stage by stage

1. **5.3 Tender response.** Complete a capability and capacity assessment and contribute to the pre-appointment BEP, especially software, skills and proposed container breakdown for your discipline.
2. **5.4 Appointment.** Review the lead appointed party EIR, agree your part of the responsibility matrix and produce your TIDP.
3. **5.5 Mobilisation.** Attend induction and training, set up software and CDE access, and take part in the naming and federation tests.
4. **5.6 Collaborative production.** Check reference information, generate information in work in progress, run the QA check, review content and share it with the right suitability code.
5. **5.7 Information model delivery.** Submit information for authorisation at each milestone and fix anything rejected by the lead appointed party or the appointing party.
6. **5.8 Project close-out.** Provide final information for archiving and handover and share lessons learned.

Task team documents by stage

Stage	Documents you produce	Documents you rely on	Good practice
5.3 Tender response	Capability & Capacity Assessment; input to the pre-appointment BEP	Invitation to tender; client EIR	Be honest about capacity and name any gaps
5.4 Appointment	Task Information Delivery Plan; acknowledgement of the lead EIR	Lead appointed party EIR; Detailed Responsibility Matrix; BEP	List dependencies on other task teams explicitly
5.5 Mobilisation	Training attendance; test containers	Mobilization Plan; Project Information Standard	Use the test to configure templates and export settings
5.6 Generate	Information containers in work in progress	Reference information; shared information from other task teams	Only use shared or published information from others
5.6 Check	Information Container QA Checklist records	Project Information Standard; CDE Workflow & Procedure	Automate naming and metadata checks where possible
5.6 Review and share	Review records; shared containers with suitability codes	TIDP; level of information need	Check content against the requirement, not just the drawing
5.7 Submit	Submission for authorisation; revised containers after rejection	MIDP; acceptance criteria	Keep a register of what was submitted at each milestone
5.8 Close-out	Final containers for archive; lessons learned input	Information Handover Strategy; AIR	Record what you would do differently next time

Before you share a container

- ☐ Name follows the project naming convention
- ☐ Metadata, status and revision are complete
- ☐ File format matches the information standard
- ☐ Content meets the level of information need in the TIDP
- ☐ Reviewed by someone other than the author
- ☐ Correct suitability code assigned
- ☐ Superseded versions are not left in the shared state

> **Work in progress is private for a reason**
> Information in work in progress is visible only to your task team. Share it only when it has passed the QA check and review, so other teams can rely on what they see.

Milestones & decision gates

Gates keep the process honest

ISO 19650-2 does not use the word gate, but its activities naturally produce decision points where one party must confirm something before the next step can start. Treating them as formal gates, with named decision makers and recorded evidence, makes audits straightforward and stops problems flowing downstream.

The production and delivery gates (G7 and G8 below) repeat at every information delivery milestone set by the appointing party.

Decision gates and the evidence they need

Gate	Clause	Decision owner	Decision	Evidence required
G1 Requirements ready	5.1.2 to 5.1.3	Appointing party	The project knows what information it needs and when	PIR traced to OIR and AIR; information delivery milestone schedule; named information management function
G2 Framework ready	5.1.4 to 5.1.8	Appointing party	Standards, methods, CDE and protocol are ready for tender	Published information standard; production methods; tested CDE; information protocol; reference information register
G3 Tender issued	5.2.1 to 5.2.4	Appointing party	Tender pack is complete and can be issued	EIR with acceptance criteria; tender response requirements; evaluation criteria; complete tender pack
G4 Tender submitted	5.3.7	Prospective lead appointed party	The response is complete and compliant	Pre-appointment BEP; capability and capacity summary; mobilisation plan; risk register; compliance check against tender requirements

Gate	Clause	Decision owner	Decision	Evidence required
G5 Appointment made	5.4.1 to 5.4.6	Appointing party	The lead appointed party is appointed on agreed terms	Evaluation record; confirmed BEP; detailed responsibility matrix; MIDP; signed appointment incorporating EIR, standard, methods and protocol
G6 Delivery team appointed	5.4.3 to 5.4.7	Lead appointed party	All appointed parties are appointed with clear scope	Lead EIR per appointed party; agreed TIDPs; signed appointed party appointments
G7 Ready to produce	5.5.1 to 5.5.3	Lead appointed party	People, IT and methods are mobilised and tested	Training records; CDE access list; methods test report; updated risk register
G8 Shared for coordination	5.6.3 to 5.6.4	Task team	Information is fit to share with other task teams	QA check record; review record; suitability code assigned in the CDE
G9 Authorised	5.6.5 and 5.7.2	Lead appointed party	The information model meets the EIR for this milestone	Federated model review; coordination report; authorisation record or rejection with reasons
G10 Accepted	5.7.4	Appointing party	The information model is accepted into the published state	Acceptance record against the EIR criteria, or rejection with reasons
G11 Closed out	5.8.1 to 5.8.2	Appointing party	Project information is archived and lessons captured	Archive record with retention and access rules; AIM update confirmation; lessons learned report

Gate owners by colour

Amber gates

Owned by the appointing party: requirements ready, framework ready, tender issued, appointment made, accepted and closed out.

Green gates

Owned by the lead appointed party: tender submitted, delivery team appointed, ready to produce and authorised.

Teal gates

Owned by task teams: shared for coordination, which happens many times for each container.

Make each gate auditable

- ☐ Name the decision maker for every gate in the BEP or assignment matrix
- ☐ Define the evidence before the gate, not on the day
- ☐ Record the decision, date and reasons in the CDE
- ☐ Link the evidence to the clause it satisfies
- ☐ Record rejections with reasons and the action required
- ☐ Review open risks at every gate

Rejection is part of the process
Both authorisation and acceptance can end in rejection. The standard expects reasons to be given so the information can be corrected and resubmitted. Plan time in the MIDP for at least one resubmission at critical milestones.

RACI by activity

Who does what

ISO 19650-2 assigns each activity to a party, and its Annex A offers an information management assignment matrix for recording responsibilities. The RACI below is a practical interpretation for a typical single-tier delivery team. Adjust it in your own assignment matrix and BEP, especially where the appointing party delegates tasks to an external information manager.

What the letters mean

R: Responsible
Does the work.

A: Accountable
Owns the outcome and signs it off. One party only per activity.

C: Consulted
Gives input before the work is finished.

I: Informed
Is told about the result.

RACI matrix for the appointment process

Activity	Appointing party	Lead appointed party	Task teams
Appoint information management function (5.1.1)	A/R	I	-
Establish project information requirements (5.1.2)	A/R	-	-
Establish milestones, standard and methods (5.1.3 to 5.1.5)	A/R	-	-
Establish reference information, CDE and protocol (5.1.6 to 5.1.8)	A/R	-	-
Establish EIR and issue invitation to tender (5.2)	A/R	I	-
Prepare pre-appointment BEP (5.3.2)	I	A/R	C
Assess capability and capacity (5.3.3 to 5.3.4)	I	A	R
Mobilisation plan and risk register (5.3.5 to 5.3.6)	I	A/R	C
Confirm BEP (5.4.1)	C	A/R	C
Detailed responsibility matrix (5.4.2)	I	A/R	C
Lead appointed party EIR (5.4.3)	I	A/R	C

Activity	Appointing party	Lead appointed party	Task teams
Task information delivery plans (5.4.4)	-	A	R
Master information delivery plan (5.4.5)	I	A/R	C
Mobilise resources, IT and test methods (5.5)	I	A/R	R
Generate, check, review and share information (5.6.1 to 5.6.4)	I	A	R

RACI matrix for delivery and close-out

Activity	Appointing party	Lead appointed party	Task teams
Federated information model review (5.6.5)	I	A/R	C
Submit information for authorisation (5.7.1)	-	A	R
Review and authorise the information model (5.7.2)	I	A/R	C
Submit authorised model for acceptance (5.7.3)	I	A/R	-
Review and accept the information model (5.7.4)	A/R	C	I
Archive the project information model (5.8.1)	A/R	C	C
Compile lessons learned (5.8.2)	A/R	R	C

Generate your own matrix
Use the AI tab to draft an information management assignment matrix for your project, then check it against your appointments and BEP before adopting it.

Common pitfalls

Pitfalls that cause the most trouble

EIR copied from another project
Requirements that do not match this project's decisions lead to wasted effort and arguments at acceptance. Derive the EIR from this project's PIR.

No acceptance criteria
If the EIR does not say how information will be accepted, the delivery team cannot test its own work and the client cannot reject fairly.

CDE set up after appointment
The CDE is meant to be ready before the invitation to tender. Setting it up late delays mobilisation and leaves tenderers guessing.

Pre-appointment BEP never updated
The confirmed BEP must reflect the real team, names and federation strategy. A tender document left unchanged becomes fiction within weeks.

Requirements not flowed down

The lead appointed party must issue an EIR to each appointed party. Without it, sub-consultants and specialists work to their own habits.

MIDP as a one-off spreadsheet

An MIDP that is never re-baselined hides slippage until a milestone is missed.

Mobilisation tests skipped

Naming, federation and export problems then surface at the first milestone, when they are most expensive to fix.

Authorisation confused with acceptance

The lead appointed party authorises; the appointing party accepts. Treating one as the other removes a quality gate.

Symptom, cause and fix

Symptom	Likely cause	Fix
Tender responses are hard to compare	No tender response requirements or evaluation criteria	Publish a response template and scoring criteria with the tender (5.2.3)
Files arrive with inconsistent names	Naming not tested during mobilisation	Test naming with sample containers and automate checks (5.5.3)
Task teams wait for each other	Dependencies missing from TIDPs	Record dependencies and need-by dates in each TIDP (5.4.4)
Client rejects authorised models	Authorisation criteria differ from the EIR	Use the EIR acceptance criteria as the authorisation checklist (5.7.2)
Handover data is incomplete	AIR not reflected in the EIR	Trace every AIR item into an EIR requirement (5.1.2 and 5.2.1)
Disputes over who owns information	Information protocol missing from lower-tier appointments	Include the protocol in every appointment (5.4.6 and 5.4.7)
Nobody knows who to ask on the client side	Information management function not appointed	Name individuals and record them in the assignment matrix (5.1.1)

Questions people often ask

Can a small project combine documents?

Yes. Proportionality applies, so a small appointment can combine, for example, the BEP and responsibility matrix. Each required output should still be identifiable within the combined document.

Is the pre-appointment BEP a contract document?

It is a tender submission. The confirmed BEP produced after appointment is the one incorporated into the appointment.

Who issues the EIR to a specialist subcontractor?

The lead appointed party, or the appointed party above it in the chain, issues an EIR to each party it appoints.

Does the appointing party have to run the CDE?

The appointing party is responsible for establishing the project CDE. It may engage a third party to host or administer it.

> **A quick health check**
> If you can open the CDE and find the current EIR, the confirmed BEP, the latest MIDP and the acceptance record for the last milestone in under five minutes, your appointment is probably in good shape.

Quiz

Test your understanding

Ten questions on the appointment process. Each answer explains the clause behind it, so you can check the diagram if you are unsure.

> **Scored less than seven?**
> Revisit the interactive diagram and click through the lead appointed party row. Most confusion comes from the difference between the pre-appointment and confirmed BEP, and between authorisation and acceptance.

Ask AI

Get help with your own appointment

Ask the tutor about any step in the process, or generate a first draft of an information management assignment matrix for your project. Generated documents are starting points: review them against your appointments, BEP and the licensed standard before use.

5. Information uses

Information uses state why information is produced. They drive the level of information need, the delivery plan and the BEP.

IU-01 Existing conditions capture Capture Plan · Design · Construct

Survey existing site, structures and services (laser scan, photogrammetry, utility survey) and model them. *Reliable baseline; fewer surprises on site.*

Inputs

- Topographic & utility survey
- Point cloud

Deliverables

- Existing conditions model
- Registered point cloud
- Survey report

Level of information need: Geometry to survey tolerance; asset identity for retained elements · **Skills:** Scan-to-BIM, Survey coordination

IU-02 Site analysis Capture Plan

Use GIS and site data to evaluate site options, access, constraints and orientation. *Better site decisions early.*

Inputs

- GIS data
- Planning constraints

Deliverables

- Site analysis report
- Massing study

Level of information need: Low geometry; rich GIS attributes · **Skills:** GIS, Urban analysis

IU-03 Programming / space planning Design & analysis Plan · Design

Validate the design against the brief: areas, adjacencies, room data. *Design meets client requirements.*

Inputs

- Client brief
- Room data sheets

Deliverables

- Area schedule
- Brief compliance report

Level of information need: Spaces with area, function and occupancy · **Skills:** Space management

IU-04 Design authoring Design & analysis Design

Create the discipline models from which drawings and schedules are produced. *Consistent drawings and data from one source.*

Inputs

- Brief
- Standards & methods

Deliverables

- Discipline models
- Drawings
- Schedules

Level of information need: Per LOIN matrix by stage · **Skills:** Authoring software, Modelling standards

IU-05 Design review & visualisation Design & analysis Design

Stakeholders review the model in 3D / VR, renderings and walkthroughs. *Faster approvals, fewer misunderstandings.*

Inputs

- Federated model

Deliverables

- Review sessions
- Renderings
- Comments (BCF)

Level of information need: Appearance and spatial geometry · **Skills:** Visualisation, Review facilitation

IU-06 Structural analysis Design & analysis Design

Analytical models for structural design and checking. *Optimised, verifiable structure.*

Inputs

- Structural model
- Loads

Deliverables

- Analysis model
- Calculations

Level of information need: Analytical geometry, materials, loads · **Skills:** Structural analysis software

IU-07 Energy & environmental analysis Design & analysis Design

Energy, daylight, thermal comfort and CFD from the model. *Lower operating cost; certification evidence.*

Inputs

- Architectural & MEP models
- Weather data

Deliverables

- Energy model
- Simulation report

Level of information need: Spaces, envelope properties, zones · **Skills:** Energy simulation

IU-08 MEP engineering analysis

Design & analysis · Design

Load calculations, sizing and system analysis from the MEP model. *Right-sized systems.*

Inputs

- MEP model
- Space data

Deliverables

- Calculations
- System schematics

Level of information need: Systems connected, flow and capacity parameters · **Skills:** MEP analysis software

IU-09 Code & regulation validation

Design & analysis · Design

Check egress, fire, accessibility and planning rules against the model. *Fewer authority comments.*

Inputs

- Discipline models
- Regulations

Deliverables

- Compliance report

Level of information need: Spaces, doors, fire ratings · **Skills:** Rule-based checking

IU-10 Sustainability certification

Design & analysis · Design · Construct

Collect model data and evidence for LEED / BREEAM / local rating systems. *Credits evidenced from model data.*

Inputs

- Models
- Product data

Deliverables

- Credit evidence
- Material schedules

Level of information need: Material and product properties · **Skills:** Sustainability assessment

IU-11 3D coordination & clash detection

Coordination · Design · Construct

Federate models, run clash tests per matrix and resolve issues through BCF. *Fewer RFIs and rework on site.*

Inputs

- Discipline models
- Clash matrix

Deliverables

- Clash reports
- Issue log (BCF)
- Coordinated model

Level of information need: Spatially correct geometry incl. clearance zones · **Skills:** Federation, Clash management

IU-12 Drawing production from model

Coordination · Design · Construct

All drawings derived from coordinated models. *Drawings always consistent with the model.*

Inputs

- Coordinated models

Deliverables

- Drawing sets
- Sheet register

Level of information need: Per drawing scale · **Skills:** Sheet & view management

IU-13 Quantity take-off

Quantities & cost · Design · Construct

Extract quantities from models by classification for BOQ and estimates. *Faster, traceable quantities.*

Inputs

- Classified models

Deliverables

- Quantity schedules
- BOQ input

Level of information need: Classified elements with measurable geometry · **Skills:** 5D / QTO, Measurement rules (e.g. NRM)

IU-14 Cost estimation (5D)

Quantities & cost · Design · Construct

Link quantities to rates for cost plans and cost checks per milestone. *Cost certainty and option comparison.*

Inputs

- Quantities
- Rates

Deliverables

- Cost plan
- Cost reports

Level of information need: Classification, quantities · **Skills:** Cost management

IU-15 Phase planning (4D)

Time & logistics · Plan · Design · Construct

Link model elements to the programme to simulate sequence. *Validated sequence; clear communication.*

Inputs

- Model
- Programme

Deliverables

- 4D simulation
- Sequence reviews

Level of information need: Elements split by pour / zone / work package · **Skills:** 4D software, Planning

IU-16 Site logistics planning

Time & logistics · Construct

Model cranes, hoarding, access, laydown and temporary works. *Safer, efficient sites.*

Inputs

- Site model
- Programme

Deliverables

- Logistics plans per phase

Level of information need: Temporary works and site equipment · **Skills:** Logistics modelling

IU-17 Construction progress monitoring

Time & logistics · Construct

Track planned vs actual using the model (status parameters, scans, site capture). *Early warning of delay; evidence for payment.*

Inputs

- 4D model
- Site captures

Deliverables

- Progress dashboard
- Progress reports

Level of information need: Status / installation date per element · **Skills:** Progress tracking, Reality capture

IU-18 Digital fabrication & prefabrication

Field & quality · Construct

Shop / fabrication models for steel, MEP spools, precast and modular units. *Less waste, higher quality.*

Inputs

- Coordinated model

Deliverables

- Fabrication models
- Spool drawings
- CNC data

Level of information need: Fabrication-level geometry (LOD 400) · **Skills:** Fabrication software

IU-19 Field layout & setting out

Field & quality · Construct

Set out from the model with total stations / robotic layout. *Accurate installation.*

Inputs

- Coordinated model
- Control points

Deliverables

- Setting-out points
- Layout reports

Level of information need: Accurate positions and grid / survey coordinates · **Skills:** Survey & layout tools

IU-20 Model-based QA/QC & inspections

Field & quality · Construct

Inspection and test plans, snags and NCRs linked to model elements. *Traceable quality records.*

Inputs

- Model
- ITPs

Deliverables

- Inspection records
- Snag lists

Level of information need: Element identity and location · **Skills:** Field BIM apps

IU-21 Health & safety planning

Field & quality · Design · Construct

Identify and communicate hazards using the model (design risk register, safety zones). *Hazards removed at design stage.*

Inputs

- Models
- Programme

Deliverables

- Design risk register
- Safety plans

Level of information need: Hazard tags, temporary works · **Skills:** CDM / H&S

IU-22 Record (as-built) modelling

Handover & assets · Construct · Operate

Update models to reflect what was built, verified by survey. *Trustworthy information for operations.*

Inputs

- Approved changes
- Site verification

Deliverables

- As-built models
- As-built drawings

Level of information need: As-built geometry and installed product data · **Skills:** As-built verification

IU-23 Asset data (COBie) delivery

Handover & assets · Construct · Operate

Deliver structured asset data per the client's AIR at each stage. *Ready-to-load CAFM data.*

Inputs

- Models
- Product data
- Client AIR / COBie standard

Deliverables

- COBie workbook
- Asset register

Level of information need: Asset attributes per AIR · **Skills:** COBie, Data validation

IU-24 Asset tagging & labelling

Handover & assets · Construct · Operate

Assign unique asset tags per the client's codification and label assets on site. *Every asset traceable between field and data.*

Inputs

- Asset register
- Tag format

Deliverables

- Tagged asset register
- Labels / barcodes / QR

Level of information need: Tag, location and system codes · **Skills:** Asset codification

IU-25 O&M and handover documentation

Handover & assets · Construct · Operate

Manuals, warranties, certificates and test records linked to assets. *Complete, searchable handover.*

Inputs

- Submittals
- Asset register

Deliverables

- O&M manuals
- Warranty register

Level of information need: Document links per asset · **Skills:** Document control

IU-26 Commissioning

Handover & assets · Construct · Operate

Plan and record testing & commissioning against systems in the model. *Systems proven before handover.*

Inputs

- Systems model
- Commissioning plan

Deliverables

- Test records
- Commissioning certificates

Level of information need: System membership per asset · **Skills:** Commissioning management

IU-27 Space management

Operations · Operate

Manage occupancy, allocation and area using the AIM. *Better space utilisation.*

Inputs

- AIM spaces

Deliverables

- Space reports

Level of information need: Spaces, zones, areas, occupancy · **Skills:** CAFM / IWMS

IU-28 Planned maintenance Operations Operate

Drive planned and reactive maintenance from AIM asset data. *Lower lifecycle cost.*

Inputs

- Asset register
- Maintenance jobs

Deliverables

- Maintenance schedules

Level of information need: Maintainable assets with jobs and spares · **Skills:** CAFM / CMMS

IU-29 Building systems analysis & BMS integration Operations Operate

Link AIM with BMS / IoT for performance monitoring. *Optimised operation; fault detection.*

Inputs

- AIM
- BMS points

Deliverables

- Integrated dashboards

Level of information need: Point and system identifiers · **Skills:** BMS / IoT integration

IU-30 Emergency management Operations Operate

Use the AIM for evacuation planning, shut-off locations and emergency response. *Faster, safer response.*

Inputs

- AIM
- Fire strategy

Deliverables

- Emergency plans

Level of information need: Fire / life safety assets and zones · **Skills:** Emergency planning

IU-31 Digital twin Operations Operate

Live AIM connected to operational data and analytics. *Data-driven decisions over the asset lifecycle.*

Inputs

- AIM
- Live data

Deliverables

- Digital twin platform

Level of information need: As per AIR plus live data identifiers · **Skills:** Data integration, Analytics

IU-32 Disposal & renovation planning Operations Operate

Use the AIM to plan refurbishment, demolition and material recovery. *Circular-economy evidence.*

Inputs

- AIM
- Material data

Deliverables

- Refurbishment options
- Material passport

Level of information need: Material composition and condition · **Skills:** Lifecycle assessment

6. Containers, naming and status

CDE states

State	Name	Meaning
WIP	Work in progress	Information being developed by its task team; not visible to other task teams.
SHARED	Shared	Approved for sharing with others for coordination; not for construction.
PUBLISHED	Published	Authorized and accepted information for use (e.g., construction, handover).
ARCHIVED	Archived	Journal of transactions and superseded information, retained for audit.

Suitability / status codes

Codes commonly used under the UK National Annex; clients may define their own scheme in the information protocol.

Code	CDE state	Meaning
S0	WIP	Initial status / work in progress
S1	SHARED	Suitable for coordination
S2	SHARED	Suitable for information
S3	SHARED	Suitable for review and comment
S4	SHARED	Suitable for stage approval
S6	SHARED	Suitable for PIM authorization
S7	SHARED	Suitable for AIM authorization
A1–An	PUBLISHED	Authorized and accepted (stage n)
B1–Bn	PUBLISHED	Partially authorized and accepted, with comments
CR	PUBLISHED	As-constructed record

Container naming fields

Fields are joined with a hyphen, e.g. PRJ01-ABC-ZZ-00-M3-A-0001. Projects can adopt a client-specific scheme where the EIR requires one.

Field	Length	Example
Project	2–6 chars	PRJ01
Originator	3–6 chars	ABC
Volume / System	1–2 chars	ZZ
Level / Location	2 chars	00

Field	Length	Example
Type	2 chars	M3
Role	1–2 chars	A
Number	4–6 digits	0001

Common type codes

- M3 – 3D model
- M2 – 2D model
- DR – Drawing
- SP – Specification
- RP – Report
- SH – Schedule
- CA – Calculation
- MI – Minutes
- PP – Presentation
- CO – Correspondence

Common role codes

- A – Architect
- C – Civil engineer
- E – Electrical engineer
- M – Mechanical engineer
- P – Public health engineer
- S – Structural engineer
- K – Client
- Q – Quantity surveyor
- W – Contractor
- X – Subcontractor
- Z – General (non-disciplinary)

7. The common data environment

The common data environment is where every information container on a project is stored, checked, shared, approved and archived. This guide explains the four CDE states and the metadata behind them, the UK National Annex suitability and revision codes, how to set up containers, permissions and approval workflows, how common CDE platforms compare, and how to run the CDE day to day through to close-out.

Fundamentals — What a CDE is

A solution and a workflow, not just software

ISO 19650 describes the common data environment as the agreed source of information for a project or asset. It is where each information container is collected, managed and shared through a controlled process.

A CDE has two parts. The first is the technology: one or more systems that store containers, hold their metadata and record every action. The second is the workflow: the agreed rules for who can create, check, share, approve and archive information, and when.

Buying a document management product does not give you a CDE. A shared drive with strict rules can work for a small project, and an expensive platform without agreed rules quickly becomes a dumping ground. The workflow is what makes the difference.

What a working CDE gives the project

One source of truth

Everyone works from the same current revision, so nobody builds or prices from an out-of-date drawing.

Controlled sharing

Information moves from one team to another only after it has been checked and given a suitability code.

Clear status

Every container shows whether it is work in progress, shared, published or archived, and what it may be used for.

Audit trail

Every upload, status change, approval and download is recorded with who did it and when.

Security

Access is limited by role and by need to know, which supports a security-minded approach.

Handover ready

Published and archived information becomes the record that feeds the asset information model.

Who provides and who uses the CDE

Party	Typical role in the CDE
Appointing party	Establishes the project CDE, or makes sure one is provided, before appointments start. Owns the published and archived record.
Lead appointed party	Uses the project CDE for its delivery team, may run a separate WIP area for its own teams, and coordinates sharing and submissions.
Task teams	Create and check information in WIP, share it for coordination and respond to comments.
Information manager	Configures containers, permissions, metadata and workflows, and monitors compliance.
Operator or facility manager	Receives the published and archived record and uses it for the asset information model.

> **The CDE should exist before the first appointment**
> The appointing party is expected to set up, or arrange, the project CDE early, so that the tender information and the first deliverables already move through it. Setting it up after design has started means importing chaos.

Fundamentals — The four states

The four CDE states: Information starts in Work in progress, is checked and moved to Shared for coordination, is authorised and accepted into Published, and is moved to Archived when it is superseded or the project closes.

What each state means

State	Who can see it	What it is for	Typical codes
Work in progress (WIP)	Only the task team that owns it	Developing information before it is checked	S0, revisions such as P01.01
Shared	The wider delivery team and, where agreed, the appointing party	Coordination, information, review and comment, and submissions for approval	S1 to S4, S6, S7, revisions such as P01
Published	Everyone who needs to use it, including the appointing party	Authorised and accepted information that can be used for construction or operation	A1 to An, B1 to Bn, CR, revisions such as C01
Archived	Read only, controlled access	A record of every superseded and final container, kept for the audit trail and the asset record	The last status and revision at the time of archiving

The normal journey of a container

1. **Create in WIP.** The task team creates the container with a compliant name and fills in the required metadata.
2. **Check, review and approve.** The task team checks the content, format and naming, and the task team lead approves it for sharing.

3. **Share.** The container moves to Shared with a suitability code and a whole revision, for example S1 and P01.
4. **Coordinate and comment.** Other teams use it as a reference, raise comments and issues, and the owner revises it in WIP as P02.01.
5. **Authorise.** The lead appointed party reviews the full information model for the milestone and authorises it for submission.
6. **Accept and publish.** The appointing party reviews and accepts it, and the container is published with an A or B code and a C revision.
7. **Archive.** When a newer revision is published, or at project close, the container moves to Archived and stays as a record.

> **Information does not go backwards**
> A shared or published container is never edited in place. Changes always start as a new WIP revision, which is then checked and shared again. This keeps the history honest.

Fundamentals — Metadata

Why metadata matters

Metadata is the information about a container rather than inside it. The CDE uses it to decide who can see the container, what it can be used for and how it is filtered, reported and found.

ISO 19650 expects each container to carry at least a status, a revision and a classification. Most projects add more fields, such as originator, title and stage, from the project information standard.

Core metadata fields

Field	What it records	Example
Status (suitability)	What the information may be used for	S2, suitable for information
Revision	Which version of the container this is	P02 when shared, P02.03 in WIP, C01 when published
Classification	What the container is about, using the agreed system	A Uniclass code and title
Unique identifier	The container name from the naming convention	PRJ01-ABC-ZZ-00-M3-A-0001
Title or description	A human-readable description	Ground floor general arrangement
Originator	The organisation responsible for the container	ABC Architects
Stage or milestone	The delivery milestone it belongs to	Stage 3 spatial coordination
Security classification	Any sensitivity or access restriction	Restricted to named roles

Metadata quality checks

- ☐ Status and revision in the metadata match the title block or file properties
- ☐ The container name follows the project naming convention exactly
- ☐ Classification uses the agreed table and is not left blank
- ☐ Mandatory fields are enforced by the CDE, not left to memory
- ☐ Pick lists are used instead of free text wherever possible
- ☐ Metadata is checked as part of the review before sharing

> **Status in the file name is a trap**
> Some teams add the status and revision to the file name. When the status changes, the name changes, links break and duplicates appear. Keep status and revision as metadata fields unless the project information standard says otherwise.

Status & revision codes — Suitability codes

UK National Annex suitability codes

The UK National Annex to ISO 19650-2 gives a set of standard suitability codes. The code tells the receiver what a container may be used for. It does not say how good it is.

Code meanings can differ between editions of the Annex and between projects. Some older lists include codes that later editions withdrew. Always follow the codes set out in your project information standard.

Suitability codes and when to use them

Code	State	Meaning	When to use it
S0	WIP	Initial status or work in progress	Any container still being developed inside the task team
S1	Shared	Suitable for coordination	Models and drawings shared so other teams can coordinate against them
S2	Shared	Suitable for information	Information others may read but should not rely on for design decisions
S3	Shared	Suitable for review and comment	Information sent to another party, often the client, for comments
S4	Shared	Suitable for stage approval	Information submitted for sign-off at the end of a stage
S6	Shared	Suitable for PIM authorisation	The project information model submitted for authorisation and acceptance at a delivery milestone
S7	Shared	Suitable for AIM authorisation	Asset information submitted for acceptance into the asset information model
A1, A2 to An	Published	Authorised and accepted, for stage n	Information formally accepted by the appointing party, for example for construction

Code	State	Meaning	When to use it
B1, B2 to Bn	Published	Partially authorised and accepted, with comments	Information accepted with minor comments that must be addressed in the next revision
CR	Published	As-constructed record	The final record of what was built, for handover and the asset record

Choosing the right code

Ask who will use it

If another team will coordinate against it, use S1. If they just need to know about it, S2 is enough.

Ask what they will do

Comments wanted means S3. A formal sign-off at the end of a stage means S4.

Milestone submissions

Use S6 for the delivery milestone package and S7 for asset information at handover.

Published codes are the client's

Only the appointing party's acceptance gives an A or B code. The delivery team never sets these itself.

> **A suitability code is not a quality stamp**
> S1 does not mean the model is good. It means it is fit to be used for coordination. Quality comes from the check, review and approve step that happens before sharing.

Status & revision codes — Revision convention

P and C revisions

The UK National Annex uses P for preliminary revisions, which cover WIP and Shared, and C for contractual revisions, which are used once information is published.

A whole number such as P01 marks a revision that has been shared. A decimal suffix such as P01.01 marks a version still in WIP. When P01.03 is checked and shared, it becomes P01. The next round of work starts as P02.01 and is shared as P02.

Revision examples through a container's life

Event	State	Revision	Status
First draft created	WIP	P01.01	S0
Further work before sharing	WIP	P01.02, P01.03	S0
Shared for coordination	Shared	P01	S1
Revised after comments	WIP	P02.01	S0
Shared again for review	Shared	P02	S3

Event	State	Revision	Status
Submitted at the milestone	Shared	P03	S6
Accepted by the appointing party	Published	C01	A1
Contractual change issued	Published	C02	A2
Superseded or project closed	Archived	Last revision kept	Last status kept

Revision rules that avoid confusion

- Never reuse a revision. Once P02 has been shared, the next change is P03.
- Only one shared revision of a container is current at a time. The previous one is marked as superseded.
- WIP suffixes never leave the task team. Nothing with a decimal suffix should appear in Shared.
- Keep the revision history on the drawing or in the file properties in step with the CDE metadata.
- Record a short reason for each revision so reviewers know what changed.

> **Follow the project's own convention**
> Some clients use different prefixes or extra letters for tender or construction issues. The project information standard and the BEP should state the convention and give worked examples.

Status & revision codes — Transitions & approvals

Who approves each transition

Transition	Activity	Who does it	Who approves
WIP to Shared	Check, review and approve for sharing	Task team author and checker	Task team lead
Shared to WIP (new revision)	Respond to comments and revise	Task team author	Task team lead, at the next sharing
Shared to Shared for submission (S6 or S7)	Review and authorise the information model	Lead appointed party information manager	Lead appointed party
Shared to Published	Review and accept the submission	Appointing party reviewers	Appointing party
Published to Archived	Supersede or close out	Information manager	Appointing party, through the agreed process
Rejected submission	Return with reasons and a resubmission date	Appointing party	Appointing party

Status transitions by party

Produce in WIP

Task team · ISO 19650-2, collaborative production of information

S0 · P01.01

Inputs

- TIDP
- Project information standard
- Shared reference information

Outputs

- Container in WIP

Activities

- Create the container with a compliant name
- Fill in status, revision and classification
- Develop the content within the task team

Tip: Set up the metadata when the container is created, not just before sharing.

Check, review and approve

Task team · ISO 19650-2, quality assurance check

Quality gate before sharing

Inputs

- Container in WIP

Outputs

- Approved container ready to share

Activities

- Check naming, metadata and format
- Review content against the requirements
- Task team lead approves for sharing

Tip: A short written checklist is quicker and more reliable than a review from memory.

Share

Task team · ISO 19650-2, information sharing

S1 to S4 · P01

Inputs

- Approved container

Outputs

- Shared container

Activities

- Move the container to Shared
- Set the suitability code and whole revision
- Notify the teams that need it

Review and authorise Lead appointed party ISO 19650-2, information model review and authorisation
S6 or S7

Inputs

- Shared containers for the milestone
- MIDP

Outputs

- Authorised submission

Activities

- Review the federated information model
- Check completeness against the MIDP
- Authorise the package for submission

Tip: Reject internally rather than let the client find the gaps.

Review and accept Appointing party ISO 19650-2, information model acceptance
A1 to An, B · C01

Inputs

- Authorised submission
- EIR acceptance criteria

Outputs

- Published containers
- Or a rejection with reasons

Activities

- Review against the acceptance criteria
- Accept, accept with comments or reject
- Publish accepted containers

Archive Appointing party ISO 19650-2, project close-out
Read-only record

Inputs

- Superseded or final containers

Outputs

- Archived record

Activities

- Move to Archived
- Keep metadata and audit trail
- Transfer the record to the asset information model where needed

Use the document flow tool
The workflow pages of this platform walk a single container through every state with the code, revision, checks and common failures at each step. Use them to train new team members.

Setting up a CDE — Structure & naming

Design the structure around the workflow

The folder or container structure should make the four states obvious and keep each task team's WIP separate. Most platforms either use folders for states or use a status field with filtered views. Either works if it is applied consistently.

Keep the structure shallow. Deep folder trees hide information and encourage people to create their own private copies. Use metadata and saved views to slice information by discipline, zone or stage.

A typical top-level structure

Area	Contents	Access
00 Project admin	Information standard, BEP, MIDP, registers, templates	Read for all, edit by the information manager
01 WIP by task team	One area per task team, for example architecture, structures, MEP	Owning task team only
02 Shared	Shared containers, often split by originator or discipline	Read for the delivery team, upload by authorised sharers
03 Published	Accepted containers grouped by milestone or purpose	Read for all who need it, upload through the acceptance workflow only
04 Archived	Superseded and final containers	Read only, controlled
05 Incoming	Information from outside the delivery team, such as surveys or client data	Information manager checks before it is shared

Naming rules to set before go-live

- Use the project naming convention for every container, for example project, originator, volume, level, type, role and number
- Agree the code lists for each field and load them as pick lists
- Decide whether the CDE enforces naming on upload or flags errors for correction
- Agree how to name containers that come from outside the delivery team
- Keep folder names short and stable, and never use them to carry status

Check naming with the platform's tool
The naming checker on this platform tests a list of container names against the project convention. Run it on the first uploads during mobilisation.

Setting up a CDE — Permissions & workflows

Permissions by role

Role	WIP	Shared	Published	Archived
Task team author	Create and edit in own area	Read; share through the approval step	Read	Read on request
Task team lead	Full control of own area	Approve sharing for own team	Read	Read on request
Other task teams	No access	Read	Read	Read on request
Lead appointed party information manager	Read for monitoring where agreed	Manage and authorise submissions	Read	Read
Appointing party	No access	Read where agreed	Accept, publish and read	Read and manage
CDE administrator	Configure	Configure	Configure	Configure

Configuring approval workflows

1. **Map the transitions.** List each state change the project needs, from WIP to Shared to Published to Archived, and who acts at each.
2. **Set the gates.** Decide which transitions need a named approver and which can be done by the author after a checklist.
3. **Build the workflows.** Configure review and approval workflows in the platform, with reviewers, time limits and outcomes.
4. **Link the metadata.** Make each workflow outcome set the status and revision automatically where the platform allows it.
5. **Notify.** Set notifications so that reviewers know when something is waiting and authors know the outcome.
6. **Test.** Run a dummy container through every workflow with real users before the project goes live.

> **Permissions follow the appointment**
> Grant access by role and organisation, not by individual favour. When someone leaves a task team, remove their access on the same day.

Setting up a CDE — Mobilisation testing

Test before you rely on it

During mobilisation, the lead appointed party is expected to test the CDE and the information production methods before real work starts. The aim is to find problems with access, naming, workflows and file exchange while they are cheap to fix.

Record the results and any changes in the BEP or the mobilisation plan, so everyone works to the tested set-up.

CDE mobilisation test checklist

- ☐ Every task team member can log in and sees only the areas they should
- ☐ A test container can be uploaded with a compliant name and full metadata
- ☐ A non-compliant name is rejected or flagged
- ☐ A container moves from WIP to Shared through the approval workflow
- ☐ The status and revision update correctly at each transition
- ☐ Other task teams can find and download the shared container
- ☐ A submission package can be authorised and sent to the appointing party
- ☐ The appointing party can accept, reject or comment and the outcome is recorded
- ☐ Superseded revisions move to Archived and remain readable
- ☐ Native and IFC models open in the viewer and federate correctly
- ☐ BCF issues can be exchanged between the CDE and the authoring tools
- ☐ Large files upload and download in an acceptable time from each office and site
- ☐ The audit trail shows every action from the test
- ☐ Backup and recovery arrangements are confirmed with the provider

> **Include the site team**
> Test from site cabins and on tablets as well as from the office. Slow connections and mobile apps are where many CDEs fail in practice.

Setting up a CDE — Audit trail

What the audit trail should record

The audit trail is the CDE's memory. It proves who did what and when, which matters for disputes, for security and for the asset record.

A good audit trail cannot be edited by users, is kept for at least as long as the contract requires, and can be exported if the project moves to another platform.

Events to record

Event	Details to capture
Upload and new revision	User, organisation, date and time, container name, revision
Status change	Old and new status, who approved it, workflow reference
Review comments	Reviewer, comment, response, outcome
Downloads and views	User and date, especially for sensitive containers
Permission changes	Who changed what access, for whom and when

Event	Details to capture
Deletion requests	Who asked, who approved and why, ideally with the container kept in Archived

Avoid deletion
Deleting containers breaks the record. Prefer to supersede or withdraw them with a reason, and keep them in Archived.

Platforms — Comparison

Read this first
This comparison is general and neutral. Products change often, features depend on licence level and configuration, and some capabilities need plug-ins or partner tools. Check the current version and the vendor's documentation, and test against your own workflow, before you choose.

Common CDE platforms at a glance

Platform	ISO 19650 workflow support	Status and revision codes	BCF	IFC viewing	APIs
Autodesk Construction Cloud (Autodesk Docs)	Configurable review and approval workflows; naming standard checks available	Custom attributes and review outcomes can carry suitability and revision	Issues can be exchanged with BCF through import, export or connected tools; check current support	Web viewer opens IFC and many native formats	Autodesk Platform Services APIs
BIM 360 (legacy)	Document management with review and approval workflows	Custom attributes for status and revision	Through plug-ins or partner tools	Web viewer for IFC and native formats	Autodesk Platform Services APIs
Bentley ProjectWise and iTwin	Strong document control with configurable states and workflows	States and attributes can be set up to match project codes	Issue exchange through iTwin services or partner tools; check current support	iTwin viewing of IFC and many formats	ProjectWise and iTwin Platform APIs
Trimble Connect	Folder permissions and approval workflows; often paired with other document control	Status and custom properties on files	Native support for BCF topics and widely used with BCF-based tools	Web and desktop viewing of IFC and Trimble formats	Trimble Connect APIs
Oracle Aconex	Strong document control, transmittals, mail and review workflows	Configurable status and revision fields	Model coordination features; check BCF support for your version	Model viewer supports IFC	Aconex APIs

Platform	ISO 19650 workflow support	Status and revision codes	BCF	IFC viewing	APIs
Viewpoint For Projects (4Projects)	Document control and approval workflows, often used on UK projects	Configurable status and revision fields	Check current support	Check current support	Check with the vendor
Asite	Configurable workflows, often set up around ISO 19650	Configurable status and revision fields	Check current support	Model viewer supports IFC	Asite APIs
Thinkproject	Document control and workflows, often set up around ISO 19650	Configurable status and revision fields	BIM tools in the product family support BCF; check current support	Model viewing available; check the product	Thinkproject APIs

What to look for, whatever the brand

Workflow fit
Can it enforce your states, codes and approval gates without manual workarounds?

Metadata control
Can mandatory fields and pick lists be enforced on upload, and checked in bulk?

Open standards
Does it handle IFC and BCF well, so teams using different tools can still coordinate?

Export and exit
Can you export containers, metadata and the audit trail in usable formats at the end of the project?

Security
Does it support role-based access, single sign-on, data residency choices and download tracking?

Usability
Will site teams and occasional users actually use it, including on mobile devices?

Platforms — Selecting a platform

A simple selection process

1. **Write the requirements.** List your states, codes, workflows, security needs, integrations and number of users from the information requirements.
2. **Shortlist.** Pick two or three platforms that appear to meet the core needs, including any your organisation already licenses.
3. **Script a demonstration.** Ask each vendor to run your own scenario: upload, check, share, authorise, accept and archive one container.
4. **Run a pilot.** Test with real users from at least two task teams and the site, using real file sizes.

5. **Check the exit.** Confirm how containers, metadata and audit records are exported at the end of the project or contract.
6. **Decide and record.** Record the decision, the configuration and any known limits in the BEP or CDE procedures.

Questions to ask every vendor

- How are WIP, Shared, Published and Archived represented, and can they be enforced?
- Can suitability and revision codes be set automatically by workflow outcomes?
- How is naming checked, and what happens to a non-compliant upload?
- Which BCF versions and BCF API features are supported, natively or through partners?
- Which IFC versions does the viewer support, and can models be federated?
- Where is data stored, and what certifications does the service hold?
- What APIs and connectors are available for reporting and integration?
- How long are audit records kept, and how are they exported?

> **More than one system is normal**
> Many projects use one platform for document control and another for model coordination. That is fine if the information standard says which is the master for each type of container and how they stay in step.

Operating the CDE — Routines, KPIs & failures

Daily and weekly routines

When	Task	Who
Daily	Clear the approval and review queues	Task team leads and reviewers
Daily	Check new uploads for naming and metadata errors	Information manager
Daily	Respond to access requests and remove leavers	CDE administrator
Weekly	Share coordinated models before the coordination meeting	Task teams
Weekly	Report container status against the TIDPs and MIDP	Information manager
Weekly	Review overdue comments and rejected submissions	Lead appointed party
Monthly	Audit permissions and a sample of the audit trail	Information manager
Monthly	Review KPIs and agree corrective actions	Project leadership

Useful CDE KPIs

Figure	Meaning	Source
% compliant	Containers with fully compliant names and metadata on first upload	
Days	Average time from submission to acceptance or rejection	

Figure	Meaning	Source
% first time	Submissions accepted without rejection	
Count	Containers overdue against the MIDP	
Count	Containers shared outside the CDE, found in audits	
Count	Open review comments older than the agreed limit	

Measure the process, not just the volume
The number of files uploaded says little. Compliance, turnaround time and first-time acceptance tell you whether the CDE is actually working.

Common failures and how to fix them

Failure	Why it happens	Fix
Information shared by email instead of the CDE	The CDE is slow or people do not trust it	Make the CDE the only route for contractual information and fix the usability problems
Two current revisions of the same container	The old revision was not superseded on sharing	Configure the platform to supersede automatically and audit weekly
WIP revisions in Shared	Containers shared without the check step	Enforce the approval workflow and block decimal revisions in Shared
Wrong suitability codes	Authors pick codes at random	Train with examples and limit the codes offered at each transition
Private copies on local drives	Deep folder trees and poor search	Simplify the structure and provide saved views
Slow approvals	No named reviewers or time limits	Set reviewers, deadlines and escalation in the workflow
Leavers keep access	No link between appointments and permissions	Make removing access part of the leaver process
Federated model uses old revisions	Links point to files, not to current shared revisions	Federate only from current shared containers and check revisions before each meeting

Most failures are behaviour, not software
Fixing the platform settings helps, but most problems come from habits. Short, regular training and visible leadership support do more than another configuration change.

Operating the CDE — Security

A security-minded CDE

ISO 19650-5 sets out a security-minded approach to information management. It starts with a security triage to decide whether the project or asset handles sensitive information, and then puts proportionate measures in place.

For the CDE this means controlling who can see what, recording access, and making sure sensitive information is not shared more widely than it needs to be.

Security measures for the CDE

- ☐ Run the security triage and record the outcome before setting up the CDE
- ☐ Grant access by role and need to know, and review it regularly
- ☐ Use single sign-on and multi-factor authentication where available
- ☐ Mark sensitive containers with a security classification and restrict them
- ☐ Track downloads and external sharing of sensitive information
- ☐ Agree where data is hosted and who the provider can share it with
- ☐ Have an incident procedure for lost devices, wrong sharing and breaches
- ☐ Remove access on the day a person leaves the project

> **Asset data can be sensitive too**
> Floor plans, security systems and asset registers can reveal how to attack a building. Apply the same controls to handover information as to design information.

Operating the CDE — Close-out & archiving

Closing out the project CDE

1. **Confirm final acceptance.** Make sure every deliverable in the MIDP has been accepted and published, or formally removed with a reason.
2. **Supersede and archive.** Move all superseded revisions to Archived and check that nothing is left in WIP or Shared without a reason.
3. **Prepare the record.** Collect the as-constructed record, with CR codes where used, and the handover information for the asset information model.
4. **Transfer.** Transfer the information needed for operation into the appointing party's asset systems in the agreed formats.
5. **Export the audit trail.** Export the audit trail and metadata so the record survives if the platform contract ends.
6. **Remove access.** Close access for delivery team members who no longer need it and record the final access list.
7. **Capture lessons learned.** Record what worked and what did not in the CDE set-up and share it for the next project.

Plan the exit at the start
Agree at mobilisation how the archive will be exported and where it will live after the project. It is much harder to negotiate at the end.

Operating the CDE — FAQ, quiz & resources

Frequently asked questions

Is a CDE a piece of software?

No. It is the combination of technology and agreed workflow. The software supports the process but does not replace it.

Who is responsible for providing the CDE?

The appointing party is expected to establish the project CDE, or make sure one is provided, before appointments start. It may ask a delivery team or a third party to host it.

Can a delivery team use its own CDE?

Yes, many lead appointed parties use their own system for WIP and internal sharing. Information shared with the appointing party and published information should still go through the project CDE.

What is the difference between Shared and Published?

Shared information is available to the delivery team for coordination, review or submission. Published information has been authorised and accepted by the appointing party for use, for example for construction.

Who sets the A and B codes?

The appointing party, through its acceptance of a submission. The delivery team never gives itself an A or B code.

What revision does a container get when it is published?

Under the UK convention it moves from a P revision to a C revision, starting at C01, with an A or B suitability code.

Should status and revision be in the file name?

Usually not. Keep them as metadata, so the name stays stable and links do not break. Follow the project information standard if it says otherwise.

What happens to superseded information?

It moves to Archived and stays readable, with its metadata and audit trail, so the history of decisions can be traced.

How do BCF issues fit into the CDE?

BCF issues reference models and objects in the CDE. Many platforms store them alongside the models, and the issue log should be managed as project information like any other.

Can a shared drive be a CDE?

For a small, simple project it can, if the states, naming, permissions and approvals are applied strictly. In practice a dedicated platform makes this far easier to enforce and audit.

Official and neutral resources

- UK BIM Framework — Guidance on ISO 19650 in the UK, including the information management process and the National Annex (https://www.ukbimframework.org/)
- buildingSMART International — Open standards such as IFC and BCF that CDE platforms use for exchange (https://www.buildingsmart.org/)
- ISO 19650-1 on iso.org — Catalogue page for the concepts and principles part of ISO 19650 (https://www.iso.org/standard/68078.html)

> **Check AI output**
> AI drafts are a starting point. The information manager should confirm states, codes, permissions and workflows against the project information standard and the chosen platform before issuing procedures.

8. Level of information need

LOD, LOI and LOIN are three ways of answering one question: how much information is enough? This hub explains the purpose-driven level of information need framework of ISO 7817-1, the BIMForum level of development scale, and the UK legacy level of information, then shows how to specify, transfer and check information requirements in real EIRs, BEPs and asset information models.

Overview

Three terms, one goal

Level of information need (LOIN) is the international framework for defining the extent and granularity of information required. It is set out in ISO 7817-1:2024, Part 1: Concepts and principles, a first edition published in June 2024. It was prepared by ISO/TC 59/SC 13 together with CEN/TC 442 under the Vienna Agreement, and its European adoption, EN ISO 7817-1:2024, replaces EN 17412-1:2020, the European standard that first introduced the concepts. ISO 19650 uses the term level of information need when it describes how information requirements should be specified.

A guidance part, ISO/TS 7817-2, is being developed to explain step by step how to specify geometrical information, alphanumerical information and documentation. At the time of writing it is still a draft, so check its status before citing it.

Level of development (LOD) is a scale published by BIMForum in its LOD Specification, currently the 2025 edition. It describes how far an element's geometry and attached information have been developed, and therefore how much other people can rely on it, using the progression 100, 200, 300, 350 and 400. LOD 500, field verification, comes from the AIA definitions and sits outside that progression.

Level of information (LOI) is a UK legacy term from the PAS 1192 era, used alongside level of model detail to describe the non-graphical content expected at each stage. PAS 1192-2 has been withdrawn and the NBS BIM Toolkit that popularised numbered LOD and LOI bands has been retired, but the terms still appear in older contracts and templates.

The three approaches at a glance

LOIN (ISO 7817-1)

Purpose-driven. Defines geometry, alphanumeric information and documentation separately for each object, purpose, milestone and actor. No fixed numbered levels.

LOD (BIMForum)

Scale-driven. Numbered levels describe how developed and reliable an element is, from conceptual to fabrication and field-verified.

LOI (UK legacy)

Stage-driven. Describes the non-graphical information expected at each stage, paired with level of model detail in the now withdrawn PAS 1192-2.

LOD, LOI and LOIN compared: LOD grades how developed and reliable an element is, LOI grades the non-graphical data expected at a stage, and LOIN states, for each purpose and milestone, exactly which geometry, data and documents are needed. LOD or LOI labels can still be used inside a LOIN requirement as shorthand, provided the EIR defines them.

Why LOIN is purpose-driven

1. **Purpose.** Start with why the information is needed: cost estimating, coordination, energy analysis, fabrication, maintenance planning.
2. **Milestone.** Decide when it is needed: which information delivery milestone or decision point.
3. **Actor.** Identify who requests it and who delivers it, so responsibility is clear.
4. **Object.** Name the object or object type the requirement applies to, using an agreed breakdown or classification.
5. **Then define the information.** Only now specify geometry, alphanumeric information and documentation that meet that purpose, at that milestone, for that actor and object.

How LOIN is specified: ISO 7817-1 works in two steps. First the prerequisites set the context: purpose, information delivery milestone, actors and the objects within a breakdown structure. Then the information itself is described in three independent parts: geometrical information (detail, dimensionality, location, appearance, parametric behaviour), alphanumerical information (identification, information content) and documentation.

> **Less is more**
> The aim of LOIN is to deliver exactly what is needed and nothing more. Over-specifying wastes effort and can create false confidence; under-specifying leaves gaps. Every requirement should trace back to a purpose.

LOIN in four words

Figure	Meaning	Source
Purpose	Why the information is needed	
Milestone	When it is needed	
Actor	Who asks and who delivers	
Object	What it is about	

LOIN explained

The framework in plain terms

ISO 7817-1 describes a framework with two parts. First come the prerequisites, which set the context: the purposes, the information delivery milestones, the actors and the objects within a breakdown structure (clause 5). Then comes the detailed description of the information required, split into geometrical information, alphanumerical information and documentation (clause 6).

The standard does not prescribe numbered levels. Instead it gives a consistent structure so that requirements can be written in the same way by any appointing party and understood by any delivery team, including in machine-readable form. It also covers verification and validation of deliveries against

the requirement (clause 7) and gives informative example methods for specifying level of information need in an annex.

One stated aim is to prevent over-delivery: information beyond what the purpose needs costs effort to produce and check, and adds no value.

Prerequisites: set the context first

Purpose

Why is the information needed? Example: quantity take-off for cost planning.

Information delivery milestone

When is it needed? Example: the end of the spatial coordination stage.

Actors

Who requests it and who provides it? Example: requested by the cost consultant, delivered by the architect.

Objects in a breakdown structure

Which objects does it apply to? Example: internal walls identified through an agreed classification system such as Uniclass or another ISO 12006-2 based system.

Information types and their aspects

Information type	Aspect	What it describes	Example range
Geometrical	Detail	How simplified or detailed the shape is	From a symbolic box to a fully detailed object with fixings
Geometrical	Dimensionality	How many dimensions are represented	0D point, 1D line, 2D surface or 3D solid
Geometrical	Location	How the object is placed and positioned	Relative to a space, or absolute in project coordinates
Geometrical	Appearance	How realistic the visual representation is	Symbolic colour, through to realistic materials and textures
Geometrical	Parametric behaviour	Whether the shape can be changed by parameters	Fixed geometry, or parametric with adjustable dimensions
Alphanumerical	Identification	How the object is identified: name, identifier, type, classification code	A type name only, through to a unique asset tag and classification
Alphanumerical	Information content	The properties required, ideally referenced to a data dictionary following ISO 23386 and data templates following ISO 23387	A few performance properties, through to full product and maintenance data
Documentation	Documents	The documents that must accompany the object	None, through to data sheets, certificates, manuals, warranties or calculations

Geometry is not the whole story
A fire damper may need only a simple 3D box for coordination, but a rich set of properties and test certificates for handover. LOIN lets you specify each of these independently instead of forcing them to rise together.

LOD 100–500

Level of development in BIMForum terms

The BIMForum LOD Specification defines levels of development that describe how much an element has been thought through and how far other people can rely on it. The emphasis is on reliability for others, not just on how detailed the geometry looks.

The current edition is the LOD Specification 2025, released at the end of 2024 in two parts: Part I with element definitions and illustrations, and Part II, a spreadsheet for recording attribute requirements. It builds on the basic LOD definitions of the AIA G202-2013 project BIM protocol form and is organised by the CSI Uniformat 2010 classification.

Levels are applied per element, not to the whole model. A model at a given stage usually contains elements at several levels.

One element through LOD 100 to 400: The same door as it develops: a symbol or opening at LOD 100, a generic placeholder of approximate size at LOD 200, a specific measurable door at LOD 300, frame fixings and interfaces with the wall at LOD 350, and fabrication detail such as hinges, seals and ironmongery at LOD 400. Field verification (LOD 500) is a check of the installed element, not a further step of detail, so it is not shown.

LOD levels and their meaning

LOD	Geometric meaning	Typical use
LOD 100	Symbol or generic representation; element shown but not geometrically defined, information may be derived from other elements	Massing, early cost per area, feasibility
LOD 200	Generic placeholder with approximate size, shape, location and orientation	Concept design, early coordination, option studies
LOD 300	Specific element with defined size, shape, location and orientation that can be measured directly	Developed and spatially coordinated design, quantities, drawings
LOD 350	As LOD 300 plus the parts needed for coordination with nearby or attached elements, such as supports and connections	Detailed multi-discipline coordination
LOD 400	Element modelled with enough detail and accuracy for fabrication, including assembly and installation information	Fabrication, prefabrication, shop drawings
LOD 500 (outside the BIMForum progression)	Defined in the AIA protocol as a field-verified representation of size, shape, location, quantity and orientation; the BIMForum specification does not define or illustrate it because it is not a further level of geometry or data	As-built record and asset management, if the contract defines it

Three levels people often misunderstand

LOD 350 is about coordination

It adds interfaces and connections with other systems, such as hangers, supports and clearance zones, so that disciplines can coordinate reliably.

LOD 400 is about fabrication

It is modelled to a level where a fabricator could make or install the element from it, often by the specialist contractor rather than the designer.

LOD 500 is about verification

It indicates field verification of the installed element, not a further step of geometric detail beyond LOD 400. BIMForum leaves it out of its definitions for that reason, so a contract that asks for LOD 500 must say what verification it expects.

Development, detail and accuracy are different

Level of development is about reliability. Level of detail describes how much geometry is shown. Level of accuracy, as used in measured survey, describes positional tolerance. Make sure everyone knows which one a contract means.

LOD 400 in Revit

What LOD 400 means in an authoring tool

Modelling to LOD 400 in Revit means elements are detailed and accurate enough for fabrication and installation. That usually involves manufacturer-specific families, real connection details, fabrication parts for MEP and structural connections for steel, and properties needed for ordering and assembly.

It is rarely sensible to model everything to LOD 400. Agree which systems and zones need it, typically prefabricated risers, plant rooms, steel connections, facade units and modular components.

Practical modelling notes

1. **Use fabrication-ready content.** Use manufacturer families or fabrication parts with real dimensions, connectors and service clearances rather than generic placeholders.
2. **Model connections and supports.** Include hangers, brackets, unistrut, fixings and steel connections where they affect fabrication or installation.
3. **Set tolerances and offsets.** Model insulation, clearances for access and maintenance, and installation tolerances explicitly.
4. **Add fabrication properties.** Add spool or assembly numbers, product codes, material grades, weights and installation sequence where the fabricator needs them.
5. **Split by assembly.** Use assemblies, worksets or separate models for prefabricated modules so they can be issued and tracked individually.

6. **Check and export.** Run clash and clearance checks, generate shop drawings and schedules, and export to the fabricator's format and to IFC as agreed.

LOD 300 versus LOD 400 in Revit

Element	Typical LOD 300 content	Typical LOD 400 content
Ductwork	Ducts and main fittings at correct size and route	Fabrication parts, joints, stiffeners, hangers, insulation and spool numbers
Pipework	Pipes at size and route with main valves	Fabrication parts, couplings, supports, welds or joints, spool sheets
Structural steel	Members at correct section and position	Connections, plates, bolts, cleats, cambers and erection marks
Precast concrete	Element outline and size	Reinforcement, lifting inserts, connection details, embeds
Curtain wall	Panels and mullions at correct layout	Unit system with brackets, fixings and manufacturer details
Plant	Generic object at correct size	Manufacturer object with connections, access zones and weights

LOD 400 readiness checks

- ☐ Scope of LOD 400 agreed by system and zone in the BEP
- ☐ Manufacturer or fabricator content approved
- ☐ Connections, supports and fixings modelled
- ☐ Clearances and maintenance access zones modelled
- ☐ Fabrication properties populated and checked
- ☐ Clash and clearance checks completed against the federated model
- ☐ Shop drawing and export outputs tested with the fabricator

> **Who models LOD 400?**
> LOD 400 is usually produced by the specialist contractor or fabricator, not the design consultant. Make the responsibility explicit in the responsibility matrix so design teams are not asked to model fabrication detail they cannot warrant.

LOI levels

Level of information in UK legacy practice

Legacy content. Under PAS 1192-2:2013, levels of definition were tied to project stages. Each level combined a level of model detail for graphical content with a level of information for non-graphical content. The NBS BIM Toolkit, launched in 2015, then published numbered LOD and LOI templates per element and stage.

PAS 1192-2 has been withdrawn and replaced by BS EN ISO 19650-1 and -2, which use level of information need, and NBS retired the BIM Toolkit in 2022. LOI still appears in older appointments and templates, so it is useful to understand and translate it, but it should not be used to write new requirements.

UK legacy levels of definition, PAS 1192-2 (withdrawn, indicative)

Level	Stage meaning	Typical non-graphical information (LOI)
1	Brief	Project requirements, performance targets, constraints and benchmarks
2	Concept	Outline performance, areas, volumes and key system choices
3	Developed design	Specified performance, generic materials and system types, cost data
4	Technical design	Specific performance values, specification references, regulatory data
5	Construction and commissioning	Manufacturer and product data, installation and commissioning records
6	Handover and close-out	Verified asset data, O&M information, warranties and spares
7	Operation and in use	Maintenance history, performance in use and condition data

Translating LOI into LOIN
When inheriting a legacy LOI requirement, ask what purpose and milestone it was meant to serve. Then restate it as LOIN: the specific properties and documents needed for that purpose, for each object type, at that milestone.

LOI questions

Is LOI still valid in contracts?

If a contract references it, it still applies to that contract. For new appointments, ISO 19650 and ISO 7817-1 terminology is the better choice.

Can LOI numbers map directly to LOD numbers?

Not reliably. They come from different frameworks and were defined with different intentions, so map them by purpose rather than by number.

Compare

LOIN compared with LOD and LOI

Criterion	LOIN (ISO 7817-1)	LOD (BIMForum)	LOI (UK legacy)
Origin	International standard ISO 7817-1:2024; its EN ISO adoption replaces EN 17412-1:2020	US industry specification published by BIMForum, 2025 edition current	UK PAS 1192 era practice and the retired NBS BIM Toolkit
Basis	Purpose, milestone, actor and object	Numbered levels of element development	Numbered levels tied to project stages

Criterion	LOIN (ISO 7817-1)	LOD (BIMForum)	LOI (UK legacy)
Fixed levels	No; requirements are described per need	Yes: 100, 200, 300, 350, 400; LOD 500 (field verified) sits outside the progression	Yes, stage-based numbers
Geometry	Described by detail, dimensionality, location, appearance, parametric behaviour	Implied by each level	Covered separately by level of model detail
Alphanumeric data	Identification and information content specified explicitly	Attached information described per level	Main focus of LOI
Documentation	An explicit, separate information type	Not a primary focus	Partly covered
Granularity	Per object type, purpose and milestone	Per element	Per stage, sometimes per element
Machine readability	Designed to be expressed in structured, machine-readable form	Usually tables in specifications	Usually tables in templates
Relation to ISO 19650	The term used by ISO 19650	Commonly referenced alongside it	Superseded with PAS 1192
Main risk	Needs effort to write well	Levels applied to whole models or misread as detail only	Outdated terminology and ambiguous meaning
Best used for	Information requirements in EIRs and BEPs	Communicating element reliability and model progression	Understanding legacy contracts

Can they work together?

LOIN as the frame

Use LOIN to structure requirements by purpose, milestone, actor and object.

LOD as shorthand for geometry

Inside a LOIN requirement, referencing an LOD level can be a convenient shorthand for the geometric aspect, if its meaning is agreed.

LOI as a legacy bridge

Translate legacy LOI requirements into explicit properties and documents within the LOIN.

Pick one language per contract

Mixing terms without definitions creates disputes. State in the information standard which framework is used, and define any LOD shorthand in terms of LOIN aspects.

By stage

How information grows through the stages

The table uses RIBA Plan of Work style stages 0 to 7 as a familiar reference. Other frameworks use different names, but the principle is the same: information grows as decisions are made, and the focus shifts from design intent to construction and then to operation.

Typical geometry and information by stage and discipline

Stage	Architecture	Structure	MEP	Information focus
0 Strategic definition	Site context, existing information	Existing structure records	Existing utilities and capacity	Business case, outcomes, OIR
1 Preparation and briefing	Area schedules, massing studies	Structural strategy assumptions	Services strategy and loads	Brief, PIR, EIR for design team
2 Concept design	Massing and generic elements	Grid, outline frame and transfer zones	Plant space allowances, main routes	Option decisions, cost plan, sustainability targets
3 Spatial coordination	Specific walls, floors, openings, coordinated	Sized members and slabs, coordinated	Main distribution coordinated with support zones	Coordinated design, planning information, cost check
4 Technical design	Specified elements with performance data	Detailed members, connections intent, reinforcement design	Full systems with sizes and specified equipment	Specifications, regulatory approval, tender information
5 Manufacturing and construction	Fabrication and installation detail where needed	Fabrication models for steel and precast	Fabrication parts and installed product data	Construction records, product data, commissioning
6 Handover	As-built record, verified spaces	As-built record	As-built systems, asset data, test results	AIM data, O&M, health and safety file
7 Use	Updated as changes occur	Updated as changes occur	Updated with maintenance and replacement	Maintenance history, performance in use

Stages are milestones, not deliverables

In ISO 19650 terms, each stage end is an information delivery milestone. LOIN should be written against those milestones, so the delivery team knows exactly what each container must contain at that point.

Element & LOIN library

A starter library

The library below shows typical geometric levels and key properties for common elements. Geometric levels use LOD numbers as shorthand. Treat it as a starting point and adjust to the purposes on your project.

Common elements: geometry by stage and key properties

Element	Concept	Coordinated design	Technical design	Handover	Key properties
Wall	LOD 100 to 200	LOD 300	LOD 300 to 350	LOD 500 where verified	Type, fire rating, acoustic rating, thermal performance, finish
Slab	LOD 200	LOD 300	LOD 350	LOD 500 where verified	Thickness, concrete grade, fire rating, loading
Column	LOD 200	LOD 300	LOD 350 to 400	LOD 500 where verified	Section, material grade, fire protection, load
Beam	LOD 200	LOD 300	LOD 350 to 400	LOD 500 where verified	Section, material grade, connection type, camber
Door	LOD 100	LOD 300	LOD 350	LOD 500 where verified	Fire rating, hardware set, access control, manufacturer, warranty
Window	LOD 100	LOD 300	LOD 350	LOD 500 where verified	U-value, glazing type, opening type, manufacturer
Duct	LOD 100	LOD 300	LOD 350 to 400	LOD 500 where verified	System, size, material, insulation, pressure class
Pipe	LOD 100	LOD 300	LOD 350 to 400	LOD 500 where verified	System, size, material, insulation, pressure rating
Air handling unit	LOD 100 space allowance	LOD 200 to 300	LOD 350	LOD 500 with product data	Airflow, duty, manufacturer, model, serial number, maintenance regime
Pump	LOD 100	LOD 200 to 300	LOD 350	LOD 500 with product data	Flow, head, power, manufacturer, model, serial number
Light fitting	Not modelled	LOD 200	LOD 300	LOD 500 with product data	Type, lumen output, power, emergency function, manufacturer
Fire damper	Not modelled	LOD 200	LOD 350	LOD 500 with product data	Fire rating, actuator type, access, test certificate, manufacturer
Lift	LOD 100 shaft allowance	LOD 200	LOD 350	LOD 500 with product data	Capacity, speed, stops, manufacturer, maintenance contract

Element	Concept	Coordinated design	Technical design	Handover	Key properties
Road pavement	Corridor surface	Typical cross-section layers	Layered model by section	As-built layers	Layer materials, thicknesses, surface course, construction date
Bridge bearing	Not modelled	Type and position	LOD 350 with set-out	Installed product data	Type, capacities, movement range, manufacturer, installation date

> **Properties should come from a dictionary**
> Key property names, definitions and units are best taken from a data dictionary that follows ISO 23386, so that the same property means the same thing across tools and projects.

Adapting the library to your project

- ☐ Remove elements that do not appear on the project
- ☐ Add elements specific to the asset type, such as process plant or rail systems
- ☐ Confirm purposes for each property and delete any without a purpose
- ☐ Align stage names with the project's milestones
- ☐ Map properties to the owner's asset data requirements

Practical specification

How to write LOIN into an EIR, BEP and MIDP

1. **1. List the purposes.** Work with the client and future operator to list the uses of information at each milestone: cost, coordination, approvals, procurement, handover, maintenance.
2. **2. Define milestones and actors.** Name each information delivery milestone and who requests and who delivers the information.
3. **3. Choose the object breakdown.** Use an agreed classification system so objects are named consistently across disciplines and tools.
4. **4. Specify geometry.** For each object and milestone, state detail, dimensionality, location, appearance and parametric behaviour, or an agreed LOD shorthand.
5. **5. Specify alphanumeric information.** List identification requirements and properties with units, referencing a data dictionary where possible.
6. **6. Specify documentation.** List the documents needed per object, such as data sheets, certificates, manuals and calculations.
7. **7. Make it checkable.** Express alphanumeric requirements in a machine-readable form such as buildingSMART IDS so models can be validated automatically.
8. **8. Put it in the EIR.** Include the LOIN matrix in the EIR, with acceptance criteria and links to the information standard.

LOIN matrix template

Object	Purpose	Milestone	Requested by	Delivered by	Geometry	Alphanumeric	Documentation
Internal wall	Cost plan	Concept	Cost consultant	Architect	Generic 3D, approximate location	Type, area	None
Internal wall	Coordination	Spatial coordination	Lead designer	Architect	Specific 3D, accurate location	Type, fire rating, thickness	None
Fire damper	Coordination	Technical design	Lead designer	MEP engineer	Simple 3D box with access zone	System, size, fire rating	None
Fire damper	Handover	Handover	Facilities manager	MEP contractor	Simple 3D, verified location	Manufacturer, model, serial number, test date	Test certificate, data sheet
Pump	Procurement	Technical design	Contractor	MEP engineer	3D with connections	Duty, power, performance requirements	Specification
Pump	Maintenance	Handover	Facilities manager	MEP contractor	3D with access zone	Manufacturer, model, serial number, maintenance regime	O&M manual, warranty

Quality checks for a LOIN specification

- ☐ Every requirement traces to a purpose
- ☐ Every requirement has a milestone and actors
- ☐ Objects use the agreed classification
- ☐ Geometry, data and documentation are specified separately
- ☐ Properties have definitions and units
- ☐ Requirements are checkable, ideally automatically
- ☐ The delivery team has confirmed it can meet the requirements

Start small
A short LOIN matrix covering the most important objects and purposes, done well, is more useful than an enormous table nobody reads. Grow it over projects as you learn what your organisation actually uses.

Worked examples

Example: a fire door's LOIN at three milestones: Fire door D-03-014 followed from concept design to handover. Geometry barely changes after technical design, while the alphanumerical information and documentation grow sharply, because the purposes shift from fire strategy and specification to procurement, certification and maintenance.

How to read the examples

Each table follows one object through three information delivery milestones and describes it in ISO 7817-1 terms: the prerequisites (purpose, milestone, actors), the five geometrical aspects, the two alphanumerical aspects and documentation. Property names follow IFC property set conventions where one exists; others are shown as project-defined properties.

All values are illustrative and invented for teaching. On a real project the purposes come from the EIR, the property names and units from the project's data dictionary, and the values from the design and the installed products.

Fire door (internal, 30-minute fire resisting with smoke seals)

Aspect	Concept design	Technical design	Handover
Purpose	Fire strategy and compartmentation check	Specification, door schedule and tender	Asset register, statutory inspection and maintenance
Actors	Requested by fire engineer; delivered by architect	Requested by lead designer and cost consultant; delivered by architect	Requested by facilities manager; delivered by main contractor with door supplier
Detail	Opening or symbol only	Leaf, frame and swing at true size; no ironmongery detail	As technical design; ironmongery not modelled in detail
Dimensionality	2D symbol in plan	3D solid	3D solid
Location	Approximate, in the correct wall	Exact position relative to grid and level	As installed, verified on site
Appearance	Symbolic	Generic material colours	Generic; finish recorded as data
Parametric behaviour	Not required	Parametric leaf width and height	Fixed to the installed size
Identification	Door type code, e.g. DT-FD30	Type mark plus unique door number D-03-014; classification code	Unique asset tag D-03-014 linked to the asset register; classification code
Information content	Pset_DoorCommon.FireRating = 30 min; IsExternal = false	FireRating = FD30S; SmokeStop = true; AcousticRating = 30 dB Rw; clear opening width 850 mm; ironmongery set IM-07	Manufacturer = (supplier name); ModelReference = (product code); SerialNumber; InstallationDate = 2026-05-12; WarrantyEndDate; inspection interval 6 months
Documentation	None	Door schedule; fire strategy drawing reference	Fire test or assessment certificate; installation certificate; O&M entry; inspection regime

Air handling unit (rooftop supply and extract unit)

Aspect	Concept design	Technical design	Handover
Purpose	Plant space allowance and structural loading	Coordination, specification and procurement	Planned maintenance and energy monitoring
Actors	Requested by architect and structural engineer; delivered by MEP engineer	Requested by lead designer and main contractor; delivered by MEP engineer	Requested by facilities manager; delivered by MEP contractor
Detail	Simple box for the space allowance	Casing, connections and access zones; no internal components	Manufacturer object with connections and maintenance zones
Dimensionality	3D box	3D solid	3D solid
Location	Approximate, within the plant area	Exact position and connection points	As installed, verified
Appearance	Symbolic	Symbolic system colour	Symbolic system colour
Parametric behaviour	Adjustable box size	Parametric by duty and size range	Fixed to the installed model
Identification	System name, e.g. AHU-01	Type, system and classification code	Unique asset tag AHU-01 linked to the CAFM system
Information content	Design airflow 4.0 m3/s; operating weight allowance 2,500 kg	Pset_UnitaryEquipmentTypeAirHandler or project-defined properties: airflow supply and extract; external static pressure 400 Pa; heat recovery efficiency; filter class; electrical supply 400 V three-phase	Manufacturer; ModelReference; SerialNumber; InstallationDate; commissioned airflow; filter change interval; WarrantyEndDate
Documentation	None	Equipment specification; schematic	Commissioning certificate; O&M manual; warranty; spares list

Structural column (steel, internal)

Aspect	Concept design	Technical design	Construction (fabrication)
Purpose	Grid layout and early cost plan	Structural analysis, coordination and tender	Fabrication and erection

Aspect	Concept design	Technical design	Construction (fabrication)
Actors	Requested by architect and cost consultant; delivered by structural engineer	Requested by lead designer; delivered by structural engineer	Requested by main contractor; delivered by steelwork contractor
Detail	Line or generic section	Correct section and orientation; connections shown as intent only	Full member with base plate, cap plate, cleats, bolts and holes
Dimensionality	1D analytical line or simple 3D	3D solid plus analytical model	3D solid
Location	On grid, approximate levels	Exact grid position, base and top levels	Exact, including erection tolerances
Appearance	Symbolic	Symbolic	Symbolic; fire protection shown by colour code
Parametric behaviour	Adjustable height	Parametric section and length	Fixed fabrication geometry
Identification	Grid reference, e.g. C3	Member mark and classification code	Piece mark and erection mark, e.g. C3-L02
Information content	Material = steel; indicative section size	Section = UC 305×305×97; SteelGrade = S355; Pset_ColumnCommon.LoadBearing = true; FireRating = 60 min	Weight; fabrication length; surface treatment; fire protection system and thickness; bolt grade
Documentation	None	Design calculations summary; general arrangement drawing	Fabrication drawings; mill certificates; erection method statement

Bridge bearing (elastomeric laminated bearing)

Aspect	Preliminary design	Detailed design	Handover
Purpose	Articulation scheme and substructure sizing	Specification, setting-out and procurement	Inspection, maintenance and replacement planning
Actors	Requested by bridge designer; delivered by structural engineer	Requested by contractor; delivered by structural engineer	Requested by asset owner; delivered by contractor with bearing supplier
Detail	Not modelled, or a point marker	Bearing outline and plinth at true size	Installed bearing outline and plinth
Dimensionality	0D point	3D solid	3D solid
Location	Approximate, per support line	Exact set-out in project coordinates	As installed, surveyed position and level

Aspect	Preliminary design	Detailed design	Handover
Appearance	Symbolic	Symbolic	Symbolic
Parametric behaviour	Not required	Parametric plan size and thickness	Fixed to the installed product
Identification	Support reference, e.g. Pier 2 bearing line	Bearing reference B-P2-01 and classification code	Unique asset identifier in the structures database
Information content	Bearing type = elastomeric; fixity (fixed, guided or free)	Design vertical load; movement range longitudinal and transverse; rotation capacity; plan size; product standard EN 1337-3	Manufacturer; product reference; InstallationDate; installed temperature; measured preset; inspection interval
Documentation	Articulation diagram	Bearing schedule; specification	Declaration of performance; installation record; inspection and replacement guidance

Patterns worth noticing
Geometry usually stops growing once coordination is done, while data and documentation keep growing towards handover. The actor who delivers also changes, from designer to contractor or supplier. Write each change down explicitly in the responsibility matrix so nobody assumes someone else is providing it.

EIR/BEP transfer

How LOIN flows through the appointment

1. **EIR (ISO 19650-2, 5.2.1).** The appointing party states its information requirements for the appointment, the level of information need for each one, the acceptance criteria and the delivery milestones.
2. **Pre-appointment BEP (5.3.2).** The prospective lead appointed party confirms how it will meet the LOIN, and flags any requirements it proposes to change.
3. **Detailed responsibility matrix (5.4.2).** Once appointed, the lead appointed party refines the high-level matrix into a detailed one: what information is produced, when, and by which task team, so each object, milestone and LOIN requirement has an owner.
4. **Lead appointed party's EIR (5.4.3).** The lead appointed party passes the relevant requirements, including LOIN, on to each appointed party in its own exchange information requirements.
5. **TIDPs (5.4.4) and MIDP (5.4.5).** Each task team lists the information containers it will produce and their milestones in a TIDP; the lead appointed party combines them into the master information delivery plan, checking every LOIN requirement is covered.
6. **Review and acceptance (5.6 and 5.7).** Containers are checked against the LOIN before sharing, reviewed and authorised by the lead appointed party, then reviewed and accepted by the appointing party against the acceptance criteria.

Example BEP responsibility matrix extract

Object or system	Milestone	Task team	Geometry	Alphanumeric	Documentation
Architectural walls and doors	Spatial coordination	Architect	Specific 3D	Type, fire rating	None
Structural frame	Spatial coordination	Structural engineer	Specific 3D	Section, grade	Design basis report
Ventilation system	Technical design	MEP engineer	Full system 3D with support zones	System, size, duty	Specification
Ventilation system	Construction	MEP contractor	Fabrication 3D	Spool and product data	Shop drawings
Ventilation system	Handover	MEP contractor	Verified 3D	Asset data	O&M manual, test certificates
Lifts	Handover	Lift contractor	Simple 3D	Asset data	O&M manual, certificates

Who owns what

Appointing party

Defines the LOIN in the EIR and accepts deliverables against it.

Lead appointed party

Confirms it can meet the LOIN, allocates it in the responsibility matrix and checks MIDP coverage.

Task teams

Produce containers to the LOIN, check them before sharing and record compliance in TIDPs.

Gaps hide at the handovers

The most common LOIN failures happen where responsibility passes from designer to contractor, or from contractor to operator. Make those transitions explicit in the responsibility matrix.

AIM guide

LOIN for handover and operation

The asset information model supports the operation, maintenance and eventual disposal of the asset. Its LOIN is driven by the asset information requirements, which in turn come from the organisation's information requirements and asset management processes.

For handover, geometry is often less important than data and documentation. Operators need reliable identifiers, locations, maintenance information and certificates, loaded into the systems they actually

use.

Typical handover LOIN by asset category

Asset category	Geometry	Alphanumeric	Documentation
Spaces	Verified boundaries and areas	Name, number, use, area, finishes	Room data sheets
Maintainable equipment	Simple 3D with access zone	Identifier, manufacturer, model, serial number, install date, warranty	O&M manual, warranty, certificates
Life safety systems	Verified location	System, rating, test dates	Test and commissioning certificates
Building fabric	As-built record	Type, materials, performance	Product data, maintenance guidance
Distribution systems	As-built routes	System, size, material, connectivity	Commissioning records
Infrastructure assets	Georeferenced point, line or 3D	Identifier, network reference, type, condition	Inspection and as-built records

AIM readiness checklist

- ☐ AIR agreed with the operator and mapped to the LOIN
- ☐ Identifiers follow the owner's convention
- ☐ Data validated against IDS or equivalent rules
- ☐ Documents linked to the correct assets
- ☐ Data loaded successfully into the owner's systems in a trial
- ☐ Responsibilities for keeping the AIM updated after handover are clear

> **Test the load before handover day**
> Ask for a sample of handover data early in construction and try loading it into the operator's systems. It is the quickest way to find naming, property and format problems while there is still time to fix them.

Quiz

Ten questions to check your understanding

The questions cover the LOIN framework, the BIMForum LOD levels, the UK legacy LOI and how requirements move through an appointment. Each answer includes a short explanation, so the quiz doubles as a revision aid.

Revisit the weak spots
If you missed questions on LOD 350 or LOD 500, revisit the LOD 100 to 500 tab. If the prerequisites or information types tripped you up, the LOIN explained tab walks through each one.

AI tools

Check AI output
AI drafts are a starting point. Check that each requirement has a purpose, milestone and actors, and that properties match your data dictionary and the owner's asset information requirements.

9. COBie and asset information

COBie is the structured, spreadsheet-friendly way to hand over the information an owner needs to run a building: spaces, assets, types, systems, documents and maintenance data. This guide explains every sheet, the colour conventions used in templates, what to deliver at each stage, how to author COBie from Revit and Archicad, the quality checks that catch most errors, and how COBie fits the ISO 19650 asset information requirements.

Overview

What COBie is

COBie (Construction Operations Building information exchange) is an information exchange specification for the managed assets of a facility. It describes what is in a building, where it is, what type it is, who made it, how it is grouped into systems and what documents and maintenance information go with it.

COBie is a data structure, not a piece of software. The same information can be carried in an IFC file using a defined model view, or in a spreadsheet workbook with one worksheet per kind of object. The spreadsheet form is the one most people meet, because it can be opened, reviewed and imported by facility and asset management systems without specialist BIM tools.

In the UK, BS 1192-4 sets out a code of practice for using COBie to fulfil the employer's information exchange requirements. It explains who provides which information, how sheets relate to each other and how classification should be applied. It remains widely referenced on projects that follow ISO 19650.

How COBie sheets link together: Spaces sit on Floors within the Facility and are grouped into Zones; Components are instances of Types, are located in Spaces and are grouped into Systems. Contact, Document, Attribute and PickLists support every other sheet.

Why owners ask for COBie

Ready for operations on day one

Asset registers, spaces and warranties arrive in a structured form that can be imported into CAFM or CMMS systems instead of being retyped from O&M manuals.

Software neutral

A workbook can be produced from any authoring tool and read by any system, so the owner is not locked into a single vendor.

Checkable

Because every sheet has defined columns and references, completeness and consistency can be tested automatically at every data drop.

Progressive

COBie is delivered in stages, so the client can check the information early, when gaps are cheap to fix, rather than at handover.

Links to documents

O&M manuals, certificates, warranties and drawings can be linked to the exact type, component or space they describe.

Supports the AIM

COBie is a convenient carrier for part of the asset information model required by the appointing party under ISO 19650-3.

The two COBie hierarchies

Hierarchy	Levels	What it answers
Spatial	Facility, then Floor, then Space; Spaces grouped into Zones	Where is it? Which room, which floor, which fire or lettable zone?
Physical	Type, then Component; Components grouped into Systems	What is it? Which product, which installed instance, which system does it belong to?
Connecting the two	Each Component references the Space it is located in	Which assets are in this room, and which room is this asset in?
Supporting information	Document, Attribute, Spare, Resource, Job, Impact, Coordinate, Issue	How is it maintained, what documents describe it, what else do we know?
People and organisations	Contact, referenced by CreatedBy and by manufacturer and warranty fields	Who created the data, who made the product, who gives the guarantee?
Shared vocabularies	PickLists, including classification codes	Which values are allowed in category and unit fields?

How the data hangs together

1. **Facility.** One row describing the building or facility, its project and site names and the units used throughout the workbook.
2. **Floor.** Each storey or level, including roof and external levels where the project needs them, with elevation and height.
3. **Space.** Each room or area, linked to its Floor by name, with description, room tag and areas.
4. **Zone.** Groups of Spaces for a purpose such as fire compartments, heating zones, security zones or lettable areas. A Space can be in several Zones.
5. **Type.** Each product type, for example a particular boiler model or door type, with manufacturer, model number, warranty and expected life.
6. **Component.** Each installed, individually managed instance of a Type, located in a Space, with serial number, installation date and tag.
7. **System.** Groups of Components that work together, such as a heating system, a fire alarm system or a lighting circuit.

Only managed assets belong in COBie

COBie is not a dump of every modelled element. Walls, slabs and ceilings are rarely needed as components; boilers, pumps, luminaires, doors, sanitaryware and fire equipment usually are. The appointing party's asset information requirements should say which product categories are in scope.

Sheets explained

Core sheets: people, places and assets

Sheet	Purpose	Key fields	Who usually fills it	When
Contact	People and organisations referenced elsewhere in the workbook	Email (the unique key), Company, Phone, Category, CreatedBy, CreatedOn	Every author adds itself; contractor adds manufacturers and installers	From the first drop, growing through construction
Facility	The single building or facility being handed over, with project, site and unit settings	Name, Category, ProjectName, SiteName, LinearUnits, AreaUnits, VolumeUnits, CurrencyUnit, AreaMeasurement	Lead designer or information manager	First drop; checked at every drop
Floor	Storeys, levels and other vertical divisions	Name, Category, Elevation, Height	Architect	Concept design onwards
Space	Rooms and areas that are managed, cleaned, let or maintained	Name, FloorName, Description, RoomTag, UsableHeight, GrossArea, NetArea, Category	Architect	Concept and spatial coordination onwards
Zone	Named groups of spaces for a purpose such as fire, HVAC, security or occupancy	Name, Category, SpaceNames	Architect and building services engineer	Spatial coordination onwards
Type	Product types, specifications and warranty information	Name, Category, Description, AssetType, Manufacturer, ModelNumber, warranty fields, ExpectedLife	Designers for generic types; contractor completes product data	Technical design and construction
Component	Individually managed installed instances of a type	Name, TypeName, Space, Description, SerialNumber, InstallationDate, WarrantyStartDate, TagNumber, BarCode, AssetIdentifier	Designers create; contractor and installers complete	Technical design onwards; serial numbers at construction
System	Groups of components that provide a service	Name, Category, ComponentNames	Building services and other system designers	Technical design onwards
Assembly	Types or components made up of other types or components	Name, SheetName, ParentName, ChildNames, AssemblyType	Specialist designers or contractor where required	Only when the requirements ask for it
Connection	Logical or physical connections between components	Name, ConnectionType, SheetName, RowName1, RowName2	Building services designers where required	Only when the requirements ask for it

Supporting sheets: operations, documents and extensions

Sheet	Purpose	Key fields	Who usually fills it	When
Spare	Spare parts, consumables and replacement items for a type	Name, Category, TypeName, Suppliers, SetNumber, PartNumber	Contractor, from manufacturer information	Construction and handover
Resource	Materials, tools and training needed for maintenance	Name, Category, Description	Contractor or maintenance planner	Handover, if required
Job	Planned maintenance, inspection and safety tasks for a type	Name, Category, Status, TypeName, Description, Duration, Start, Frequency (each with its unit), TaskNumber, Priors, ResourceNames	Contractor with manufacturers and FM team	Handover
Impact	Environmental, cost or other impacts over the life cycle	Name, ImpactType, ImpactStage, SheetName, RowName, Value, ImpactUnit	Specialists where the client asks for it	Rarely used; only when specified
Document	References to documents that describe any row in another sheet	Name, Category, ApprovalBy, Stage, SheetName, RowName, Directory, File	Every author for its own information; contractor for O&M	Throughout; heavily at handover
Attribute	Extra properties for any row, beyond the standard columns	Name, SheetName, RowName, Value, Unit, Category	Designers and contractor, as listed in the requirements	Throughout, matched to the stage
Coordinate	Locations and bounding points for floors, spaces and components	Name, Category, SheetName, RowName, CoordinateXAxis, CoordinateYAxis, CoordinateZAxis	Usually generated by the export tool	When location data is required
Issue	Known risks, problems or deviations attached to rows	Name, Type, Risk, Chance, Impact, SheetName1, RowName1, Description, Mitigation	Any author	As needed; often used for exceptions
PickLists	Allowed values for category, unit and type fields, including classification	Columns per list, for example categories, units and asset types	Information manager sets it up from the template	Before the first drop

Fields you will see on nearly every sheet

Name

The unique key on most sheets. Other sheets refer to rows by this name, so it must be unique and stable between drops.

CreatedBy

The email of the person or organisation that created the row. It must exist on the Contact sheet.

CreatedOn

The date and time the row was created, in a consistent, machine-readable date format.

Category

Classification, usually written as code and title from the agreed system such as Uniclass, for example a code followed by a colon and its description.

External fields

ExtSystem, ExtObject and ExtIdentifier record the authoring tool, object type and GUID the row came from, so it can be traced back to the model. The Facility sheet uses longer names, such as ExternalSystem and ExternalFacilityIdentifier, for project, site and building.

Description

A short, human-readable description that makes sense to the facility manager, not an internal family or layer name.

What the portal's COBie checker looks for

The checker treats Contact, Facility, Floor, Space, Type and Component as required worksheets, and checks Zone, System, Document and Attribute when they are present. It tests mandatory columns, unique names, a single Facility row, valid dates, classification in code: title form, and cross-references such as Space to Floor, Component to Type and Space, System to Component and CreatedBy to Contact.

Sheet-by-sheet reference

Every sheet, every column

This reference lists each COBie 2.4 worksheet in workbook order, from Contact to PickLists. For every sheet it explains what one row represents, who usually fills it in and when, and then describes each standard column with an example value and, for reference columns, the sheet or picklist the value must match.

You can also download a blank COBie workbook with the same sheets and column headers to use as a starting point. Always check it against the template and requirements your client has issued, which take precedence.

Reading the column colours

Column kinds follow the common template convention: yellow for required values, salmon or orange for references to another sheet or a picklist, purple for external identifiers written by the export tool, and green for optional or as-specified columns that become mandatory when the requirements ask for them. Clients may add their own columns or attributes and may make optional columns mandatory, so the project's template and responsibility matrix always win.

Colour coding

Why COBie templates are colour coded

Most COBie spreadsheet templates colour the column headers so that people filling them in can see at a glance which fields must be completed, which must match a value somewhere else and which are produced by software.

The colours are a convention used in widely shared templates, not a requirement of the data format. Different templates and tools use slightly different shades and meanings, so always check the legend in the template your project has adopted.

Common header colour convention

Colour (typical)	Meaning	Examples	What to do
Yellow	Required: a value must be provided	Name, CreatedOn, Description, ModelNumber	Always fill in; use n/a only where the requirements allow it
Orange or salmon	Reference: the value must match another sheet or a picklist	CreatedBy, FloorName, TypeName, Space, Manufacturer, Category, units	Pick from the referenced sheet or picklist; do not type free text
Purple or lilac	External: generated by software to trace the source	ExtSystem, ExtObject, ExtIdentifier	Leave to the export tool; do not edit by hand
Green	Optional or as specified: only needed if the requirements ask	Optional attributes, some warranty or area fields	Fill in when listed in the AIR or EIR; otherwise leave blank or n/a
Blue or another template-specific colour	Project or client extension columns added to the standard sheets	Client asset codes, maintenance contract references	Follow the client legend; these are defined in the project requirements
Grey or uncoloured	Not used for this project or secondary information	Rarely used sheets and columns	Ignore unless the project states otherwise

Reading the colours in practice

Yellow means stop and check

A blank yellow cell is the most common reason a drop is rejected. Run a filter for blanks on every required column before issuing.

Orange means look it up

If a reference field does not match exactly, including spaces and case in some tools, the relationship breaks and the asset becomes an orphan.

Purple means hands off

External identifiers link the row to a model object. Editing them by hand breaks traceability between the model and the workbook.

Green means read the brief

Optional fields become mandatory when the client asks for them. The AIR or the project's COBie responsibility matrix decides.

Colours do not survive every tool
Exporters, CSV conversions and some viewers drop cell formatting. Do not rely on colour to carry meaning in the delivered data; rely on the column definitions and the checker instead.

Delivery timing

Progressive delivery, not a handover surprise

COBie is best delivered in stages, often called data drops, aligned to project milestones. Each drop adds detail to the same workbook structure, and each is checked before the next stage begins.

The stage names below follow the RIBA Plan of Work, which many UK projects use. Your project's information delivery milestones in the MIDP and the client's information requirements define the actual content and dates.

What is typically delivered at each stage

Stage	Typical COBie content	Main contributors	Client check focus
Stage 2 Concept Design	Contact, Facility, Floor and an initial Space list with names, descriptions and target areas	Architect, information manager	Structure set up correctly, naming and classification agreed
Stage 3 Spatial Coordination	Full Space schedule, Zones, generic Types for main asset categories, first Systems	Architect, building services and structural engineers	Space and zone completeness, classification of types
Stage 4 Technical Design	Specified Types with performance attributes, Components placed in Spaces, Systems with components, design Documents	All designers, specialist designers	Every in-scope asset modelled, located and grouped
Stage 5 Manufacturing and Construction	Manufacturer, model number, serial numbers, installation dates, tags and barcodes; Spare and Job data begins	Contractor, subcontractors, manufacturers	Product data replaces generic data; installed reality matches
Stage 6 Handover	Complete and verified workbook with warranties, O&M documents, Jobs, Spares and final as-built attributes	Contractor, commissioning manager, information manager	Acceptance against the AIR and import into the client's systems
Stage 7 Use	Updates for replacements, refurbishment and changes to the asset register	Client FM team and maintenance contractors	The AIM stays current as assets change

Running a data drop

1. **Confirm the scope.** Check the responsibility matrix for which sheets, columns and attributes are due at this milestone.
2. **Export from the models.** Each task team exports its COBie data using the agreed tool and settings, with consistent names and classification.

3. **Merge.** The lead appointed party merges task team outputs into one workbook, removing duplicates and resolving references between teams.
4. **Self-check.** Run the COBie checker and the project's own rules, fix errors and record known exceptions on the Issue sheet.
5. **Share and review.** Upload to the CDE in the shared state, then the appointing party reviews it against the information requirements for that stage.
6. **Accept or reject.** Accepted drops are published; rejected drops come back with a clear list of issues and a date for resubmission.

> **Early drops are the cheapest**
> The structure, names and classification set up at stage 2 are reused in every later drop. Getting them right early avoids renaming thousands of rows during construction, when time is shortest.

Authoring from Revit/Archicad

Authoring COBie from models

Most COBie data starts in the authoring models. Spaces come from rooms or zones, types from family types or object types, and components from placed instances. Parameters on those objects map to COBie columns and to the Attribute sheet.

The key decision is where each field is maintained. Geometry-related fields such as floor, space and area are best taken directly from the model. Product data such as serial numbers and warranty dates is often added later in a spreadsheet, database or CAFM-preparation tool and merged in.

Revit and Archicad in general terms

Revit: rooms and spaces

Rooms (and MEP spaces) become COBie Spaces. Level names become Floors. Room number and name should follow the agreed naming so that Space names are unique and meaningful.

Revit: types and components

Family types become Types and placed instances become Components. Shared parameters hold COBie fields such as description, asset type, manufacturer, model number and warranty values.

Revit: export tools

Autodesk provides a COBie extension and the IFC exporter can include COBie-related property sets. Third-party tools also offer mapping and checking. Availability varies by version.

Archicad: zones and objects

Zones become Spaces, stories become Floors, and objects and library parts become Types and Components. Classification and properties are managed in the classification and property manager.

Archicad: export

COBie data can be produced through IFC with appropriate translators and property mapping, or through schedule exports and translator templates provided for COBie.

Other tools

Plant, structural and MEP tools often export COBie through IFC. Specialist COBie tools can merge, validate and enrich data from several sources.

Typical model-to-COBie mapping

COBie field	Typical model source	Tips
Floor.Name	Level or story name	Use the agreed level naming; exclude reference and working levels from export
Space.Name	Room or zone number	Keep unique across the whole facility, not just per floor
Space.Description	Room or zone name	Use names a facility manager recognises
Space.Category	Room classification parameter	Code and title from the agreed table, for example Uniclass spaces
Type.Name	Family and type name, or object type	Avoid generic names such as Type 1; make them readable
Type.Category	Classification parameter on the type	Use the products or systems table agreed in the information standard
Type.Manufacturer	Shared parameter on the type	Enter the manufacturer's Contact email, not just the company name
Component.Name	Mark, tag or generated asset name	Unique and stable; often type name plus a sequence number
Component.Space	Room or space containing the instance	Check items in ceiling voids, risers and plant areas resolve to a space
Component.SerialNumber	Added during construction	Usually managed outside the design model and merged in
System.ComponentNames	MEP system or a system parameter	List component names exactly as they appear on the Component sheet

Set up before you model

- ☐ Agree which object categories are in COBie scope and flag them in the model
- ☐ Load the shared parameter file or property set that holds COBie fields
- ☐ Agree naming rules for floors, spaces, types and components
- ☐ Load the agreed classification tables and assign codes to rooms and types
- ☐ Configure the export tool settings once and share them with every task team
- ☐ Exclude non-managed elements such as walls, floors and generic annotation from the Component export
- ☐ Test a small export at mobilization and run the checker on it

Features vary by software version
Export tools, extensions and settings change between releases of Revit, Archicad and third-party products. Treat the descriptions here as general guidance and confirm the exact workflow for the versions your project uses.

Quality checks

COBie quality checklist

- ☐ All required sheets are present: Contact, Facility, Floor, Space, Type and Component
- ☐ Facility sheet has exactly one row
- ☐ Units on the Facility sheet are filled in and match the model
- ☐ Every Name is unique within its sheet
- ☐ Every Contact email is unique and well formed
- ☐ Every CreatedBy value exists on the Contact sheet
- ☐ CreatedOn values are valid dates in a consistent format
- ☐ No required (yellow) cell is blank; n/a used only where permitted
- ☐ Every Space.FloorName exists on the Floor sheet
- ☐ Every Zone.SpaceNames entry exists on the Space sheet
- ☐ Every Component.TypeName exists on the Type sheet
- ☐ Every Component.Space entry exists on the Space sheet
- ☐ Every System.ComponentNames entry exists on the Component sheet
- ☐ Category values use the agreed classification in code: title form
- ☐ Type.Manufacturer references a Contact email
- ☐ Every Type has at least one Component, or the exception is explained
- ☐ Document and Attribute rows point to a real SheetName and RowName
- ☐ Only managed assets are included; no walls, slabs or annotation as components
- ☐ Serial numbers, installation and warranty dates are complete at handover
- ☐ Spot checks on site confirm that tags and locations match the workbook

Common errors and how to fix them

Error	Likely cause	Fix
Duplicate space names	Room numbers repeated on different floors	Include a floor prefix or use facility-wide numbering
Orphan components	Component.Space blank or not matching a space name	Make sure items are within a room boundary or assign the space parameter manually
Component references a missing type	Type renamed after components were exported	Re-export both sheets together, or update TypeName references
CreatedBy not found in Contact	Exporter used a login name instead of an email	Set the author email in the export settings and add it to Contact

Error	Likely cause	Fix
Category in the wrong format	Only a code or only a title was entered	Use code and title together from the agreed table
Thousands of irrelevant components	Every modelled element exported	Filter export by category and the in-scope flag
Manufacturer as company name	Template expects a Contact reference	Add the manufacturer to Contact and reference its email
Dates rejected	Mixed regional date formats or text values	Use one consistent ISO-style date format throughout
Systems with no components	System names created but components not assigned	Assign components in the model or the system parameter, then re-export
Generic data at handover	Design-stage types never updated to installed products	Update Type and Component data from approved submittals and site records
Broken document links	Files renamed or moved after the Document sheet was made	Freeze file names in the CDE before final export and check every link

Three layers of checking

Format checks

Sheets, columns, required values, dates and unique names. Fully automatic and the first gate for every drop.

Content checks

Classification, naming rules, references between sheets and project-specific attribute rules. Mostly automatic with a rule set.

Reality checks

Sampling on site and against approved submittals to prove the data describes what was actually installed. Needs people.

> **Valid is not the same as useful**
> A workbook can pass every format check and still be useless to an FM team if descriptions are cryptic, the wrong assets are included or serial numbers are placeholders. Involve the people who will use the data in the review.

ISO 19650 link

Where COBie sits in ISO 19650

ISO 19650 does not mandate COBie. It requires the appointing party to define its information requirements and expects information to be delivered in a structured, open and checkable way. COBie is one widely used means of meeting the asset-related part of those requirements, especially in the UK.

The chain runs from the organisation's needs to the project deliverables: organisational information requirements inform the asset information requirements, which inform the exchange information requirements for each appointment, which are delivered through the master information delivery plan and end up in the asset information model.

COBie and the ISO 19650 information requirements

ISO 19650 concept	What it means	How COBie relates
OIR (organisational information requirements)	Information the organisation needs for its strategic objectives	Explains why asset data is needed, for example maintenance planning or compliance
AIR (asset information requirements)	Information needed to manage the assets, defined under ISO 19650-3	Specifies which asset categories, attributes and documents COBie must carry
EIR (exchange information requirements)	Requirements for a specific appointment, including acceptance criteria	States COBie format, version, scope, data drop content and checking rules
BEP (BIM execution plan)	How the delivery team will meet the requirements	Confirms the COBie tools, responsibilities, export settings and checks
MIDP and TIDPs	Delivery plans for information containers by milestone	List each COBie drop as a deliverable with an owner and a date
PIM (project information model)	Information produced during delivery	COBie drops build up within the PIM during design and construction
AIM (asset information model)	Information supporting operation and maintenance	The accepted handover COBie is transferred into the AIM and kept current
CDE (common data environment)	Managed environment for information containers	COBie workbooks move through WIP, shared, published and archived states like any container

Parts of ISO 19650 that matter most

ISO 19650-1

Concepts and principles: the information requirements hierarchy, the PIM and AIM, and the role of open, structured information.

ISO 19650-2

The delivery phase: where EIRs, BEPs, MIDPs and the acceptance of COBie drops are managed for each appointment.

ISO 19650-3

The operational phase: defines AIR and how the AIM is maintained, which is where handover COBie is used and kept up to date.

ISO 19650-4

Information exchange: sets out criteria for reviewing and exchanging information, which apply to each COBie submission and acceptance decision.

ISO 19650-5

Security-minded information management: asset registers can reveal sensitive information, so apply the security assessment before sharing.

Write COBie into the requirements, not just the contract

A clause that simply says provide COBie leads to disputes. The EIR should name the version, the in-scope asset categories, the attributes by stage, the classification system, the checker and the acceptance criteria.

Delivery playbook

Who does what

Appointing party (client)

Defines AIR and EIR, provides the template and picklists, names the receiving system and accepts each drop.

Client FM or asset manager

Confirms which assets and attributes are really needed and tests imports into CAFM or CMMS systems.

Lead appointed party information manager

Sets up export settings, merges task team data, runs checks and submits drops.

Design task teams

Create spaces, zones, types, components and systems in the models with correct names and classification.

Main contractor and subcontractors

Replace generic data with installed product data, serial numbers, dates, spares and maintenance jobs.

Commissioning manager

Confirms systems, components and tags match what has been commissioned and tested.

COBie RACI by activity

Activity	Appointing party	Lead appointed party	Design teams	Contractor	FM team
Define AIR and COBie scope	A	C	I	I	R
Provide template, picklists and classification	A	R	I	I	C
Set up export and naming rules	I	A	R	C	I
Spatial data (Floor, Space, Zone)	I	A	R	I	C
Design types and components	I	A	R	C	I

Activity	Appointing party	Lead appointed party	Design teams	Contractor	FM team
Product, serial and warranty data	I	A	C	R	I
Spares and maintenance jobs	C	A	I	R	C
Merge and check each drop	I	A	C	R	I
Review and accept drops	A	I	I	I	R
Import into CAFM and maintain the AIM	A	I	I	C	R

Acceptance process for a COBie drop

1. **Submission.** The lead appointed party shares the workbook in the CDE with the right status, revision and a short submission note listing known exceptions.
2. **Automated check.** The client runs the agreed checker and rule set and records the result against the acceptance criteria.
3. **Content review.** FM and asset representatives sample descriptions, classification and attributes for usefulness.
4. **Test import.** Where possible, the workbook is imported into a test instance of the receiving system to prove it loads cleanly.
5. **Decision.** The drop is accepted and published, or rejected with an issue list, owners and a resubmission date.
6. **Lessons learned.** Recurring errors are fed back into export settings and modelling guidance so the next drop is better.

> **Agree a COBie responsibility matrix**
> Columns change hands during a project. A matrix that lists each sheet and column, the stage it is due, and who provides it removes most arguments about missing data.

FAQ

Frequently asked questions

Is COBie mandatory under ISO 19650?

No. ISO 19650 requires clear information requirements and structured delivery, but does not name COBie. Many clients, especially in the UK, choose COBie to meet their asset information requirements.

Which COBie version should we use?

COBie 2.4 is the version most widely used with BS 1192-4 and by common tools. Whatever you choose, state it in the EIR and make sure the export tools and the checker use the same version.

Do we have to fill every sheet?

No. Contact, Facility, Floor, Space, Type and Component form the core. Other sheets are completed only when the client's requirements ask for them.

Should walls and floors be components?

Usually not. COBie is for managed assets. Include building fabric only if the client manages it as an asset, for example for specific finishes or roofs with warranties.

What does n/a mean in a COBie cell?

It is commonly used to show that a required field does not apply to that row. Agree with the client when it may be used, because overuse hides genuinely missing data.

Who provides serial numbers?

The contractor or installer, from the installed equipment. They are not known at design stage and are added during construction or commissioning.

Can COBie be delivered as IFC instead of a spreadsheet?

Yes. COBie can be carried in IFC using an appropriate model view. Many clients still ask for the spreadsheet because it is easy to review and import.

How do we handle items outside any room, such as external lighting?

Create external spaces or areas on an external floor so every component has a valid location. Agree the approach in the naming rules.

What classification should be used?

The one stated in the project information standard. In the UK this is usually Uniclass. Use code and title together in the Category fields.

How big should a COBie workbook be?

As big as the in-scope assets require and no bigger. A sudden jump in components usually means non-managed elements are being exported.

Who owns the COBie data after handover?

Ownership and licence terms are set in the appointment and information protocol. In practice the appointing party receives it into its AIM and maintains it.

Can we fix COBie errors directly in the spreadsheet?

For product data maintained outside the model, yes. For model-derived data, fix the model and re-export, otherwise the next drop will reintroduce the error.

> **Still unsure?**
> Use the AI tools tab to ask a specific question about your project, or run your workbook through the COBie checker to see exactly which rows need attention.

Quick answers by role

Client

Define scope and attributes in the AIR, provide the template and picklists, and check every drop, not just the last one.

Designer

Get names, classification, spaces and types right in the model; most later errors start here.

Contractor

Plan how product, serial and warranty data will be collected from subcontractors well before handover.

FM team

Say early what your CAFM or CMMS system needs and test imports with each drop.

Quiz

Revise these before you start

- ☐ The spatial hierarchy: Facility, Floor, Space, Zone
- ☐ The physical hierarchy: Type, Component, System
- ☐ Which sheets reference Contact, Floor, Space and Type
- ☐ The common header colour convention
- ☐ What is delivered at each stage
- ☐ How AIR, EIR and AIM relate to COBie

Learn from the answers

Each answer has a short explanation. If you get one wrong, revisit the matching tab before trying again.

AI tools

Check your workbook with the COBie checker

The COBie checker is under AI tools, then COBie checker. Upload your workbook to see missing sheets and columns, blank required fields, duplicate names, broken references and classification problems, with a score for each drop.

Check AI output

AI drafts are a starting point. The information manager and the client's FM team should confirm asset scope, attributes, stages and acceptance criteria before a schedule is issued.

10. Issues and coordination (BCF)

BCF is the open format for passing model-based issues between tools. Instead of sending screenshots and spreadsheets, a BCF issue carries the camera view, a snapshot, the affected objects and the full conversation. This hub explains what BCF contains, how the versions and the BCF API differ, how issues move from creation to closure, how BCF supports ISO 19650 coordination, how it works in common tools, and how to run a clash coordination routine that actually closes issues.

BCF basics — What BCF is

Issues that point at the model

BCF, the BIM Collaboration Format, is an open standard from buildingSMART for exchanging issues about building models. It was created so that people using different authoring and checking tools could talk about the same problem in the same place in the model.

A BCF issue, called a topic, does not contain the model. It refers to it. It stores where the camera was, what the viewer saw, which objects are involved and what people have said about the problem. The receiver opens the topic in their own tool and is taken straight to the problem.

Objects are identified by their IFC GUID. As long as GUIDs are kept stable between model versions, an issue raised on one export can still find the right objects in the next.

What a BCF topic carries

Part	What it holds	Why it matters
Topic	Title, type, status, priority, labels, assignee, due date, stage and description	Lets the issue be filtered, assigned and tracked
Comments	A dated conversation with the author of each comment	Keeps the decision history with the issue instead of in email
Viewpoint	Camera position, direction and field of view, plus clipping planes	Takes the receiver to exactly the same view
Snapshot	An image of the view when the issue was raised	Shows the problem even if the model is not open
Components	The IFC GUIDs of selected, visible, hidden or coloured objects	Highlights the affected elements in the receiver's tool
Related files and documents	References to the models and documents the topic relates to	Shows which model version the issue was raised on

Why teams use BCF

Tool neutral
A Solibri check result can be opened in Revit, Archicad, Tekla or a web viewer without retyping.

Precise
The viewpoint and GUIDs take everyone to the same objects, so there is no doubt about which pipe or beam is meant.

Traceable
Status, assignee and comments show who must act and what was decided.

Measurable
Issue counts by status, age and discipline give a clear picture of coordination progress.

BCF depends on stable GUIDs
If an authoring tool or export setting changes IFC GUIDs between versions, old issues can no longer find their objects. Check GUID stability during mobilisation by exporting the same model twice.

BCF basics — Versions & BCF API

BCF versions compared

Version	Main points	In practice
BCF 1.0	The original file-based format, introduced by tool vendors and then adopted by buildingSMART	Rarely used now; most tools have moved on
BCF 2.0	Added richer topic fields and the idea of a server-based exchange alongside the file	Still read by many tools; some older projects use it
BCF 2.1	Refined 2.0 with clearer viewpoints, visibility and colouring of components, and better consistency	The most widely supported version for file exchange; the platform's BCF studio writes 2.1
BCF 3.0	Cleaner schema, an extensions file that lists allowed types, statuses, priorities, labels and users, and better handling of documents	Growing support; check that every tool on the project can read and write it before choosing it

File-based or server-based

BCF-XML (file)
Topics are zipped into a .bcf or .bcfzip file and sent or uploaded. Simple, works offline, but every exchange is a copy and status can drift between copies.

BCF API (server)
A RESTful web service where tools read and write topics on a shared server. Everyone sees the same live issue, with no file round trips.

Which to use

Use the API where your tools and platform support it well. Fall back to files for tools that do not, and agree one master issue list either way.

Agree these in the BEP

- The BCF version to use for file exchange
- Whether a BCF API server is used, and which platform hosts it
- Where the master issue list lives
- The allowed values for type, status, priority and labels
- How issue files are named and stored in the CDE when files are used

The BCF studio reads all three

This platform's BCF studio opens BCF 2.0, 2.1 and 3.0 files, writes BCF 2.1, and converts issues to and from spreadsheets. It is useful when two tools disagree on versions.

BCF basics — BCF and ISO 19650

Where BCF fits

ISO 19650 does not mention BCF by name. It expects task teams to coordinate their information during collaborative production, to review it before sharing and to resolve problems through the CDE. BCF is a practical, open way of doing that for model-based problems.

Clash coordination, design reviews, model checking against requirements and site snagging can all produce BCF topics. The issue log becomes part of the project information and should be managed like any other information.

BCF in the ISO 19650 activities

Activity	How BCF helps
Collaborative production of information	Teams raise and answer issues on shared models instead of by email
Quality assurance check before sharing	Model checking tools export rule failures as topics for the author to fix
Review of shared information	Reviewers place comments on the exact view and objects
Information model review and authorisation	The lead appointed party checks that critical issues are closed before submission
Review and acceptance by the appointing party	Client comments can be returned as topics linked to the submitted models
Mobilisation	BCF exchange between tools and the CDE is tested before production starts

An issue is not a change of status
Closing a BCF topic does not move a model to Shared or Published. The fix is made in a new WIP revision, which is then checked and shared through the CDE in the normal way.

Issue lifecycle — From open to closed

The issue lifecycle

1. **Create.** Raise the topic on the current shared model, with a clear title, a snapshot, the viewpoint and the affected objects selected.
2. **Assign.** Give it to one responsible person or task team, with a priority and a due date.
3. **Comment.** Discuss the solution in the topic. Add new viewpoints where they help explain a proposal.
4. **Resolve.** The assignee fixes the model in WIP and marks the topic as resolved, saying which revision contains the fix.
5. **Verify.** The person who raised the issue, or the coordinator, checks the fix in the next shared model.
6. **Close.** If the fix is confirmed, the topic is closed. If not, it is reopened with a comment explaining why.

Who acts at each step

Raise the issue BIM coordinator ISO 19650-2, collaborative production of information
Status: Open

Inputs

- Current shared models
- Clash test or model check results

Outputs

- Open BCF topic

Activities

- Group related clashes into one problem
- Create a topic with a snapshot and viewpoint
- Select the affected objects

Tip: One topic per problem, not one per clash.

Assign and prioritise BIM coordinator ISO 19650-2, collaborative production of information
Status: Open or In progress

Inputs
- Open topic

Outputs
- Assigned topic

Activities
- Assign one responsible party
- Set priority and due date
- Add labels for discipline and zone

Fix in WIP Responsible task team ISO 19650-2, collaborative production of information
Status: Resolved

Inputs
- Assigned topic
- Own WIP model

Outputs
- Resolved topic
- Revised model shared through the CDE

Activities
- Agree the solution in comments
- Change the model in WIP
- Mark resolved and name the revision with the fix

Tip: Resolved means fixed in a model others can see, not just fixed on your screen.

Verify BIM coordinator ISO 19650-2, review of shared information
Status: Closed or reopened

Inputs
- Resolved topic
- New shared model

Outputs
- Closed topic

Activities
- Re-run the clash test or check
- Confirm the fix in the viewpoint
- Close or reopen with a comment

Report and authorise Lead appointed party ISO 19650-2, information model review and authorisation
Before milestone submission

Inputs

- Issue log
- Federated model

Outputs

- Issue status report with the submission

Activities

- Review open critical issues
- Agree what can remain open with a reason
- Authorise the submission

A simple status set

Status	Meaning	Who sets it
Open	Raised and waiting for action	Issue author or coordinator
In progress	The assignee is working on it	Assignee
Resolved	Fixed in a shared model and ready to check	Assignee
Closed	Fix verified, no further action	Issue author or coordinator
Reopened	The fix did not work or caused a new problem	Issue author or coordinator
Not an issue	Accepted as it is, with a reason recorded	Coordinator, with agreement of the teams

Issue lifecycle — Clash coordination

From clash results to BCF issues

Clash detection tools often find hundreds or thousands of clashes. Most are repeats of the same problem or are not real problems at all. Sending every clash as an issue buries the team in noise.

The coordinator's job is to turn raw clashes into a short list of real problems, each with a clear owner. BCF is the format that carries those problems back into the authoring tools.

Turning clashes into useful issues

1. **Set up clash tests.** Define tests by discipline pair and zone, with tolerances and rules that ignore known acceptable overlaps.
2. **Run on current shared models.** Use only the current shared revisions so results reflect what everyone can see.
3. **Group.** Group clashes by cause, for example one duct run hitting a row of beams, and treat the group as one problem.
4. **Filter.** Remove false positives such as insulation touching, and approved penetrations.

5. **Create topics.** Create one BCF topic per problem, with the clash test name and group in the description or labels.
6. **Re-run and close.** After each model update, re-run the tests and close the topics whose clashes have gone.

> **Link topics to the clash test**
> Put the clash test name and the group reference in the topic. When the test is re-run, the coordinator can quickly see which topics are cleared.

Issue lifecycle — Coordination meeting

A weekly clash coordination meeting

1. **Before the meeting.** Teams share updated models in the CDE by an agreed cut-off. The coordinator federates them, runs the clash tests and updates the issue list.
2. **Review the numbers.** Start with open, resolved and closed counts by discipline and zone, and the oldest open issues.
3. **Verify resolved issues.** Go through resolved topics and close or reopen each one on screen.
4. **Work the priorities.** Discuss high-priority open issues in the model, agree solutions and owners, and record decisions as comments.
5. **Raise new issues.** Add new topics from this week's tests, assign them and set due dates.
6. **After the meeting.** Export or sync the updated topics, share the issue report and confirm the next model cut-off date.

Meeting readiness checklist

- ☐ All teams have shared models by the cut-off
- ☐ Models are the current shared revisions and federate correctly
- ☐ Clash tests have been run and grouped
- ☐ Resolved topics from last week are ready to verify
- ☐ The issue report is prepared with counts and ageing
- ☐ Each discipline has someone present who can make decisions

> **Decide in the room**
> A coordination meeting that only reviews issues and decides nothing makes the list longer. Agree an owner and a date for every high-priority issue before moving on.

BCF by software — Tool comparison

General guidance only
Tools, add-ins and versions change often. The notes below describe common workflows in general terms. Check the current version and the vendor's documentation, and test the round trip on your own project.

BCF workflow in common tools

Tool	Typical role	How BCF is used	Notes
Autodesk Revit	Authoring	Through add-ins such as BCF managers from several vendors, or through issue connectors from coordination platforms	Check that the add-in matches your Revit version and maps GUIDs to Revit elements correctly
BIMcollab Zoom and BIMcollab	Model checking, issue management	Creates and reads BCF topics; the BIMcollab service can act as a shared issue server with connectors for authoring tools	Often used as the central issue list across tools
Autodesk Navisworks	Clash detection, federation	Clash results are usually turned into BCF through add-ins or by moving them to a coordination platform	Group clashes before exporting, or the list becomes unmanageable
Solibri	Model checking, clash detection	Exports and imports BCF topics from checking results and presentations	Strong for rule-based checking; one of the original BCF tools
Revizto	Coordination, issue tracking	Supports BCF import and export alongside its own issue tracker	Check how statuses map between Revizto and BCF
Autodesk Construction Cloud Issues	Platform issue tracking	Issues can be exchanged with BCF through import, export or connected tools; check current support	Decide whether ACC or another tool holds the master issue list
Trimble Connect	CDE, viewing, issue tracking	Works natively with BCF topics and connects to tools that use BCF	Useful when teams use Tekla, SketchUp or other Trimble tools alongside others
Archicad and Tekla	Authoring	Both offer BCF import and export, built in or through add-ons	Check GUID handling when exchanging IFC with other tools

Choosing a set-up

One master list
Pick one place where the official status of each issue lives. Every other tool reads from it or syncs to it.

Prefer live connections
Where a BCF API or a direct connector is available, use it. File round trips create copies with conflicting status.

Map the fields

Check how type, status, priority and labels map between tools before production starts.

Test the round trip

Raise a topic in the checking tool, open it in each authoring tool, comment, resolve and close it, and check nothing is lost.

BCF by software — On this platform

Open the BCF studio

The BCF studio at /bim-tools/bcf opens BCF 2.0, 2.1 and 3.0 files, shows topics with snapshots, comments and viewpoints, lets you edit them, writes BCF 2.1 and converts issues to and from spreadsheets or clash reports.

Track issues on the project Issues page

Inside a project, the Issues (BCF) page under Model and coordination keeps the project issue list, with status, priority, labels and assignees, and can import and export BCF.

Good uses of the studio

- Check a BCF file from another team before importing it into your tool
- Convert a spreadsheet of clashes into BCF topics
- Export topics to a spreadsheet for a meeting report
- Convert a newer BCF file to 2.1 for a tool that cannot read 3.0
- Clean up titles, labels and assignees in bulk before issuing

Best practice — Writing good issues

Topic fields and how to use them

Field	Good practice	Example
Title	Say what and where in a few words, starting with the location	L02 grid C4: duct clashes with primary beam
Description	Say what is wrong, why it matters and what is proposed	Duct passes through the beam web. Reroute below or agree an opening with structures.
Type	Use the agreed list, such as clash, design query, request for information or model quality	Clash
Status	Use the agreed status set and keep it current	Open
Priority	Use a short scale with clear meanings	High: blocks the next milestone
Labels	Add discipline, zone and the clash test name	MEP, Zone B, Duct vs Structure

Field	Good practice	Example
Assigned to	One person or team, never a list	Mechanical task team lead
Due date	A real date tied to the next model share or milestone	Before next Thursday's model cut-off

Rules for good topics

- One issue per problem. Do not put three unrelated problems in one topic.
- Always include a snapshot and a viewpoint with the affected objects selected.
- Write titles that make sense in a list without opening the topic.
- Raise issues on current shared models, never on someone else's WIP.
- Record decisions as comments in the topic, not in meeting minutes only.
- Close topics only after the fix has been seen in a shared model.

> **Keep the lists short**
> Agree short lists for type, status, priority and labels at mobilisation. Long free-text lists make filtering and reporting impossible. BCF 3.0 can carry these allowed values in its extensions file.

Best practice — KPIs & reporting

Issue KPIs worth tracking

Figure	Meaning	Source
Open vs closed	Issues opened and closed each week; the gap shows whether coordination is keeping up	
Ageing	How long open issues have been open, grouped by age band	
By discipline	Open issues assigned to each task team	
By priority	High-priority issues open against the next milestone	
Reopen rate	Share of resolved issues that had to be reopened	
By zone	Open issues by building zone or level, to focus effort	

A weekly issue report

Section	What to show
Summary	Total open, resolved and closed, with the change since last week
Ageing	Open issues by age band, such as under a week, one to four weeks and over four weeks
By discipline	Open and overdue issues for each task team
Top risks	High-priority issues that could delay the next milestone

Section	What to show
Decisions needed	Issues waiting for a client or design decision

Watch the trend, not just the total
A total that stays flat can hide many new issues and many closures. Track opened and closed each week, and the age of what is left.

Best practice — Common mistakes

Common mistakes and fixes

Mistake	Effect	Fix
Every clash sent as a separate issue	Thousands of topics nobody reads	Group clashes into problems before creating topics
Vague titles such as Clash 145	Nobody can tell what the issue is from the list	Use location plus a short description
Issues assigned to several people	Nobody feels responsible	Assign one owner and mention others in comments
Closed without checking	Problems come back on site	Close only after the fix is seen in a shared model
Different status sets in each tool	Statuses get lost or mapped wrongly	Agree one status set and map it in each tool
Issues raised on old models	Teams fix problems that are already gone	Federate only current shared revisions
GUIDs change between exports	Topics no longer highlight the objects	Fix export settings and test GUID stability
Several copies of the issue list	Conflicting statuses and arguments in meetings	Agree one master list, ideally on a BCF API server

BCF is not a substitute for talking
Complex problems often need a short call or a meeting. Use BCF to record the problem and the decision, not to hold a long argument in comments.

FAQ, quiz & resources — FAQ

Frequently asked questions

Does a BCF file contain the model?

No. It contains issues that refer to the model through viewpoints and IFC GUIDs. The receiver needs the model in their own tool to see the objects.

Which BCF version should we use?

BCF 2.1 is the safest choice for file exchange because most tools support it. Use BCF 3.0 only if every tool on the project reads and writes it reliably.

What is the BCF API?

A RESTful web service defined by buildingSMART that lets tools read and write topics on a shared server, so everyone works on the same live issue list.

Can BCF be used without IFC?

Topics can still carry snapshots, viewpoints and comments, but object references rely on IFC GUIDs. Tools that map their own element IDs to IFC GUIDs can still use them.

Why do my issues not highlight any objects?

Usually the GUIDs in the topic do not match the model you opened, because export settings changed or a different model version is loaded.

Should every clash become a BCF topic?

No. Group clashes into real problems and create one topic per problem. Ignore false positives and approved overlaps.

Who should close an issue?

The person who raised it or the coordinator, after checking the fix in a shared model. The assignee marks it resolved, not closed.

Where should BCF files be stored?

In the CDE, named and managed like other project information, unless a BCF API server holds the live issues. Either way, agree one master list.

Does closing an issue change the model's status in the CDE?

No. The fix goes into a new WIP revision, which is checked and shared through the CDE in the normal way.

Can the client raise issues in BCF?

Yes. Review comments on submitted models can be returned as BCF topics, which makes them easy to locate and track.

FAQ, quiz & resources — Resources & AI

buildingSMART BCF resources

- BCF-XML on GitHub — Schemas, documentation and examples for the file-based format (https://github.com/buildingSMART/BCF-XML)
- BCF API on GitHub — Specification of the RESTful BCF web service (https://github.com/buildingSMART/BCF-API)
- buildingSMART International — The organisation behind IFC, BCF and other open standards (https://www.buildingsmart.org/)
- UK BIM Framework — Guidance on ISO 19650 information management, including collaborative production (https://www.ukbimframework.org/)

Practise in the BCF studio
Open a sample BCF file in the BCF studio at /bim-tools/bcf, edit a topic and export it again. Then try the same round trip with your own tools.

Check AI output
AI drafts are a starting point. Confirm statuses, priorities, tools and responsibilities with the BIM coordinators and the BEP before issuing a procedure.

11. Handover and the asset information model

Handover is the moment the project information model becomes the owner's asset information model. Done well, the facility team opens its systems on the first day of operation and finds every asset, document and warranty ready to use. Done badly, they inherit boxes of PDFs and months of retyping. This guide covers the handover strategy and timeline, what to hand over, acceptance and status codes, data quality checks, loading into CAFM and CMMS systems, roles, a twelve-week plan, a readiness checklist, KPIs, and how the AIM stays alive after handover.

Strategy & timeline — Overview

Handover starts at the beginning

Under ISO 19650 the appointing party sets out what it needs to operate the asset in its asset information requirements. The delivery team builds the project information model to meet those requirements, and at the end the relevant parts are checked, accepted and transferred into the asset information model. Handover is the last step of that chain, but it is decided at the first.

The information handover strategy is the document that makes this concrete. It says what will be handed over, in what format, when, how it will be checked, which system it will be loaded into and who is responsible for each step. It is written by the client's information manager with the facility manager and agreed with the delivery team.

The best handovers follow a soft landings mindset: the people who will operate the building are involved from early design, asset data is collected while installation happens, and support continues after occupation until the building and its information settle down.

What good handover looks like

Planned from day one

The handover strategy is agreed before construction and priced into the appointments, not invented at practical completion.

Progressive

Asset data arrives in staged drops tied to installation and commissioning, so gaps are found when they are cheap to fix.

Structured

Asset data is in a structured, checkable format such as COBie or the client's template, with documents linked to the assets they describe.

Checked against the AIR

Acceptance uses measurable criteria from the asset information requirements, not a judgement of complete and correct.

Loaded and tested

Data is trial loaded into the receiving CAFM or CMMS before the final drop, so the import works on handover day.

Owned afterwards

A named person owns the AIM after the project team leaves, and trigger events keep it up to date.

Handover at a glance

Figure	Meaning	Source
PIM → AIM	Relevant project information is accepted and transferred into the asset information model	
S7	Common suitability code for information submitted for AIM authorisation	
CR	Common status for accepted as-constructed record information	
Staged	Asset data is delivered in drops during construction, not once at the end	

Read with the HOS guide and the COBie hub
The platform's document guide for the Information Handover Strategy (HOS), at /doc-guides/handover-strategy, explains each section of the document with examples. The COBie hub explains the asset data format in detail. This hub ties them together into a working handover process.

Strategy & timeline — Timeline

Early planning, staged delivery

Handover activities run through the whole project. Most of the work happens in the last third of construction, but the decisions that make it possible are taken in briefing and design. The table shows a typical timeline; your project's MIDP sets the actual dates.

Handover activities through the project

Project stage	Handover activity	Lead	Output
Briefing	Define organisational and asset information requirements with the operations team	Client information manager	OIR and AIR
Tender	Issue the handover strategy and COBie template with the EIR so the effort is priced	Client information manager	HOS and EIR issued to tenderers
Appointment and mobilisation	Confirm deliverables, formats, responsibilities and the receiving system	Lead appointed party	BEP and MIDP with handover containers
Design	Build spaces, types, systems and classification into the models	Design task teams	Design-stage COBie drops
Early construction	Agree how product, serial and warranty data will be collected from subcontractors	Main contractor	Data collection plan and trial drop
Installation and commissioning	Collect installed data progressively by area and system; trial load into CAFM	Main contractor and specialists	Progressive drops and a trial import
Pre-completion	Final drop, verification, site sampling, H&S file and O&M documents	Main contractor with client FM	Handover package submitted as S7

Project stage	Handover activity	Lead	Output
Handover	Acceptance, transfer to the AIM and load into the live systems	Client information manager and FM	Accepted AIM and live systems
Aftercare	Resolve defects in data, review after occupation and record lessons	Client and delivery team	Updated AIM and lessons learned

Soft landings style handover

1. **Involve operations early.** Bring the facility manager into briefing and design reviews so the AIR reflects how the building will really be run.
2. **Set expectations.** Agree the handover outcomes, formats and receiving systems before construction and write them into the appointments.
3. **Collect data as you build.** Capture product, serial and installation data at the time of installation, not from memory at the end.
4. **Rehearse the handover.** Trial load early drops into the receiving system and walk the FM team through the data and documents.
5. **Hand over with support.** Keep delivery team contacts available for an agreed aftercare period to resolve data and performance issues.
6. **Review after occupation.** Review how the information is being used after occupation, fix gaps and feed lessons into the next project's requirements.

Phased and sectional handover
Large or phased projects often hand over by building, section or system. Plan a separate set of drops and acceptance for each, so the operations team can take over part of the asset while the rest is still being built.

Strategy & timeline — Roles

Who does what at handover

Client information manager
Writes the handover strategy, sets acceptance criteria, accepts the information and transfers it into the AIM.

Facility manager
Confirms which assets and data operations need, reviews drops, tests imports into CAFM or CMMS and owns the AIM afterwards.

Lead appointed party information manager
Plans handover containers in the MIDP, merges task team data, runs checks and submits the handover package.

Main contractor
Collects product, serial, warranty and installation data from subcontractors and suppliers and compiles O&M information.

Specialist subcontractors

Provide product data, commissioning records, test certificates and maintenance requirements for their systems.

Health and safety lead

Compiles the health and safety file from residual risks and as-built information and issues it to the client.

Handover RACI

Activity	Client IM	Facility manager	Lead appointed party IM	Main contractor	Specialists
Write the handover strategy	R	C	C	I	I
Define acceptance criteria	A	R	C	I	I
Plan handover containers in the MIDP	I	I	A	R	C
Collect product and installation data	I	I	A	R	R
Compile O&M manuals and warranties	I	C	A	R	R
Run COBie and naming checks	C	I	A	R	C
Site sample of asset tags	A	R	C	R	I
Trial import into CAFM or CMMS	A	R	C	C	I
Accept handover information	A	R	I	I	I
Maintain the AIM after handover	A	R	I	I	C

Name people, not companies

Handover tasks fall between organisations. Name the role and the person for each task in the strategy, and make sure the facility manager is a named contributor from the start.

Content & acceptance — What to hand over

The content of the asset information model

Not everything in the project information model belongs in the asset information model. Design options, coordination models, superseded drawings and correspondence are archived with the PIM. What transfers is the information the owner needs to operate, maintain, alter and eventually dispose of the asset, as set out in the AIR.

Typical handover content

Content	What it includes	Typical format	Usually provided by
Record models	As-constructed geometry of the building and its systems, at the level of information need set in the AIR	IFC plus native files where agreed	Design task teams and contractor
Asset data	Spaces, zones, types, components, systems, serial numbers, tags and installation dates	COBie workbook or the client's asset template	Contractor, merged by the lead appointed party
O&M documents	Operation and maintenance manuals, maintenance schedules, spare parts lists	PDF linked to assets, plus structured job and spare data	Contractor and specialists
Warranties and guarantees	Product and workmanship warranties with start dates, periods and providers	Warranty fields in asset data plus certificates	Contractor and suppliers
Test and commissioning records	Commissioning sheets, test certificates and witness records	PDF linked to systems and components	Commissioning manager and specialists
Health and safety file	Residual risks, hazardous materials, safe access and information for future works	Structured file as required by the client	Health and safety lead
As-constructed drawings	Record drawings where the client still needs drawings	PDF and native where agreed	Design task teams and contractor
Statutory and compliance certificates	Fire, lift, electrical, gas and other certificates	PDF linked to systems and spaces	Contractor and specialists
Training records	Who was trained on which systems and when	Register or document	Contractor

Usually archived with the PIM, not transferred

- Design options and feasibility models
- Coordination and clash detection models
- Superseded revisions of drawings and models
- Correspondence, meeting minutes and requests for information
- Programme and cost information not needed in operation
- Work in progress containers that were never shared

Trace every deliverable to the AIR
If a handover deliverable cannot be traced to an asset information requirement, ask whether it is needed. If an AIR requirement has no deliverable, the strategy has a gap.

Content & acceptance — Acceptance & status

How handover information is accepted

Handover information moves through the CDE like any other information container, but its last step is different. Instead of being authorised for the next stage of design or construction, it is submitted for acceptance into the asset information model.

Under the common UK convention, containers submitted for AIM acceptance carry the suitability S7. When the client accepts them, they are given an authorised status, and record information is published as CR, as-constructed record. Use the codes your project's information standard defines.

Acceptance flow for a handover container

1. **Work in progress.** The task team prepares the as-constructed model, asset data or document set in its WIP area and runs its own checks.
2. **Shared for review.** The container is shared with the lead appointed party, who checks it against the AIR, the naming rules and the other task teams' data.
3. **Submitted for AIM acceptance.** The lead appointed party submits the handover package to the client with the S7 suitability, a revision and a submission note listing any exceptions.
4. **Client review.** The client information manager and facility manager check it against the acceptance criteria: COBie checks, site samples, trial import and document links.
5. **Accepted or rejected.** Accepted containers receive an authorised status. Rejected containers return with an issue list, owners and a resubmission date.
6. **Published as record.** Accepted record models and drawings are published as CR and transferred into the AIM; the rest of the PIM is archived.

Common statuses around handover

Code	Meaning	Used for
S6	Suitable for PIM authorisation	Information submitted at the end of a design or construction stage
S7	Suitable for AIM authorisation	Handover information submitted for acceptance into the AIM
A1, A2 and so on	Authorised and accepted	Containers accepted by the client
B1, B2 and so on	Partially authorised, with comments	Containers accepted subject to minor changes
CR	As-constructed record	Accepted record models and drawings

> **Acceptance must be measurable**
> Write acceptance criteria that anyone can check: for example zero errors in the COBie checker, a ten per cent site sample of asset tags with no mismatches, every O&M document linked to an asset, and a clean trial import.

Content & acceptance — Documents & H&S file

Documents that make the data usable

Structured asset data tells the facility team what they own. Documents tell them how to look after it. The two must be linked: each O&M manual, certificate and warranty should be attached to the type, component, system or space it describes, so it can be found from the asset record.

The health and safety file is a special case. It holds the information needed to carry out future work safely, such as residual risks, hazardous materials and safe access. It should be structured, kept up to date and easy to find from the spaces and assets it relates to.

Linking documents to assets

Document	Link to	Tip
Product data sheet	Type	One sheet per product type, not per installed item
O&M manual section	Type or System	Split large manuals so each section links to what it describes
Warranty certificate	Type or Component	Record the start date and provider as data, not only in the PDF
Commissioning record	System or Component	Name the file with the system or asset tag
Test certificate	Component or Space	Link statutory certificates to the asset or area they cover
Health and safety file entry	Space, Component or Facility	Record residual risks against the location where they apply
Record drawing	Floor, Space or System	Use the same names as the asset data so drawings can be found from it

Document handover checks

- ☐ Every in-scope asset type has its product data and O&M information
- ☐ Documents are linked to assets in the COBie Document sheet or the client's register
- ☐ File names follow the project naming standard and are frozen before the final export
- ☐ Warranty start dates and periods are recorded as data
- ☐ Commissioning and test records are complete for every system
- ☐ The health and safety file is issued and its entries are linked to locations
- ☐ Hazardous materials and residual risks are recorded where they apply

Data quality & systems — Data quality checks

Check early, check often

Handover data quality is built up drop by drop. Each drop should pass the same checks, so that by the final drop only the installed data is new. Checks fall into three layers: format, content and reality.

Handover data quality checks

Check	What it tests	How	Who
COBie structure	Required sheets, columns, unique names and valid dates	COBie checker	Lead appointed party, then client
References	Components to types and spaces, systems to components, contacts to CreatedBy	COBie checker	Lead appointed party
Naming	Asset, space and file names follow the project naming standard	Rule set or script	Lead appointed party
Classification	Every type and space has the agreed classification in the right form	Rule set	Lead appointed party
Completeness	Every in-scope asset has the attributes listed in the AIR	Attribute matrix against the AIR	Client information manager
Asset tags	Tags in the data match labels on site	Site sample, often ten per cent	Facility manager with contractor
Installed data	Serial numbers, install dates and warranty dates are real values, not placeholders	Spot checks against submittals and site	Facility manager
Document links	Every linked document exists and opens	Link check	Lead appointed party
Model to data	Assets in the record model match the asset data	Comparison report	Lead appointed party

Signs that data is not ready

Placeholder values

Serial numbers like TBC or 0000 and identical install dates for every asset show data was filled to pass a check.

Sudden jumps in count

A large rise in components between drops often means non-managed elements are being exported.

Orphan assets

Components without a valid space or type cannot be found in the CAFM system.

Documents not linked

A folder of PDFs with no links to assets is an archive, not part of the AIM.

Run the COBie checker on every drop

The platform's COBie checker, at /ai-tools/cobie, tests sheets, required columns, unique names, dates, classification and references between sheets, and scores each drop. Run it before every submission and share the report with the drop.

Data quality & systems — Loading into CAFM/CMMS

From workbook to working system

The handover is only finished when the data works in the systems the facility team uses every day: computer-aided facility management for space and services, a computerised maintenance management system for planned and reactive maintenance, and sometimes a document system and a building management system.

Each system has its own data structure. The handover strategy should name the receiving system early, so the mapping from COBie or the client template to that system can be agreed and tested before the final drop.

Loading asset data into the receiving system

1. **Name the receiving system.** Record the CAFM or CMMS product and version, and who administers it, in the handover strategy.
2. **Agree the field mapping.** Map each COBie sheet and column, or each template field, to the receiving system's fields. Get the mapping signed off by the system owner.
3. **Prepare reference data.** Load sites, buildings, floors, classification codes and asset categories into the system before asset data arrives.
4. **Trial import.** Import an early drop into a test instance, record errors and fix the template, mapping or source data.
5. **Final import.** Import the accepted final drop into the live system and reconcile counts of spaces, types and components.
6. **Set up maintenance.** Turn job and spare data into planned maintenance schedules and check them with the maintenance team.
7. **Train and hand over.** Train the facility team on the data and how to keep it current, and confirm who owns updates.

Typical mapping from COBie to a maintenance system

COBie source	Receiving system field	Notes
Facility and Floor	Site, building and level hierarchy	Load before assets so locations exist
Space	Location or room	Space name usually becomes the location code
Type	Asset class or model	Holds manufacturer, model and warranty defaults
Component	Asset	Name or tag becomes the asset number
System	System or asset group	Used for planned maintenance and fault reporting
Job	Planned maintenance task	Frequency and duration drive the schedule
Spare	Inventory item	Linked to the asset class
Document	Attachment or document link	Check the system can store or link the files

Do the trial import early
A trial import of an early construction drop is the single most effective handover check. It finds mapping, template and naming problems while there is still time to change how data is collected.

Data quality & systems — KPIs

Measuring handover quality

Agree a small set of handover KPIs in the strategy and report them at every drop. They give the client an early warning and give the delivery team a clear target. Targets below are examples; set your own in the strategy.

Handover KPIs

KPI	How it is measured	Example target
Data completeness	Required AIR attributes filled, as a percentage of all required attributes for in-scope assets	100% at final drop; agreed percentage at earlier drops
COBie errors	Errors reported by the checker per drop	Zero at final drop
Tag accuracy	Site sample of tags that match the data	100% of the sample
Document coverage	In-scope types with linked O&M information	100% at final drop
Data defects	Open data issues raised by the client after each drop	Falling drop on drop; none open at acceptance
Drops on time	Drops submitted by the MIDP date	All drops on time
First-time acceptance	Drops accepted without resubmission	Rising through construction
Import success	Records imported into CAFM or CMMS without error	100% at final import
Post-handover data defects	Data issues found by the FM team in the first months of operation	Tracked and closed within the aftercare period

Example dashboard at the final drop

Figure	Meaning	Source
98%	Data completeness against the AIR (example)	
0	COBie checker errors (target)	
10%	Asset tags sampled on site (typical)	
12	Open data defects to close before acceptance (example)	

Plan & operation — 12-week plan

Twelve weeks before handover

1. **Week 12: confirm the scope.** Review the handover strategy with the client, confirm deliverables, formats, the receiving system and acceptance criteria, and check the MIDP dates.
2. **Week 11: chase product data.** Issue a data request to every subcontractor and supplier for missing product, warranty and maintenance information, with a due date.
3. **Week 10: progressive drop.** Submit a progressive asset data drop for completed areas and run the COBie checker. Share the report with the client.
4. **Week 9: trial import.** The facility manager trial imports the drop into a test instance of the CAFM or CMMS, and the team fixes mapping and data issues.
5. **Week 8: O&M structure.** Agree the O&M document structure and file naming, and start linking documents to assets.
6. **Week 7: tag on site.** Fix asset labels on site as systems are commissioned, using the tag format in the AIR.
7. **Week 6: H&S file draft.** The health and safety lead issues a draft health and safety file for client review.
8. **Week 5: record models.** Task teams update models to as-constructed and share them for review against the asset data.
9. **Week 4: final data drop.** Submit the near-complete asset data with serial numbers, install dates and warranties, and run all checks.
10. **Week 3: site sample.** The facility manager and contractor check a sample of asset tags and installed data on site and record mismatches.
11. **Week 2: close defects.** Fix data defects from the checks and the site sample, freeze file names and complete document links.
12. **Week 1: submit as S7.** Submit the handover package for AIM acceptance with a submission note, train the FM team and confirm the transfer and archive plan.

> **Twelve weeks is the end, not the start**
> This plan assumes the strategy, template and data collection were set up at mobilisation and that progressive drops have been made through construction. If not, start immediately and expect handover data to lag behind practical completion.

Plan & operation — Readiness checklist

Handover readiness checklist

- ☐ Handover strategy agreed, current and known to the whole delivery team
- ☐ Every deliverable traced to the AIR with an owner and a due date
- ☐ Receiving CAFM or CMMS named, with a signed-off field mapping
- ☐ Trial import completed and issues closed
- ☐ Final COBie drop passes the checker with zero errors

- ☐ Every in-scope asset has a tag, a space and a type
- ☐ Serial numbers, install dates and warranty dates are real values
- ☐ Site sample of asset tags completed with no open mismatches
- ☐ O&M documents complete and linked to assets
- ☐ Test, commissioning and statutory certificates complete
- ☐ Health and safety file issued to the client
- ☐ Record models and drawings updated to as-constructed
- ☐ File names frozen and document links checked
- ☐ Facility team trained on the systems and the data
- ☐ AIM owner named and trigger events defined
- ☐ PIM archive plan agreed with access and retention rules

> **Use the checklist as a gate**
> Agree with the client that the handover package is only submitted for acceptance when every item is ticked or has an agreed exception recorded in the submission note.

Plan & operation — After handover

Keeping the AIM alive

The asset information model is only valuable if it stays true to the building. ISO 19650-3 describes the operational phase as a series of information exchanges started by trigger events: something happens to the asset, and information must be delivered to update the AIM.

The client information manager defines the trigger events and the information each must deliver, and puts those requirements into maintenance and works contracts. The facility manager checks the returned information and accepts it into the AIM.

Typical trigger events

Trigger event	Information to return	Usually provided by
Planned maintenance	Task completion, readings and any defects found	Maintenance contractor
Asset replacement	New product data, serial number, install date and warranty; old asset retired	Maintenance or works contractor
Fit-out or refurbishment	Updated spaces, assets, record models and documents	Fit-out appointed party
Failure or incident	Fault, cause, repair and any change to the asset	Maintenance contractor
Survey or inspection	Condition data and survey records	Surveyor or specialist
Change of use or ownership	Updated spaces, zones, responsibilities and access	Client information manager
End of warranty or defects period	Defect closure records and warranty status	Client and contractor

Post-occupancy review

1. **Gather feedback.** After a few months of occupation, ask the facility and maintenance teams which information they use, what is missing and what is wrong.
2. **Measure.** Count data defects raised since handover and check how many assets have had their data updated.
3. **Fix.** Close data defects within the aftercare period with the delivery team.
4. **Improve the requirements.** Update the OIR, AIR and handover strategy template so the next project asks for the right information.

> **Write data updates into every contract**
> If maintenance and works contracts do not require information to be returned, the AIM drifts out of date within months. Include information requirements and acceptance in every contract that touches the asset.

Plan & operation — Common mistakes

Common handover mistakes

Mistake	Result	How to avoid it
Asking for full COBie with no template or field list	Disputes about scope and data nobody can use	Issue the client template and AIR field list with the strategy
All asset data due at practical completion	Rushed, incomplete data and late acceptance	Plan progressive drops tied to installation and commissioning
Facility manager involved only at the end	Data in the wrong structure for operations	Make the FM a contributor from briefing and in every drop review
No receiving system named	Data that cannot be imported	Name the CAFM or CMMS and plan a trial import
Acceptance criteria written as complete and correct	Endless argument about whether data is good enough	Use measurable checks with thresholds and a named checker
O&M manuals as one large PDF	Information that cannot be found from the asset	Split documents and link them to types, systems and components
Tags in the data differ from labels on site	Maintenance staff cannot find the asset	Use one tag format from the AIR and sample on site
Placeholder values to pass checks	Data passes format checks but is useless	Spot check installed data against submittals and site
No AIM owner after handover	The AIM drifts out of date	Name the owner in the strategy and define trigger events

> **Handover is a process, not an event**
> Almost every handover failure comes from leaving the work to the end. Plan it at the start, deliver it in drops and rehearse it with the people who will use it.

FAQ, quiz & AI — FAQ

Frequently asked questions

Who writes the information handover strategy?

The client's information manager, working with the facility manager. It is agreed with the delivery team and ideally issued with the tender documents so the effort is priced.

Is the PIM handed over in full?

No. The relevant parts of the PIM, as set out in the AIR, are accepted and transferred into the AIM. The rest is archived with access and retention rules.

What does S7 mean?

Under the common UK convention it means suitable for AIM authorisation: the container is submitted for acceptance into the asset information model. Use the codes in your project's information standard.

What does CR mean?

As-constructed record. It is commonly used for accepted record models and drawings that describe the asset as built.

Is COBie required for handover?

Not by ISO 19650 itself. The client decides the format in its requirements. COBie is widely used because it is structured, open and can be checked automatically.

When should the first asset data drop be made?

During design for spaces and types, and early in construction for installed data. A trial drop early in construction finds template and mapping problems while they are cheap to fix.

Who checks asset tags on site?

Usually the facility manager with the contractor, on an agreed sample such as ten per cent of tagged assets. Any mismatch should trigger a wider check.

What is a trigger event?

Something that happens to the asset during operation, such as maintenance, replacement or fit-out, that requires information to be delivered to update the AIM.

What if the receiving CAFM system is chosen late?

Deliver to a neutral structured format such as COBie, keep the mapping step in the plan and do the trial import as soon as the system is known.

How long should aftercare last?

It is set in the appointments. Many projects keep delivery team support available through the first year of operation, or until the end of the defects period.

Quick answers by role

Client information manager

Write the strategy early, make acceptance measurable and name the AIM owner.

Facility manager

Say what you need, review every drop and run the trial import yourself.

Main contractor

Collect data at installation, chase subcontractors early and tag assets as you commission.

Lead appointed party information manager

Put every handover container in the MIDP and run the checks before the client does.

FAQ, quiz & AI — Quiz

Learn from the answers

Each answer has a short explanation. If you get one wrong, revisit the matching tab before trying again.

FAQ, quiz & AI — AI tools

Check your COBie with the checker

Open the COBie checker at /ai-tools/cobie to test each drop for missing sheets and columns, blank required fields, duplicate names, broken references and classification problems.

Read the HOS document guide

The Information Handover Strategy guide at /doc-guides/handover-strategy explains each section of the strategy, with examples, a step-by-step method, a checklist and common mistakes.

Related tools on this platform

- COBie checker — Check each asset data drop before submission (/ai-tools/cobie)
- Handover strategy guide — How to write the HOS, section by section (/doc-guides/handover-strategy)
- COBie hub — Every COBie sheet, delivery timing and quality checks (/hubs/cobie)

Check AI output

AI drafts are a starting point. The client information manager and the facility manager should confirm scope, formats, acceptance criteria and responsibilities before the strategy is issued.

12. Implementing ISO 19650 in an organisation

How organisations make information management according to ISO 19650 the normal way of working: strategy and business case, maturity assessment, people and change, process, technology, standards and the KPIs that show whether it is working. Use it alongside the adoption guide, which tracks the individual steps.

Strategy

Start from the business, not the software

BIM implementation succeeds when it is framed as a change in how the organisation specifies, produces, checks and uses information. Software is an enabler. Begin by stating which business problems information management should solve: rework and late changes, poor handover data, client mandates, risk, or the cost of running assets.

The adoption guide in this portal sets out the individual steps. This hub explains the decisions behind them: how to phase the roadmap, how to make the case for investment, and how to govern the programme so it survives beyond the first pilot.

Phased roadmap

1. **Phase 0: Mobilise (first weeks).** Appoint a sponsor and implementation lead, agree scope (appointing party, delivery organisation or both) and baseline current maturity.
2. **Phase 1: Foundation (first few months).** Publish the organisational information requirements, information standard, CDE procedure and house templates. Configure the CDE.
3. **Phase 2: Pilot (one or two projects).** Run the full ISO 19650 process on selected projects with close support. Measure results against the baseline and capture lessons.
4. **Phase 3: Scale (following year).** Mandate the process for new projects above a threshold, train by role, extend to the supply chain and automate checks.
5. **Phase 4: Optimise (ongoing).** Audit, review KPIs, connect project information to asset information and operations, and update standards annually.

Building the business case

Benefits

Less rework, fewer RFIs and site clashes, faster approvals, better handover data, lower operating cost, stronger bids where clients require ISO 19650.

Costs

Software and CDE licences, hardware, training, consultancy, internal time for standards and templates, and the productivity dip during transition.

Risks of not acting

Losing work from clients who require ISO 19650, disputes over information quality, and asset data that has to be recreated after handover.

Evidence

Use your own baseline data where possible: rework logs, RFI counts, handover defects. Internal evidence persuades boards more than generic industry claims.

Governance structure

Body or role	Purpose	Typical frequency
Executive sponsor	Owns the business case, removes blockers, secures budget	Monthly review
Steering group	Approves standards, priorities and funding; tracks KPIs	Monthly or quarterly
Implementation lead	Runs the programme day to day and reports progress	Continuous
Standards working group	Maintains the information standard, templates and methods	Monthly
Champions network	Discipline and office champions support projects and feed back issues	Fortnightly or monthly
Project information managers	Apply the process on projects and report lessons	Per project milestone
Internal audit	Checks compliance and effectiveness on live projects	Quarterly or per milestone

Protect the pilot

Choose pilot projects with a supportive client and team, a realistic programme and enough time to learn. A pilot that fails for unrelated commercial reasons can set implementation back by years.

Maturity

Information management maturity levels

Level	Characteristics	Typical next step
1. Ad hoc	Practice varies by project and person; no shared standard; CAD and email-based exchange	Appoint a lead and write a basic information standard
2. Emerging	Some projects use models and a CDE; templates exist but are not consistent	Publish house templates and a CDE procedure; train key roles
3. Defined	Organisation-wide standards, roles and CDE workflow aligned with ISO 19650	Pilot the full process and measure results
4. Managed	Process mandated and audited; KPIs tracked; supply chain assessed before appointment	Automate checks and connect to asset information
5. Optimising	Continuous improvement from data; information flows into operations and decision-making	Integrate with digital twins and portfolio analytics

How this relates to the older BIM levels

Many organisations still describe themselves using the UK BIM maturity levels (0 to 3) from the Bew and Richards diagram. ISO 19650 replaced that language: in the UK the phrase BIM according to the ISO 19650 series is now used instead of BIM Level 2. The levels above are a practical internal scale, not a formal classification in the standard.

How to assess maturity

1. **Choose dimensions.** Assess requirements, standards and methods, CDE, roles and responsibilities, competence, technology, supply chain, and asset information.
2. **Gather evidence.** Review documents and live projects, not only questionnaires. Look at real EIRs, BEPs, MIDPs and CDE audit trails.
3. **Score consistently.** Use defined criteria for each level so different assessors reach the same score. Record evidence for each score.
4. **Prioritise gaps.** Rank gaps by business impact and effort. Fix foundations, such as the information standard and CDE workflow, before advanced uses.
5. **Reassess.** Repeat at least annually and after major changes, using the same criteria to show progress.

Capability and capacity under ISO 19650

What the standard asks

During tender response (activity 5.3), each task team's capability and capacity is assessed and summarised for the delivery team. Appointing parties use this when evaluating tenders.

Capability

Skills, knowledge and experience of the people, and the suitability of the information technology, for the information management tasks required.

Capacity

Whether the team has enough people and IT resources available when the work is needed, alongside its other commitments.

Use it internally

Run the same assessment on your own organisation and supply chain before tenders. It doubles as a practical maturity check.

People

Roles and core competencies

Role	Focus	Core competencies
Executive sponsor	Strategy and investment	Business case, change leadership, client expectations
Implementation or digital lead	Running the programme	ISO 19650, programme management, stakeholder management
Information manager (appointing party)	Requirements and acceptance	OIR, AIR, EIR, information protocol, review and acceptance

Role	Focus	Core competencies
Lead information manager (delivery team)	BEP, MIDP and federation	BEP authoring, planning, CDE workflow, coordination
Task information manager	Task team compliance	Information standard, naming, metadata, TIDP, QA
BIM coordinator	Model coordination	Federation, clash detection, issue management, BCF
CDE administrator	Platform configuration	Permissions, workflows, metadata, audit trail
Modeller or author	Producing information	Authoring tools, modelling standards, LOIN, IFC export
Asset or FM manager	Using information in operation	AIR, COBie, CAFM integration, ISO 19650-3

Training paths by audience

1. **Everyone.** Short awareness session: why the organisation is adopting ISO 19650, the CDE states, naming and where to find standards.
2. **Project leaders and directors.** Requirements, appointments, responsibilities and how information management affects risk, fees and programme.
3. **Information managers.** Practitioner-level training on the full process, authoring EIRs and BEPs, planning and acceptance, ideally with assessed certification.
4. **Modellers and coordinators.** Software skills tied to the house templates, LOIN, model QA and IFC exchange; coordination workflow and BCF.
5. **Supply chain.** Briefings on your standards and templates, with capability assessment before appointment.

Change management practices that work

- Explain the reasons for change in terms each audience cares about
- Visible sponsorship: leaders use the CDE and ask for the outputs
- Champions in each office or discipline who others can ask
- Early wins publicised internally, with real project examples
- Time and budget allowed for learning on pilot projects
- Simple, well-maintained templates that make the right way the easy way
- Feedback channels so issues reach the standards group quickly
- Recognition of good practice in reviews and appraisals
- Consistent enforcement once the process is mandated

> **Resistance is information**
> When staff resist, find out why. Often the template is too complicated, the software is too slow, or the process duplicates work. Fixing the cause works better than insisting on compliance.

Process

Standard methods and procedures

Information standard
Naming, status and revision codes, classification, metadata, units, coordinates, drawing and model standards.

Production methods
How information is captured, generated, reviewed, approved, secured and delivered, written once and reused on every project.

CDE procedure
States, transitions, who can approve each transition, permissions and how transmittals and audit trails work.

Review and acceptance
Checks before sharing, authorization by the lead appointed party and acceptance by the appointing party, with defined criteria.

House template library

- ☐ Organisational and asset information requirements (OIR, AIR)
- ☐ Exchange information requirements (EIR) with clause library
- ☐ Pre- and post-appointment BIM execution plans
- ☐ Responsibility matrix
- ☐ Task and master information delivery plans (TIDP, MIDP)
- ☐ Information protocol wording for appointments
- ☐ Capability and capacity assessment forms
- ☐ Model QA and IFC export checklists
- ☐ Coordination meeting agenda and minutes
- ☐ Lessons learned log

CDE workflow in practice

1. **Work in progress.** Task team develops information in its own area. Not visible to other teams.
2. **Check and share.** Task team checks against the information standard and suitability, then shares. Other teams can now use it as reference.
3. **Review and authorize.** Lead appointed party reviews shared information against the EIR and authorizes it for delivery.
4. **Accept and publish.** Appointing party reviews and accepts; the container is published for construction, operation or other agreed use.
5. **Archive.** Superseded and close-out information is archived with its audit trail for future reference.

Quality assurance layers

Layer	Who	What is checked
Self-check	Author	Content, naming, metadata, model health
Task team check	Task information manager	Compliance with information standard and LOIN

Layer	Who	What is checked
Coordination check	BIM coordinator	Clashes, spatial coordination, federation
Authorization	Lead appointed party	Completeness against the MIDP and EIR
Acceptance	Appointing party	Acceptance criteria in the EIR
Audit	Internal or third party	Process compliance across projects

Technology

Technology selection criteria

Criterion	Questions to ask
Fit to process	Does it support the CDE states, approvals and metadata your standard requires?
Openness	Does it read and write IFC, BCF and other open formats reliably?
Interoperability	Does it integrate with authoring, coordination, CAFM and document tools you already use?
Security	Does it support ISO 19650-5 controls, access levels, audit trails and data residency needs?
Scalability	Can it handle your largest projects, data volumes and number of external users?
Usability	Can occasional users, including clients and contractors, use it without heavy training?
Total cost	Licences, storage, integration, training, administration and exit costs
Vendor stability	Roadmap, support, user community and data export if you leave

The technology stack

CDE

The single source of information with states, metadata, workflows and audit trail. Choose this carefully; it shapes daily practice.

Authoring

Discipline tools for architecture, structure, MEP and civil. Standardise templates and object libraries rather than forcing a single vendor.

Coordination

Federation, clash detection and issue management tools, ideally exchanging issues with BCF.

Checking and validation

Rule-based model checking, IFC validation and information delivery specification checks to automate QA.

Asset information

CAFM, asset registers and digital twin platforms that receive structured handover data.

openBIM and interoperability

openBIM means exchanging information through open, vendor-neutral standards so that teams can use different tools and clients are not locked in. The main ones are IFC for models, BCF for issues, IDS for machine-readable information requirements and bSDD for shared data dictionary content, all promoted by buildingSMART International.

Interoperability does not happen by default. Agree IFC versions and model view definitions, map properties from authoring tools to IFC property sets, and test exports during mobilization rather than at the first deadline.

> **Avoid tool-first implementation**
> Buying a CDE or authoring tool before the information standard and workflows are agreed usually means configuring the tool twice. Define the process, then configure the technology to it.

Standards

Core standards for BIM implementation

Standard	Subject	Why it matters for implementation
ISO 19650-1	Concepts and principles	Shared vocabulary and the requirements and information model hierarchy
ISO 19650-2	Delivery phase	The eight-activity process (5.1 to 5.8) your project procedures follow
ISO 19650-3	Operational phase	Managing the asset information model and trigger events in operation
ISO 19650-4	Information exchange	Criteria and process for each exchange and CDE transition
ISO 19650-5	Security-minded approach	Assessing sensitivity and applying proportionate security controls
ISO 7817-1	Level of information need	Method for specifying geometry, information and documentation by purpose and milestone
ISO 12006-2 and Uniclass	Classification framework	Uniclass is a UK classification system aligned with the ISO 12006-2 framework
ISO 16739 (IFC)	Industry Foundation Classes	Open data schema for exchanging models; IFC 4.3 extends to infrastructure
BS 1192-4 (COBie)	Facility information exchange	Structured handover data for operation and maintenance
ISO 23386	Properties in data dictionaries	Methodology to describe, author and maintain properties consistently
ISO 23387	Data templates	Structuring product data templates linked to data dictionaries
National annexes and frameworks	Local application	For example the UK National Annex to ISO 19650-2 and the UK BIM Framework guidance

Using standards sensibly

Adopt, then adapt

Use the international standards as written and record any local choices in your information standard, rather than rewriting them.

Buy the licensed text

Guidance like this hub paraphrases the standards. Teams who write procedures need the licensed editions.

Check editions

Standards are revised. Record which edition each procedure references and review when new editions are published.

Follow national guidance

National annexes and frameworks fill in details such as naming fields and roles. Apply the one for the project's jurisdiction.

Standards questions

Do we need to comply with every part of ISO 19650?

No. Parts 1 and 2 are central for delivery. Part 3 matters if you own or operate assets, Part 5 when information is sensitive. Apply what your role and projects require.

Is LOD the same as level of information need?

No. LOD schemes describe model development. Level of information need, as set out in ISO 7817-1, covers geometry, alphanumeric information and documentation together, tied to purpose and milestone.

Is COBie still required?

Many clients still ask for COBie or COBie-like data for handover. Specify what the client actually needs in the AIR and EIR, whether COBie, IFC or a direct CAFM import.

KPIs

Implementation KPIs

KPI	Definition	How measured	Target guidance
Process adoption	Share of eligible projects using the mandated ISO 19650 process	Project register against policy threshold	Increase steadily towards all eligible projects
Information standard compliance	Share of containers passing naming and metadata checks first time	Automated CDE or checker reports	Rising trend; set internal targets from baseline
First-time acceptance	Share of deliverables accepted by the appointing party without rework	CDE review and acceptance records	Improve on baseline each quarter
Overdue deliverables	Containers delivered after their MIDP date	MIDP against CDE dates	Low and falling
Clash and issue closure	Average time to close coordination issues	Issue tracker or BCF records	Shorter cycle time than baseline

KPI	Definition	How measured	Target guidance
RFIs and rework	RFIs or rework items attributable to information quality	Project logs, categorised	Reduction against comparable past projects
Training coverage	Staff in each role who completed role-based training	Training records	Near complete for key roles before mandate
Supply chain capability	Appointed parties assessed and meeting minimum capability	Capability assessment records	All key appointments assessed
Handover data quality	Asset data complete and valid against the AIR at handover	Validation reports against AIR	High completeness before practical completion
Maturity score	Overall internal maturity assessment	Annual assessment with fixed criteria	Agreed improvement per year

Setting targets without inventing benchmarks

Figure	Meaning	Source
Baseline first	Measure two or three recent projects before setting targets	
Trend over point	Judge KPIs by direction of travel rather than one-off values	
Few and owned	A small set of KPIs, each with a named owner, beats a long dashboard	

Measure what you can automate
KPIs that depend on manual data collection tend to lapse. Prefer measures you can pull from the CDE, issue tracker and checking tools.

Common mistakes

Mistakes and fixes

Mistake	Consequence	Fix
Treating BIM as a software purchase	Tools configured without a process; low adoption	Define standards and workflows first, then configure tools
No executive sponsor	Programme stalls when budgets tighten	Name a sponsor accountable for the business case
Generic, copied EIRs	Requirements that do not reflect what the client needs	Derive EIRs from OIR, AIR and project purpose
BEP written only to win the tender	Plan not followed during delivery	Review the BEP at every stage gate and keep it current
Over-specifying information	Wasted modelling effort and cost	Specify the level of information need by purpose and milestone
Ignoring the supply chain	Inconsistent quality from consultants and contractors	Brief and assess capability before appointment

Mistake	Consequence	Fix
Training once, then stopping	Skills fade and new starters miss out	Role-based training with refreshers and onboarding
Too many KPIs	Nobody acts on them	Track a small set with owners and review monthly
Late interoperability testing	IFC exchanges fail at deadlines	Test exchanges in mobilization
Skipping lessons learned	The same problems repeat	Hold lessons sessions and update templates each time

Health check for your implementation

- ☐ Sponsor and lead are named and active
- ☐ Information standard and CDE procedure are published and current
- ☐ House templates are used on live projects
- ☐ Role-based training records are up to date
- ☐ Supply chain capability is assessed before appointment
- ☐ KPIs are reviewed by the steering group
- ☐ Lessons learned feed back into templates

Simplicity wins
If a procedure is too complex for project teams to follow under pressure, it will be ignored. Keep standards short, link to detail, and remove steps that nobody uses.

AI tools

Make it yours
Use the generated strategy as a first draft. Replace generic statements with your own business drivers, baseline data, roles and timescales before it goes to the steering group.

13. AI management (ISO/IEC 42001)

ISO/IEC 42001:2023 sets the requirements for an AI management system (AIMS): the policies, roles, risk processes and controls an organisation uses to develop, provide or use AI responsibly. This hub explains the clauses and Annex A controls in plain terms, shows how they apply to a BIM consultancy using AI for design checks and document drafting, and links each step to the platform templates that help you produce the evidence.

Overview — What an AIMS is

An AI management system in plain terms

An AIMS is the set of rules, people and processes that keeps your use of AI under control. It answers simple questions: which AI systems do we use, what could go wrong, who is responsible, and how do we know it is working?

ISO/IEC 42001 follows the same high-level structure as ISO 9001 and ISO/IEC 27001: clauses 4 to 10 set the management system requirements, and Annex A lists controls you select based on your risks.

It is certifiable. An accredited certification body can audit your AIMS and issue a certificate, usually valid for three years with surveillance audits in between.

Key facts

Figure	Meaning	Source
2023	Year ISO/IEC 42001 was published	ISO/IEC
7	Requirement clauses (4 to 10)	ISO/IEC 42001
38	Annex A controls in 9 control areas	ISO/IEC 42001 Annex A
3 years	Typical certification cycle	Certification practice

> **It is about management, not the algorithm**
> ISO/IEC 42001 does not tell you how to build a model. It asks you to show that AI is used for a defined purpose, its risks and impacts are assessed, controls are in place, and people stay accountable.

Overview — Who it is for

AI roles an organisation can hold

AI provider

Offers an AI product or service to others, such as a software vendor selling an automated model checker.

AI producer or developer

Designs, trains or fine-tunes AI systems, such as a firm building its own clash classification model.

AI user

Uses AI systems in its own work, such as a BIM consultancy using a language model to draft BEP sections.

AI partner or customer

Supplies data, tools or integration, or receives AI-assisted outputs, such as a client receiving AI-checked models.

A typical BIM consultancy

Most design and BIM firms are AI users, and some are also developers when they build scripts, plug-ins or connectors around AI models.

Typical systems in scope are general-purpose AI assistants for writing, AI features in authoring and checking tools, computer vision on site photos and in-house automations that call AI through an API or an MCP connector.

Overview — Related standards

Standards that work with ISO/IEC 42001

Standard	What it covers	How it relates
ISO/IEC 27001:2022	Information security management system	Same structure; many firms run one integrated system covering both
ISO/IEC 23894:2023	Guidance on AI risk management	Helps you design the AI risk assessment in clauses 6.1.2 and 8.2
ISO/IEC 42005:2025	Guidance on AI system impact assessment	Helps with the impact assessment in clauses 6.1.4 and 8.4 and Annex A.5
ISO/IEC 22989:2022	AI concepts and terminology	Defines the vocabulary used in the AIMS
ISO/IEC 42006	Requirements for bodies that audit and certify an AIMS	Relevant when choosing a certification body
ISO 19650 series	Information management for built assets	AI outputs entering the CDE still follow ISO 19650 review and approval

Regulation is separate
Certification to ISO/IEC 42001 does not by itself prove legal compliance, for example with the EU AI Act. It gives you a structured way to collect the evidence that regulation may ask for.

Clauses 4–10 — Context and leadership

Clauses 4 and 5

Clause	What it asks	What auditors look for	Platform template
4.1 Context	Understand internal and external issues, the purpose of your AI systems and your AI roles	A context analysis that names your roles (user, developer, provider) and real issues	AI-CTX
4.2 Interested parties	Identify who cares about your AI use and what they require	A register with clients, staff, regulators and suppliers and their requirements	AI-CTX
4.3 Scope	Set the boundaries of the AIMS	A scope statement that matches the AI inventory, with justified exclusions	AI-CTX, AI-INV
4.4 AIMS	Establish, run and improve the system	Evidence the processes link together and are maintained	All templates
5.1 Leadership	Top management leads and resources the AIMS	Directors in reviews, budget and decisions on record	AI-MR
5.2 AI policy	Set an AI policy that frames objectives	An approved, communicated and reviewed policy	AI-POL
5.3 Roles	Assign responsibilities and authorities	A RACI and named owners for each AI system	AI-RACI

Clauses 4–10 — Planning

Clause 6

Clause	What it asks	What auditors look for	Platform template
6.1.1 Risks and opportunities	Plan actions on risks and opportunities and set AI risk criteria	Documented risk criteria: likelihood, consequence and acceptance levels	AI-RISK
6.1.2 AI risk assessment	Identify, analyse and evaluate AI risks with a repeatable method	A risk register per AI system using the agreed criteria	AI-RISK
6.1.3 AI risk treatment	Choose treatments and controls, compare with Annex A, produce the SoA	A treatment plan and a Statement of Applicability with justifications	AI-RISK, SoA
6.1.4 Impact assessment	Assess consequences for individuals, groups and society	Impact assessments for systems that affect people	AISIA

Clause	What it asks	What auditors look for	Platform template
6.2 AI objectives	Set measurable objectives and plan how to meet them	Objectives with targets, owners, dates and results	AI-OBJ
6.3 Planning of changes	Change the AIMS in a planned way	Change records for new tools or major updates	AI-CAPA

Risk and impact are different
The risk assessment looks at risks to your organisation and its objectives. The impact assessment looks outward, at the consequences for people and society. Auditors expect both.

Clauses 4–10 — Support and operation

Clauses 7 and 8

Clause	What it asks	What auditors look for	Platform template
7.1 Resources	Provide what the AIMS needs	Budget, tools and time allocated	AI-MR
7.2 Competence	Make sure people affecting AI performance are competent	Training records and a competence matrix	AI-RACI
7.3 Awareness	People know the policy and their part	Staff can explain the AI policy and how to report concerns	AI-POL
7.4 Communication	Decide what is communicated, to whom and when	A communication plan, including to clients	AI-CTX
7.5 Documented information	Control AIMS documents and records	Version control, approval and retention	All templates
8.1 Operational control	Plan and control AI processes, including outsourced ones	Life cycle procedures followed in practice	AI-LC
8.2 Risk assessment	Repeat risk assessments at intervals or on change	Dated reassessments after tool or model changes	AI-RISK
8.3 Risk treatment	Carry out the treatment plan and check it works	Evidence controls are operating	AI-RISK
8.4 Impact assessment	Repeat impact assessments at intervals or on change	Updated assessments when use or data changes	AISIA

Clauses 4–10 — Evaluation and improvement

Clauses 9 and 10

Clause	What it asks	What auditors look for	Platform template
9.1 Monitoring	Decide what to measure, how and when, and evaluate results	KPIs such as error rates in AI-drafted text and review coverage	AI-OBJ
9.2 Internal audit	Audit the AIMS at planned intervals	An audit programme, independent auditors and reports	AI-AUD
9.3 Management review	Top management reviews the AIMS	Minutes with inputs, decisions and actions	AI-MR
10.1 Continual improvement	Keep improving the AIMS	Improvements tracked over time	AI-CAPA
10.2 Nonconformity	Fix problems, find causes and prevent repeats	A corrective action log with root causes and closure	AI-CAPA

Evidence auditors commonly ask for

- ☐ AIMS scope statement and AI system inventory
- ☐ Approved AI policy and evidence it was communicated
- ☐ RACI with named owners
- ☐ AI risk criteria, risk register and treatment plan
- ☐ Statement of Applicability with reasons for each control
- ☐ Impact assessments for systems affecting people
- ☐ AI objectives with measured results
- ☐ Training records
- ☐ Internal audit report and management review minutes
- ☐ Corrective action log

Annex A controls — Control areas

Annex A control areas

Control area	Objective in short	Example evidence in a BIM firm
A.2 Policies related to AI	A documented AI policy aligned with other policies and reviewed	AI policy referencing the information security policy and the BIM information standard
A.3 Internal organization	Clear AI roles and a way to report concerns	RACI; a concerns channel on the intranet
A.4 Resources for AI systems	Document data, tools, computing and people used	Inventory entries listing the model, data sources, hosting and competent users
A.5 Assessing impacts of AI systems	A process to assess and record impacts on people and society	Impact assessment for site computer vision covering worker privacy

Control area	Objective in short	Example evidence in a BIM firm
A.6 AI system life cycle	Responsible design, verification, deployment, monitoring and logging	Validation tests for a clash triage model before use; event logs kept
A.7 Data for AI systems	Data management, quality, provenance and preparation	Records of which project models trained a classifier and with whose permission
A.8 Information for interested parties	Information for users, external reporting and incident communication	Note to clients on where AI was used in deliverables; incident contact
A.9 Use of AI systems	Responsible use within the intended purpose	Rules that AI-drafted BEP text is reviewed by a named author
A.10 Third-party and customer relationships	Allocate responsibilities with suppliers and customers	AI clauses in software contracts and appointments

Controls are selected, not automatic
You compare your risk treatment with Annex A and record in the Statement of Applicability which controls apply and why. Excluding a control is allowed when you can justify it.

Annex A controls — Key controls in detail

Controls that matter most for BIM firms

A.4.3 Data resources
Know which project data goes into each AI tool, including prompts and uploaded models.

A.5.4 Impact on individuals or groups
Assess effects on workers, occupants and the public, for example from site monitoring.

A.6.2.4 Verification and validation
Define tests and acceptance criteria before an AI tool is relied on.

A.6.2.8 Event logs
Decide when to log AI use so you can trace an output back to its source.

A.7.5 Data provenance
Record where training and reference data came from and who owns it.

A.9.4 Intended use
Use AI only for the purposes recorded in the inventory and the supplier's documentation.

A.10.3 Suppliers
Check AI vendors' terms on data retention, training on your data and hosting location.

A.8.4 Communication of incidents
Plan how you tell users and clients when an AI tool fails or leaks data.

AI in BIM — Use cases and risks

AI use cases, risks and controls

Use case	Main risks	Controls	Annex A
Generative design options	IP ownership of outputs; options that ignore constraints	Contract terms on IP; designer approves every option	A.10.2, A.9.2
Clash triage and grouping	Real clashes hidden as false positives; over-reliance	Sample-check AI groupings; human decides status changes	A.6.2.4, A.9.4
Document review and drafting (EIR, BEP)	Hallucinated requirements or clause references; data leakage	Named author checks every clause reference; approved tools only	A.9.2, A.4.3
Automated model checking	Rules applied wrongly; missed errors	Validate rule sets against known test models	A.6.2.4, A.6.2.6
MCP connectors to authoring tools	Unintended changes to models; broad data access	Least-privilege access; read-only by default; logged actions	A.6.2.8, A.4.4
Site photo analysis	Worker privacy; bias in detection	Impact assessment; consent and retention rules	A.5.4, A.7.4

Five risks to name in every assessment

Data leakage

Project or client data sent to a public AI service that keeps or trains on it.

Hallucinated requirements

Plausible but wrong clauses, standards or quantities in drafted documents.

Intellectual property

Unclear ownership of AI outputs and of the data used to produce them.

Bias

Models that perform worse for some building types, regions or groups of people.

Over-reliance

Staff accepting AI output without the checks they would apply to a colleague's work.

AI in BIM — Inventory example

AI system inventory (example, AI-INV)

System	Purpose	Our role	Data used	Owner	Risk level
Enterprise AI assistant	Drafting and summarising BEP, EIR and meeting notes	User	Project documents, no personal data	BIM manager	Medium

System	Purpose	Our role	Data used	Owner	Risk level
Clash triage add-in	Grouping and ranking clashes in federated models	User	Federated coordination models	Lead BIM coordinator	Medium
In-house IDS checker with AI explanations	Explaining failed checks to modellers	Developer	IFC models and IDS files	Digital lead	Low
MCP connector to authoring tool	Querying and batch-editing model parameters	Developer and user	Live authoring models	Digital lead	High
Site progress vision service	Comparing site photos with the 4D model	User	Site photos that may show workers	Project director	High

Start with the inventory
Your scope, risk assessment, impact assessments and Statement of Applicability all depend on knowing which AI systems you have. Include AI features inside existing software, not only stand-alone tools.

AI in BIM — Impact assessment example

Worked example: site progress vision service (AISIA)

1. **Describe the system.** Cameras and phone photos are compared with the 4D model to report progress each week. A vendor hosts the model.
2. **Identify who is affected.** Site workers in the photos, subcontractors whose progress is reported and the client relying on the reports.
3. **Assess impacts on individuals.** Workers could be identified and monitored without knowing. Wrong progress data could affect subcontractor payments.
4. **Assess wider impacts.** Low societal impact, but trust on site could suffer if monitoring feels covert.
5. **Decide treatments.** Blur faces before upload, tell workers through site induction, keep photos for a limited period and have a person confirm any payment-related report.
6. **Record and review.** Record the assessment, link it to the risk register and review it when the vendor, cameras or use change.

Impact summary

Impact	Severity before	Treatment	Severity after
Worker privacy	High	Face blurring, notice, limited retention	Low
Unfair payment decisions	Medium	Human confirmation of reports	Low
Loss of trust on site	Medium	Clear communication and a concerns channel	Low

Implementation — 12-step roadmap

Implementing an AIMS in 12 steps

1. **1. Get leadership backing.** Agree why you want an AIMS, the budget and a named AIMS lead.
2. **2. Build the AI inventory.** List every AI system and AI feature in use or planned (AI-INV).
3. **3. Analyse context and parties.** Record issues, AI roles and interested parties (AI-CTX).
4. **4. Set the scope.** Decide which offices, services and systems the AIMS covers.
5. **5. Write the AI policy.** Approve and communicate the policy (AI-POL).
6. **6. Assign roles.** Name owners and complete the RACI (AI-RACI).
7. **7. Assess AI risks.** Set criteria, then assess each system (AI-RISK).
8. **8. Assess impacts.** Complete impact assessments where people may be affected (AISIA).
9. **9. Select controls and write the SoA.** Compare treatments with Annex A and justify each decision (SoA).
10. **10. Set objectives and run controls.** Set measurable objectives (AI-OBJ) and follow the life cycle procedure (AI-LC).
11. **11. Audit and review.** Run an internal audit (AI-AUD) and a management review (AI-MR).
12. **12. Correct and certify.** Close nonconformities (AI-CAPA), then book the stage 1 and stage 2 audits.

Implementation — Statement of Applicability

Writing a good SoA

The Statement of Applicability lists every Annex A control, says whether it applies, explains why and states whether it is implemented.

Link each decision to the risk treatment plan. A control that applies should point to evidence, such as a procedure, a record or a contract clause.

Keep it current. Update it when you add an AI system, change a supplier or close a corrective action.

SoA example rows

Control	Applies?	Justification	Status
A.6.2.4 Verification and validation	Yes	We rely on AI clash triage; outputs must be validated	Implemented
A.7.2 Data for development	Yes	We fine-tune a classifier on project data	In progress
A.6.2.5 Deployment	Yes	In-house tools are deployed to staff	Implemented
A.8.3 External reporting	Yes	Clients need a way to report problems with AI-assisted deliverables	Planned

> **Avoid blanket exclusions**
> Excluding whole control areas because 'we only use AI' is a common audit finding. Users still need controls on data, intended use, suppliers and information for interested parties.

Implementation — Audit and certification

From internal audit to certificate

1. **Internal audit.** Audit all clauses and the applicable controls using auditors who do not audit their own work.
2. **Management review.** Top management reviews audit results, objectives, risks and resources and records decisions.
3. **Choose a certification body.** Pick an accredited body that offers ISO/IEC 42001 certification.
4. **Stage 1 audit.** The auditor reviews documents and readiness, often remotely, and raises any concerns.
5. **Stage 2 audit.** The auditor tests that the AIMS works in practice through interviews, records and sampling.
6. **Certificate and surveillance.** After nonconformities are closed, the certificate is issued, with surveillance audits each year and recertification after three years.

Readiness checklist

- ☐ Scope approved and consistent with the inventory
- ☐ AI policy approved, published and understood by staff
- ☐ Risk and impact assessments complete for all in-scope systems
- ☐ SoA complete with justifications and evidence references
- ☐ Objectives set and at least one round of results measured
- ☐ Training delivered and recorded
- ☐ Internal audit completed across all clauses
- ☐ Management review held and minuted
- ☐ Corrective actions raised and on track
- ☐ Supplier terms for AI tools reviewed

FAQ, quiz & AI — FAQ

Frequently asked questions

Do we need ISO/IEC 42001 if we only use AI tools?

It is voluntary, but users are in scope. If clients ask how you control AI in deliverables, an AIMS gives you a clear answer.

Can we combine it with ISO/IEC 27001?

Yes. Both share the same structure, so policy, document control, audit and management review can be shared.

Is ChatGPT-style use in scope?

Yes. General-purpose AI assistants used for work belong in the inventory and the risk assessment.

What is the difference between the risk assessment and the impact assessment?

The risk assessment covers risks to the organisation and its objectives. The impact assessment covers consequences for people and society.

Do we have to apply every Annex A control?

No. You decide based on your risks and record the reasons in the Statement of Applicability.

How long does implementation take?

For a small or medium firm, several months is typical. It depends on the number of AI systems and existing management systems.

Who should own the AIMS?

A named AIMS lead with top management support. Each AI system also needs an owner.

Does certification prove we meet the EU AI Act?

No. It helps you organise evidence, but legal obligations must be checked separately.

How does it fit with ISO 19650?

AI outputs that enter the CDE still go through ISO 19650 checking, review and approval. The AIMS adds controls on how the AI was used.

Should AI-drafted documents be labelled?

Many firms record AI assistance in the revision notes. Agree the approach with clients and set it in your AI policy.

What about MCP connectors?

Treat each connector as an AI system or a component of one. Limit its permissions, log its actions and assess the risk of unintended model changes.

FAQ, quiz & AI — Ask AI & generate

Official sources

- ISO/IEC 42001:2023 — ISO page for the AI management system standard (https://www.iso.org/standard/81230.html)
- NIST AI Risk Management Framework — Voluntary AI risk framework that maps well to the AIMS (https://www.nist.gov/itl/ai-risk-management-framework)
- EU AI Act — Regulation (EU) 2024/1689 (https://eur-lex.europa.eu/eli/reg/2024/1689/oj)

14. Glossary

ISO 19650 process

Appointment
The agreed instruction for a party to provide information, works, goods or services, whether or not a formal contract exists.

Assessment and need
The first activity, where the appointing party sets up its information requirements, standards and CDE before any appointment.

Asset life cycle
The whole life of an asset from the first idea through design, construction and operation to disposal.

BIM Level 2
A former UK maturity label for managed, file-based collaboration, now superseded by the ISO 19650 terminology.

BS 1192
The former British standard for collaborative production of information, whose naming and CDE ideas fed into ISO 19650.

Capability and capacity assessment
An assessment of whether each task team has the skills, resources and technology needed to deliver its information.

Collaborative production of information
The activity in which task teams produce, check and share information containers in the CDE.

Delivery phase
The part of the asset life cycle in which the asset is designed, built and commissioned.

Federation
Combining separate information models, usually one per discipline, so they can be viewed and checked together without merging their ownership.

Federation strategy
The agreed approach to splitting the information model into containers and combining them for coordination.

Information container breakdown structure
The hierarchy used to divide project information into containers by location, system, discipline or other criteria.

Information delivery milestone
A planned date by which a defined set of information must be delivered.

Information exchange
The act of passing information from one party to another at an agreed point and in an agreed form.

Information management
The planned way of specifying, producing, checking, sharing and storing information across the life of an asset.

Information management function
The set of information management duties an organisation takes on, which may be carried out in-house or by a third party.

Information management process
The eight-activity sequence in ISO 19650-2 that runs from assessment and need through to project close-out.

Information model
The set of structured and unstructured information containers for an asset, such as models, drawings, schedules and documents.

Information model delivery
The activity in which the lead appointed party authorises the information model and submits it to the appointing party for acceptance.

Invitation to tender stage
The activity in which the appointing party prepares the tender package, including the EIR and reference information.

ISO 19650
The international series of standards for managing information over the whole life of a built asset using building information modelling.

ISO 19650-1
Part 1 of the series, setting out the concepts and principles of information management.

ISO 19650-2
Part 2, describing the information management process during the delivery phase of assets.

ISO 19650-3
Part 3, covering information management during the operational phase of assets.

ISO 19650-4
Part 4, setting out the process and criteria for information exchange between parties.

ISO 19650-5
Part 5, describing a security-minded approach to information management.

ISO 19650-6
Part 6, covering the collection and use of health and safety information across the asset life cycle.

Key decision point (KDP)
A moment in the project when the appointing party must make a significant decision, which drives what information is needed.

Lessons learned
Knowledge gathered during a project about what worked and what did not, recorded for reuse on future work.

Mobilisation
The activity in which the delivery team gets its people, technology and procedures ready and tests them before production starts.

Operational phase
The part of the asset life cycle in which the asset is used, operated and maintained.

PAS 1192-2
The withdrawn UK specification for information management in the capital delivery phase, replaced by ISO 19650-2.

PAS 1192-3
The withdrawn UK specification for information management in the operational phase, replaced by ISO 19650-3.

PAS 1192-5
The withdrawn UK specification for security-minded BIM, replaced by ISO 19650-5.

PAS 1192-6
The UK specification for sharing structured health and safety information using BIM, a forerunner of ISO 19650-6.

Plain language question (PLQ)
A question written in everyday terms that the client needs answered at a key decision point, used to derive information requirements.

Post-appointment BEP
The BEP confirmed and developed after appointment, forming part of the agreed appointment documents.

Pre-appointment BEP
The BEP submitted with a tender to show how the prospective delivery team would meet the EIR.

Project close-out
The final activity, where the PIM is archived and lessons learned are captured for future projects.

Review and accept
The appointing party's review of a submitted information model against the agreed requirements and acceptance criteria.

Review and authorise
The lead appointed party's check of the federated information model before it is submitted to the appointing party.

Tender response
The activity in which a prospective lead appointed party prepares its pre-appointment BEP and capability assessment.

UK BIM Framework
The UK's free guidance and standards framework for implementing ISO 19650, which replaced the earlier BIM Level 2 material.

UK National Annex (NA)
The UK annex to ISO 19650-2 that adds detail such as naming fields and status codes.

Information requirements

Acceptance criteria
The measurable conditions an information deliverable must meet before the appointing party accepts it.

Alphanumerical information
The LOIN facet covering the identification and properties of objects, expressed as text and numbers.

Asset information requirements (AIR)
The information needed to operate and maintain an asset, which shapes what the AIM must contain.

Documentation
The LOIN facet listing the documents, such as certificates or manuals, that must accompany an object.

Employer's information requirements
The older UK term for the client's information requirements in a tender, now replaced by exchange information requirements.

EN 17412-1
The European standard that first set out the level of information need concepts, later adopted as ISO 7817-1.

Exchange information requirements (EIR)
The appointment-specific requirements setting out what information is to be delivered, when, in what format and to what level of need.

Geometrical information
The LOIN facet covering detail, dimensionality, location, appearance and parametric behaviour of objects.

Information delivery specification requirement
An information requirement written in machine-readable form so that software can check models against it automatically.

Information purpose
The reason a piece of information is needed, such as cost planning or maintenance, which sets the level of information need.

Information requirement
A statement of what information is needed, when, how and to what level, for a given purpose.

Information requirement schedule
A table listing each required item of information with its purpose, owner, format and milestone.

Information use
A defined way in which information supports a project or asset task, also called a BIM use, such as clash detection or quantity take-off.

ISO 7817
The international standard for specifying the level of information need, which replaced EN 17412-1.

Level of accuracy (LOA)
A specification of how closely measured or modelled geometry must match the real conditions, common in survey and scan-to-BIM work.

Level of definition
A UK umbrella term combining level of geometry and level of information for a model element.

Level of detail
An older label for how much graphical detail a model element shows; often confused with level of development.

Level of development (LOD)
A widely used scale (such as 100 to 500) describing how far a model element's geometry and information have been developed and can be relied on.

Level of geometry (LOG)
The amount of geometric detail and dimensional accuracy expected in a model element.

Level of information (LOI)
The amount of non-graphical data, such as properties and attributes, attached to a model element.

Level of information need (LOIN)
A framework for defining just enough geometry, alphanumeric data and documentation for each purpose and milestone, and no more.

Organisational information requirements (OIR)
The information an organisation needs to meet its high-level strategic and business objectives.

Performance requirement
A requirement stated as the outcome or behaviour an asset or element must achieve rather than how it is made.

Project information production methods and procedures
The agreed methods for producing, checking, reviewing and approving information on a project.

Project information requirements (PIR)
The information the appointing party needs to make decisions while the asset is being delivered.

Project information standard
The project-specific rules for how information is named, structured, classified and exchanged.

Reference information
Existing information, such as surveys or as-built records, shared with the delivery team to inform its work.

Shared resources
Common items such as templates, object libraries and title blocks given to all task teams for consistency.

CDE & status

Access control
Rules in the CDE that decide who can view, edit, share or publish each container.

Archived
The CDE state holding a record of superseded and historical information containers for audit and reference.

As constructed record (CR)
Status code for published information that records the asset as actually built, for the AIM.

Audit trail
The automatic history of who did what to an information container, and when.

Authorised and accepted (A1)
Status code family (A1, A2, A3…) for published information accepted for use at a given stage, such as construction.

CDE solution
The software platform or set of tools used to run the CDE workflow.

CDE workflow
The agreed process that moves information containers between states, with checks and approvals at each transition.

Check, review and approve
The task team's quality step before information moves from work in progress to shared.

Common data environment (CDE)
The agreed place, combining software with an agreed workflow, where project or asset information is gathered, controlled and shared, with each container moving through set states.

Contractual revision (C01)
A revision code (C01, C02…) used for published information that has been contractually accepted.

Document control
The discipline of managing the numbering, revision, approval and distribution of project documents.

Information container
Any uniquely named set of information, such as a file, model, drawing or table, that can be stored and found again in a storage system.

Information container ID
The unique identifier of a container, made by joining the naming fields with hyphens.

Initial status (S0)
Status code for information still in work in progress and not yet suitable for sharing.

Level or location code
The naming field identifying the floor level or location the information relates to.

Markup
Comments, clouds and sketches drawn over a drawing or model view during review.

Metadata
Data that describes an information container, such as its status, revision, author and classification.

Naming convention
The agreed rule for building unique container names from fields such as project, originator, volume, level, type, role and number.

Number field
The sequential naming field that makes each container ID unique.

Originator code
The naming field identifying the organisation that produced the information container.

Partially accepted (B1)
Status code family (B1, B2…) for published information accepted with minor comments still to be resolved.

Preliminary revision (P01)
A revision code (P01, P02…) used for shared information before it is contractually accepted; minor versions may use P01.1.

Project code
The first naming field, a short code that identifies the project.

Published

The CDE state for authorised information that may be used for its stated purpose, such as construction or handover.

Revision
A recorded change to an information container, identified by a revision code so that everyone knows which edition they hold.

Role code
The naming field identifying the discipline or role responsible, such as A for architect or S for structural engineer.

Shared
The CDE state for checked information released to other task teams for coordination or comment, but not for construction.

Status code
A short code, defined in the UK National Annex, recording a container's suitability, such as S1 or A1.

Suitability
Metadata showing what an information container may be used for, such as coordination, information or review.

Suitable for AIM authorisation (S7)
Status code for information submitted so that the AIM can be authorised at handover.

Suitable for coordination (S1)
Status code for shared information that others may use to coordinate their own work.

Suitable for information (S2)
Status code for shared information provided for reference only.

Suitable for PIM authorisation (S6)
Status code for information submitted so that the PIM can be authorised at an information delivery milestone.

Suitable for review and comment (S3)
Status code for shared information issued so that others can review it and return comments.

Suitable for stage approval (S4)
Status code for shared information submitted for the client's approval at the end of a stage.

Superseded
Describes a container revision that has been replaced by a newer one and should no longer be used.

Transmittal
A record that accompanies issued information, stating what was sent, to whom, when and for what purpose.

Type code
The naming field describing the kind of container, such as M3 for a 3D model or DR for a drawing.

Version
An iteration of an information container while it is being worked on, usually tracked automatically by the CDE.

Volume or system code

The naming field for the functional or spatial breakdown of the project, such as a zone, block or system.

Work in progress (WIP)
The CDE state for information being developed by its task team and not yet visible to others.

Documents & plans

BIM execution plan (BEP)
The delivery team's plan explaining how it will meet the information management requirements of the appointment.

BIM meeting minutes (MOM)
The record of decisions and actions from a BIM coordination or progress meeting.

BIM progress report (BPR)
A periodic report on information delivery progress, model quality, issues and risks.

Brief
The client's statement of what the project must achieve, including functions, budget and quality.

Clash report
A report listing detected clashes with their location, elements involved, status and responsible party.

Detailed responsibility matrix (DRM)
A matrix allocating each information deliverable to the task team responsible for producing it.

Document register
The master list of project documents with their current revision and status.

Drawing register
A list of all drawings issued on a project, with numbers, titles, revisions and status.

Handover plan
A plan setting out how and when information and the asset will be transferred to the operator.

Information container QA checklist
A checklist used to confirm that a container meets naming, content and format rules before it is shared.

Information delivery risk register (RR)
A register of risks to delivering information on time and to standard, with owners and mitigations.

Information exchange schedule
A timetable of the information exchanges between parties and the formats to be used at each one.

Information standard
A document setting out the naming, classification, formats and quality rules that information must follow.

Information uses schedule (IU)
A list of the agreed information uses for the project and the stages at which each applies.

Lessons learned report (LL)
A report capturing what went well and what should change, prepared at project close-out.

Master information delivery plan (MIDP)

The delivery team's combined schedule of all TIDPs, showing what information will be produced, by whom and when.

Mobilisation plan (MOB)
The plan for setting up and testing the delivery team's people, CDE and technology before production starts.

Model audit report
A report on a model's health and compliance with the project's standards, such as naming, warnings and coordinates.

Model production and delivery table (MPDT)
A table setting out which model elements each party produces and to what level at each project stage.

Model responsibility matrix (MRM)
A matrix showing which discipline models each element or system, and to what level, at each stage.

Operation and maintenance manual (O&M)
The handover documents explaining how to operate, maintain and replace the installed systems and products.

Project execution plan (PEP)
The overall plan for delivering a project, covering scope, organisation, programme, cost and controls.

RACI matrix (RACI)
A matrix marking each party as Responsible, Accountable, Consulted or Informed for each task.

Responsibility matrix
A table that assigns information management tasks or deliverables to parties, at high level first and then in detail.

Risk register
A log of identified risks with their likelihood, impact, owner and planned response.

Schedule of accommodation
A list of the rooms and spaces a building must provide, with their areas and requirements.

Shop drawing
A detailed drawing prepared by a contractor or supplier showing how an element will be fabricated and installed.

Specification
A written description of the materials, products, workmanship and standards required for the works.

Task information delivery plan (TIDP)
A task team's schedule of the information containers it will deliver and their due dates.

Technical query (TQ)
A formal question raised by a contractor or designer to clarify a technical matter in the design.

Models & data

3D model (3D)
A model representing the geometry of an asset in three dimensions.

4D BIM (4D)

Linking model elements to the construction programme to simulate and check sequence over time.

5D BIM (5D)
Linking model quantities to cost data for estimating and cost control.

6D BIM (6D)
Using model data for sustainability and energy analysis, such as embodied carbon; some regions use 6D for facility management instead.

7D BIM (7D)
Using model data to support facility and asset management after handover.

Asset information model (AIM)
The information model that supports the operation and maintenance of an asset during its operational phase.

Attribute
A data field describing an object or entity; often used interchangeably with property.

BIM authoring tool
Software used to create and edit information models, such as Revit, Archicad or Tekla.

BIMcollab
A cloud issue-management service that uses BCF to track model issues across different BIM tools.

Building information modelling (BIM)
The use of a shared digital representation of a built asset to support design, construction and operation decisions.

Central model
A shared model file that several users work on at once through local copies and synchronisation.

Clash avoidance
Designing in coordinated zones and sequences so that clashes are prevented rather than found later.

Clash detection
Automatically finding places where elements from different models overlap or come too close.

Clash matrix
A table setting which pairs of disciplines or systems are tested against each other and with what tolerance.

Classification
Grouping information and objects using an agreed coded system, such as Uniclass or OmniClass.

Computer-aided design (CAD)
Using software to create 2D drawings and 3D geometry, generally without the rich object data of BIM.

Coordinate reference system (CRS)
A defined system, such as a national grid, for locating positions on the earth.

Coordination model
A federated model prepared specifically for checking interfaces and resolving clashes between disciplines.

Data dictionary
A controlled list of properties with agreed names, definitions, units and data types.

Discipline model

A model produced by one discipline, such as architecture, structure or MEP.

DWG
A widely used proprietary file format for CAD drawings.

Fabrication model
A highly detailed model used to manufacture components directly, often by a specialist contractor.

Family
In Revit, a reusable group of elements with shared parameters and graphics, such as a door or pump type.

Federated model
A combined view of several discipline models used for coordination and review, while each stays owned by its author.

Geographic information system (GIS)
Software and data for capturing, analysing and mapping location-based information.

Georeferencing
Placing a model in a real-world coordinate reference system so it aligns with maps and other data.

Globally unique identifier (GUID)
A long code that identifies a model object uniquely so it can be tracked across exports and versions.

Hard clash
Two physical elements occupying the same space, such as a duct passing through a beam.

Light detection and ranging (LiDAR)
A survey method that measures distances with laser pulses to capture accurate 3D geometry.

Linked model
A model referenced into another model for context or coordination without being merged into it.

Model checking
Running rule-based checks on a model to find errors, missing data or non-compliance.

Model coordination
The regular process of combining models, finding conflicts and agreeing fixes between task teams.

Model element
A part of an information model that represents a component, system or space, with geometry and data.

Model validation
Confirming that a model meets the agreed information requirements, standards and formats.

Navisworks
Autodesk software for federating models, running clash detection and 4D simulation.

Object
A digital representation of a real-world thing in a model, carrying geometry and properties.

Object library
A collection of reusable model objects, such as families or types, shared across a project or organisation.

Parameter
A named value attached to a model element, such as width, fire rating or manufacturer.

Photogrammetry
Deriving 3D measurements and models from overlapping photographs, often taken by drone.

Point cloud
A large set of measured 3D points captured by laser scanning or photogrammetry, representing existing conditions.

Project base point
A reference point in a model, usually at a grid intersection, used to position the building.

Project information model (PIM)
The information model built up during the delivery phase, covering design, construction and handover information.

Property
A characteristic of an object, such as material or U-value, stored as a name and value.

Quantity take-off (QTO)
Measuring quantities of materials and elements, increasingly extracted directly from the model.

Record model
A model updated to reflect the asset as built, handed over for use in operations.

Revit
A widely used BIM authoring application from Autodesk for architecture, structure and MEP.

Scan to BIM
Creating an information model of an existing asset from point cloud survey data.

Shared coordinates
An agreed coordinate system and origin used by all models so they line up when federated.

Shared parameter
A parameter defined once in a shared file so that the same data field can be used across families and projects.

Soft clash
An element intruding into a clearance or access zone around another element, also called a clearance clash.

Solibri
A model checking application that tests IFC models against rule sets and information requirements.

Space
A bounded area or volume, such as a room, used to organise model data and track use.

Survey point
A model point tied to a real-world survey coordinate so the model sits correctly on site.

Tolerance
The acceptable deviation in size, position or clash distance before something is treated as a problem.

Workflow clash
A clash in time or sequence, such as two trades scheduled in the same space at once, found through 4D review.

Workset
A subdivision of a shared Revit model that controls editing ownership and visibility.

Zone
A grouping of spaces sharing a common characteristic, such as a fire compartment or HVAC zone.

Open standards (IFC, BCF, IDS, COBie)

BCF API
The web service form of BCF that lets tools sync issues through a shared server instead of files.

BIM Collaboration Format (BCF)
An open format for exchanging model issues, with viewpoints and comments, between different BIM tools.

BS 1192-4
The UK code of practice that defines COBie for meeting employers' information exchange requirements.

buildingSMART Data Dictionary (bSDD)
An online service hosting classifications and property definitions so they can be linked from models.

buildingSMART International
The not-for-profit body (bSI) that develops openBIM standards such as IFC, BCF, IDS and bSDD.

CityGML
An open data model for 3D city and landscape models, often used alongside IFC in GIS work.

COBie Component
The COBie sheet listing each individual installed item, with its location, serial number and tag.

COBie Type
The COBie sheet describing each type of product or equipment, such as a model of pump, with shared data.

Construction Operations Building information exchange (COBie)
A structured, usually spreadsheet-based, format for handing over asset data such as spaces, types, components and systems.

Coordination View 2.0 (CV 2.0)
The common IFC2×3 model view for exchanging models between design disciplines for coordination.

Data template
A standard structure listing the properties a type of product should carry, as described in ISO 23387.

Design Transfer View (DTV)
An IFC4 model view intended for handing over editable model content between authoring tools.

E57
An open, vendor-neutral file format for storing point cloud and 3D imaging data.

Exchange requirement
The set of information that must be passed in one specific exchange, as defined in an IDM.

gbXML
An open XML schema for transferring building geometry and data to energy analysis software.

IFC entity
A class in the IFC schema, such as IfcWall or IfcDoor, that defines a type of object and its attributes.

IFC2×3
An older but still common IFC release, mostly used with the Coordination View for building models.

IFC4
The IFC release (IFC4 ADD2 TC1) published as ISO 16739-1:2018, adding better geometry, properties and reference views. That edition has been replaced by ISO 16739-1:2024 (IFC 4.3).

IFC4.3
The IFC release that extends the schema to infrastructure such as roads, railways, bridges and ports.

IfcGUID
The compressed 22-character form of a GUID used to identify each object in an IFC file.

Industry Foundation Classes (IFC)
The open, vendor-neutral data schema for exchanging building and infrastructure models between software.

Information Container for linked Document Delivery (ICDD)
A standard container format (ISO 21597) for bundling documents and models together with the links between them.

Information Delivery Manual (IDM)
A method for documenting a business process and the information exchanged within it, published as ISO 29481.

Information Delivery Specification (IDS)
A buildingSMART XML format for writing information requirements that software can check IFC models against automatically.

ISO 12006-2
The framework standard for building classification systems, on which Uniclass and OmniClass are based.

ISO 16739
The international standard that publishes the IFC schema.

ISO 23386
The standard describing how to author and maintain properties in interconnected data dictionaries.

ISO 23387
The standard for data templates that describe construction objects in a consistent, machine-readable way.

ISO 29481
The international standard for the Information Delivery Manual methodology.

LandXML
An open XML format for exchanging civil engineering and survey data such as surfaces and alignments.

Linked data
Publishing data with web identifiers and defined relationships so that datasets can be joined and queried.

Model view definition (MVD)
A defined subset of the IFC schema for a particular exchange purpose, such as coordination.

openBIM
An approach to BIM built on open, vendor-neutral standards so data can flow between any software.

Product data sheet (PDS)
A completed product data template holding the actual values for a specific product.

Product data template (PDT)
A template defining the data a manufacturer should provide about a product type.

Property set (Pset)
A named group of properties attached to an IFC object, such as Pset_WallCommon.

Quantity set
A named group of measured quantities, such as area or volume, attached to an IFC object.

Reference View
An IFC4 model view for referencing models in other tools for coordination, not for editing.

STEP
The ISO 10303 family of standards for exchanging product model data, on whose modelling language IFC was built.

Roles & parties

Appointed party
A provider of information about works, goods or services, appointed by a lead appointed party.

Appointing party
The client or asset owner who appoints others and receives information about works, goods or services from them.

Asset manager
The person responsible for planning and optimising how assets are operated, maintained and renewed.

Asset owner
The organisation that owns an asset and is accountable for its performance over its life.

BIM coordinator
The person who manages model quality and coordination for a task team or across disciplines.

BIM manager
The person leading BIM strategy, standards and delivery within an organisation or project.

BIM modeller
A technician who creates and maintains model content to the project's standards.

Clerk of works
An inspector who checks on site, on the client's behalf, that work matches the design and specification.

Client
The organisation commissioning and paying for the project; in ISO 19650 terms usually the appointing party.

Delivery team
The lead appointed party together with all of its appointed parties.

Design manager

The person who manages the design process, programme and information from the design team.

Facility manager
The person who runs the day-to-day operation and maintenance of a building or estate.

Information approver
The person authorised to approve information for sharing or publication.

Information author
The person who produces an information container and is responsible for its content.

Information manager
The person who carries out the information management function, such as setting up the CDE and checking compliance.

Information reviewer
The person who checks an information container for technical content and standards compliance.

Interested party
A person or organisation that can affect, be affected by, or believe itself affected by a decision or activity.

Lead appointed party
The party appointed directly by the appointing party that coordinates the information of its own appointed parties.

Lead designer
The person or firm responsible for coordinating the design across all design disciplines.

Main contractor
The contractor appointed by the client to build the works and manage subcontractors.

Project manager (PM)
The person responsible for planning and controlling the project to meet its time, cost and quality goals.

Project team
The appointing party and all the delivery teams working on the project.

Quantity surveyor (QS)
The professional who manages construction cost, measurement and contracts.

Subcontractor
A company appointed by the main contractor to carry out a specific part of the works.

Supplier
A party providing products or materials, who may also supply product data for the model.

Task information manager
The person in a task team responsible for its TIDP, information quality and use of the CDE.

Task team
A group of people brought together to perform a specific task, such as a design discipline or trade package.

The Engineer
In FIDIC contracts, the party appointed by the employer to administer the contract and certify work.

Top management
The person or group who directs and controls an organisation at the highest level.

Security & H&S

Accountable person
Under the Building Safety Act, the party responsible for managing building safety risks in an occupied higher-risk building.

Aggregation risk
The risk that combining pieces of harmless information reveals something sensitive.

Building Safety Act (BSA)
The UK law of 2022 that introduced stricter safety duties and information requirements for buildings, especially higher-risk ones.

Built asset security information requirements (BASIR)
Security requirements written into appointments so that every party handles sensitive information correctly.

Built asset security management plan (BASMP)
The plan describing the security policies, processes and controls applied to an asset and its information.

Built asset security manager
The person responsible for overseeing the security-minded approach on behalf of the appointing party.

Built asset security strategy (BASS)
The high-level strategy for managing security risks to an asset and its information.

Construction (Design and Management) Regulations (CDM)
The UK regulations that set health and safety duties for clients, designers and contractors on construction projects.

Construction phase plan (CPP)
The principal contractor's plan for managing health and safety risks while the works are carried out.

Fire strategy
The document describing how a building's design protects people from fire, including escape routes and compartments.

Golden thread of information
The accurate, up-to-date digital record of a higher-risk building's safety information, required under the UK Building Safety Act.

Hazard
Anything with the potential to cause harm, such as working at height or live electrics.

Health and safety file
The record of safety information needed for future maintenance, alteration or demolition of the asset.

Health and safety information
Information about hazards and risks that people need to work on or use an asset safely.

Higher-risk building (HRB)
Under the Building Safety Act, a residential or similar building at least 18 m or seven storeys tall.

Information security
Protecting the confidentiality, integrity and availability of information.

ISO/IEC 27001
The international standard for information security management systems.

Need-to-know
The principle that people get access only to the information they need for their role.

Permit to work
A formal written authorisation needed before high-risk work, such as hot works or confined spaces, can start.

Pre-construction information (PCI)
Information about the site and project, provided by the client, that helps others plan for health and safety.

Principal contractor
Under CDM, the contractor who plans and coordinates health and safety during the construction phase.

Principal designer
Under CDM, the designer who plans and coordinates health and safety during the pre-construction phase.

Residual risk
The risk left over after design and control measures have been applied, which must be communicated to others.

Risk assessment
Identifying hazards, judging how likely and serious harm would be, and deciding on controls.

Risk assessment and method statement (RAMS)
A document setting out the hazards of a task and the safe step-by-step method for doing it.

Security breach management plan
The plan for discovering, containing, reporting and recovering from a security breach or incident.

Security triage process
The initial assessment that decides whether an asset or project needs a security-minded approach and how strong it should be.

Security-minded approach
Managing information so that sensitive details about an asset are protected without stopping legitimate collaboration.

Sensitive information
Information whose loss or misuse could harm the safety, security or resilience of an asset or its users.

Sensitivity assessment
Working out whether information about an asset could be misused to cause harm, and how serious that harm could be.

Operations & assets

As-built
Information that records what was actually constructed, including any changes from the design.

Asset

An item, thing or entity that has potential or actual value to an organisation, such as a building, bridge or pump.

Asset information
Data, documents and models about an asset needed to operate, maintain and manage it.

Asset management
The coordinated activity of an organisation to get value from its assets over their whole life.

Asset register
A structured list of an organisation's assets with their identifiers, locations and key data.

Asset tag
A unique code, often shown on a physical label, that links an installed item to its record in the asset system.

Building management system (BMS)
A control system that monitors and runs building services such as HVAC, lighting and alarms.

Commissioning
Testing and adjusting installed systems so they work as designed before handover.

Computer-aided facility management (CAFM)
Software for managing space, assets, maintenance and services in buildings.

Computerised maintenance management system (CMMS)
Software that schedules and records maintenance work orders, assets and spare parts.

Enterprise asset management (EAM)
Organisation-wide software and practice for managing physical assets across their life cycle.

Facility management (FM)
The services that keep a building working for its users, such as maintenance, cleaning and security.

Handover
The transfer of the completed asset and its information from the delivery team to the client or operator.

Integrated workplace management system (IWMS)
An enterprise platform combining real estate, space, maintenance and sustainability management.

Internet of Things (IoT)
Networked sensors and devices that collect and share data about the physical world.

ISO 55000
The international series of standards for asset management systems.

Life cycle costing (LCC)
Estimating and comparing the costs of design options over the life of an asset.

Planned preventive maintenance (PPM)
Maintenance carried out on a schedule to prevent failure, rather than after a breakdown.

Post-occupancy evaluation (POE)
A review of how a building performs in use compared with what was intended.

Reactive maintenance
Repairs carried out in response to a fault or breakdown.

SFG20

The UK standard library of maintenance task schedules for building services.

Soft Landings (GSL)
An approach, including Government Soft Landings, that keeps designers and builders involved after handover so the asset performs as intended.

Space management
Planning and tracking how rooms and areas are allocated and used.

Spare parts
Replacement components held to keep equipment running, often recorded in COBie and the CMMS.

Trigger event
A planned or unplanned event during operation, such as a refurbishment or failure, that changes the asset and requires information to be exchanged.

Warranty
A manufacturer's or installer's promise to repair or replace a product that fails within a stated period.

Whole life cost (WLC)
The total cost of an asset over its life, including capital, operation, maintenance and disposal.

AI & digital

AI management system (AIMS)
The policies, objectives and processes an organisation uses to develop, provide or use AI responsibly.

AI risk
The effect of uncertainty on objectives arising from developing or using an AI system.

AI system
An engineered system that produces outputs such as predictions, content or decisions for human-defined objectives.

AI system impact assessment (AISIA)
An assessment of the possible effects of an AI system on individuals, groups and society.

Application programming interface (API)
A defined way for software systems to exchange data and commands with each other.

Artificial intelligence (AI)
Computer systems that perform tasks normally needing human judgement, such as recognising patterns or generating text.

Automation
Using software to perform repetitive tasks, such as checks, exports or reminders, without manual effort.

Bias
Systematic error in data or an AI model that leads to unfair or skewed results.

Cloud computing
Using remote servers on the internet to store, manage and process data.

Computer vision
AI that interprets images and video, for example to track site progress or detect safety issues.

Continual improvement
Recurring activity to enhance performance, often through the plan-do-check-act cycle.

Data governance
The rules, roles and processes that keep an organisation's data accurate, secure and properly used.

Data provenance
A record of where data came from and how it has been changed, so its reliability can be judged.

Digital twin
A live digital representation of a physical asset, connected to real data so that its state can be monitored and predicted.

Energy modelling
Simulating a building's energy use and thermal performance to inform design decisions.

Explainability
The extent to which people can understand why an AI system produced a particular output.

Generative AI (GenAI)
AI that creates new content such as text, images or design options from a prompt.

Generative design
Using algorithms to produce and rank many design options against set goals and constraints.

Hallucination
When an AI model produces confident but false or unsupported information.

Human oversight
Keeping a competent person able to review, correct or stop AI outputs and decisions.

ISO/IEC 42001
The international management system standard for artificial intelligence.

Large language model (LLM)
An AI model trained on large amounts of text that can answer questions, summarise and draft documents.

Machine learning (ML)
A branch of AI in which systems learn patterns from data instead of following hand-written rules.

Nonconformity
A failure to meet a requirement of a standard, contract or the organisation's own procedures.

Parametric design
Design in which geometry is driven by rules and parameters, so changing an input updates the whole model.

Prompt
The instruction or question given to a generative AI model.

Retrieval-augmented generation (RAG)
A technique where an AI model first retrieves relevant documents, such as project records, and then answers from them.

Software as a service (SaaS)
Software delivered over the internet on subscription rather than installed locally.

Statement of Applicability (SoA)

The document listing which controls apply, why, and whether they have been implemented.

Visual programming
Building scripts by connecting graphical nodes, as in Dynamo or Grasshopper, to automate modelling tasks.

Contracts & procurement

Bill of quantities (BoQ)
A priced list of measured work items used for tendering and valuing work.

CIC BIM Protocol
A UK contract appendix that brings BIM and information management obligations into appointments.

Defects liability period (DLP)
The period after completion during which the contractor must put right any defects that appear.

Design and build (D&B)
A procurement route in which one contractor is responsible for both design and construction.

Early contractor involvement (ECI)
Bringing the contractor into the team during design to improve buildability, cost and programme.

Employer's requirements (ER)
The client's statement of what a design-and-build contractor must design and deliver.

FIDIC
The international federation of consulting engineers, whose standard contracts are common in the Middle East.

Framework agreement
An umbrella agreement setting terms under which a client can place several future appointments.

Information protocol
Contract terms covering the obligations, liabilities, licences and limits on use of information, built into each appointment.

Intellectual property (IP)
Legal rights over creations such as designs and models, and the licences that allow others to use them.

Invitation to tender (ITT)
The package sent to bidders asking them to price and propose for a defined scope.

Joint Contracts Tribunal (JCT)
The body that publishes a widely used family of UK building contracts.

Key performance indicator (KPI)
A measurable value used to track how well a project or service is meeting its targets.

Licence
Permission granted by the owner of information or software for another party to use it on stated terms.

New Engineering Contract (NEC)
A family of UK contracts, now NEC4, built around collaboration, early warnings and programme management.

Practical completion (PC)

The contract milestone when the works are complete enough for the client to take possession.

Pre-qualification questionnaire (PQQ)
A questionnaire used to shortlist suitable bidders before inviting tenders.

Professional indemnity insurance (PII)
Insurance covering consultants against claims of negligent advice or design.

Provisional sum
An allowance in a contract for work that cannot yet be fully defined.

Request for information (RFI)
A formal request, usually from the contractor, for clarification or missing information.

Retention
A percentage of payments held back by the client as security until defects are corrected.

Scope of services
The description of the services a consultant or contractor is appointed to provide.

Service level agreement (SLA)
An agreement defining the standard of service to be provided, such as response times.

Tender
A formal offer to carry out work or supply services for a stated price and terms.

Traditional procurement
A route in which the client appoints designers first and then tenders the construction separately.

Two-stage tender
A tender where a contractor is chosen early on preliminary terms and the final price is agreed later.

Value engineering (VE)
Reviewing a design to deliver the required function at lower cost or better value.

Variation
A formal change to the scope, design or terms of a contract, also called a change order.

General construction

BREEAM
A UK-originated sustainability assessment and certification method for buildings.

British Standards Institution (BSI)
The UK national standards body.

Buildability
How easily and safely a design can be constructed.

Critical path method (CPM)
A scheduling method that finds the longest chain of dependent activities, which sets the shortest project duration.

Design for manufacture and assembly (DfMA)
Designing components so they are easy to make in a factory and quick to assemble on site.

Embodied carbon

The greenhouse gas emissions from making, transporting, installing and disposing of building materials.

Estidama
Abu Dhabi's sustainability programme, which includes the Pearl Rating System for buildings.

Heating, ventilation and air conditioning (HVAC)
The systems that control temperature, air quality and humidity in a building.

Infrastructure
Networks and structures such as roads, railways, utilities and bridges that support society.

International Organization for Standardization (ISO)
The global body that publishes international standards, including ISO 19650.

ISO 9001
The international standard for quality management systems.

Issued for construction
A drawing or model status meaning it may be used to build from; not to be confused with IFC, the data schema.

Last Planner System (LPS)
A collaborative planning method in which the people doing the work commit to short-term tasks.

Leadership in Energy and Environmental Design (LEED)
A US Green Building Council rating system for sustainable buildings, widely used in the Gulf.

Lean construction
Applying lean principles to construction to remove waste and improve flow and value.

MasterFormat
The North American standard for organising specifications by work results into numbered divisions.

Mechanical, electrical and plumbing (MEP)
The building services disciplines covering HVAC, power, lighting, water and drainage.

Method statement
A document describing step by step how a piece of work will be carried out safely.

Milestone
A significant point or event in a programme, such as the end of a stage.

Modern methods of construction (MMC)
Approaches such as off-site manufacture and modular building that improve speed, quality and safety.

NBS
A UK provider of construction specification and classification tools, including Uniclass.

New Rules of Measurement (NRM)
The RICS rules for measuring and estimating building work, from early cost plans to bills of quantities.

Off-site manufacture
Making building components or modules in a factory before transporting them to site.

OmniClass
A North American classification system for the built environment, organised into 15 tables.

Operational carbon

The emissions caused by the energy a building uses while in operation.

Programme
The time plan for a project, showing activities, durations, dependencies and milestones.

Publicly Available Specification (PAS)
A fast-track standard-like document, often published by BSI, that may later become a full standard.

Quality assurance (QA)
Planned activities that give confidence that information and work will meet the required standards.

Quality control (QC)
Inspecting and testing outputs to find and correct defects.

RIBA Plan of Work (RIBA)
The Royal Institute of British Architects' framework of stages 0 to 7 for organising building projects.

Snagging
Identifying and fixing small defects before handover; known in North America as a punch list.

Stakeholder
Anyone with an interest in, or affected by, a project.

Uniclass
The UK unified classification system for the construction industry, covering everything from complexes to products.

UniFormat
A North American classification that organises building information by elements and systems, useful for early cost plans.

Work breakdown structure (WBS)
A hierarchy that breaks project scope into manageable packages of work.